



Decentralized Crypto World Bank

A Blockchain-Based Banking Platform
with AI-Enhanced Security and Assistance

by

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A pre-thesis 1 report submitted to the Department of Computer Science and Engineering
in partial fulfillment of the requirements for the degree of
B.Sc. in Computer Science

Department of Computer Science and Engineering
BRAC University
February 2026

Declaration

It is hereby declared that

1. The report submitted is our own original work while completing degree at BRAC University.
2. The report does not contain material previously published or written by a third party, except where this is appropriately cited through full and accurate referencing.
3. The report does not contain material which has been accepted, or submitted, for any other degree or diploma at a university or other institution.
4. We have acknowledged all main sources of help.

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Ethics Statement

This project operates exclusively on public blockchain test networks using test tokens with no real monetary value. No real financial transactions, personal banking data, or user financial records are involved at any stage of development or evaluation. All transaction data used for Artificial Intelligence and Machine Learning (AI/ML) model training and evaluation is either synthetically generated or sourced from publicly available anonymized datasets. The platform prototype does not process Know Your Customer (KYC) or Anti-Money Laundering (AML) data in its current scope, and all wallet addresses used during testing are disposable testnet addresses. We acknowledge the ethical considerations inherent in AI-assisted financial decision-making and have incorporated explainability mechanisms (SHapley Additive exPlanations, or SHAP) to ensure that automated risk assessments remain transparent and auditable by human reviewers. Future phases plan to incorporate a privacy-preserving ZKP-based identity compliance layer in which users prove KYC status without exposing personal data on-chain; the ethical implications of this design will be evaluated in the final thesis.

Abstract

Global development finance relies on multilayered institutional structures to distribute capital across borders and communities, yet these systems often struggle with fragmented transparency, procedural friction, and uneven access to credit. Although decentralized finance has demonstrated that programmable infrastructures can automate lending and enhance auditability, existing models largely employ flat architectures that do not reflect institutional hierarchies or structured governance. As a result, there remains limited exploration of how tiered financial systems might be represented within decentralized environments while preserving oversight and adaptability. Here we present *Crypto World Bank*, a prototype framework that models a multi-tier institutional lending architecture on a programmable blockchain infrastructure. We show that hierarchical capital flows, role-based governance, and data-informed risk analytics can be coordinated within a unified smart contract environment, enabling transparent state visibility alongside adaptive decision support. The Crypto World Bank implements a complete institutional banking architecture across six functional domains—deposit mobilization, credit allocation, payment settlement, risk intermediation, liquidity management, and ancillary financial services—within a four-tier hierarchical governance structure. This hybrid design illustrates how institutional finance and decentralized systems may converge, demonstrating that a complete, hierarchically governed banking institution can be implemented on open blockchain infrastructure.

Keywords: Blockchain, Decentralized Finance, Institutional Architecture, Financial Inclusion, Smart Contracts, AI-Augmented Governance, Group Lending, Deposit Mobilization, Reserve Management, Financial Sustainability

Dedication

Dedicated to our families for their unwavering support throughout our academic journey.

Acknowledgment

We would like to express our sincere gratitude to our panel members for their guidance and support throughout this project. We also thank our supervisor, Mr. Annajiat Alim Rasel, Senior Lecturer at the Department of Computer Science and Engineering, BRAC University, for his guidance and support. Finally, we thank the Department of Computer Science and Engineering at BRAC University for providing us with the resources and academic environment to pursue this work.

Contents

Declaration	2
Approval	3
Ethics Statement	5
Abstract	6
Dedication	7
Acknowledgment	8
List of Tables	v
List of Figures	viii
List of Formulas	xii
List of Abbreviations	xiii
1 Introduction	1
1.1 Background	1
1.2 Rationale of the Study	2
1.3 Problem Statement	2
1.4 Objectives	4
1.5 Research Questions	4
1.6 Research Contribution	5
1.7 Blockchain Justification	6
1.7.1 Blockchain Fundamentals and the EVM Execution Model	7
1.8 Proposed Solution	8
1.8.1 Banking Functions of the Platform	8
1.8.2 Service Delivery Across Scale	9
1.8.3 Cross-Tier Lending System	9
1.9 Methodology in Brief	11
1.10 Scopes and Challenges	12
1.10.1 Economic Sustainability: Interest Revenue Versus Gas Costs	12
1.10.2 Resistance to Institutional Capture	13
1.10.3 Stablecoin-First Lending and Supply Scalability	13
1.11 Market Analysis and Partnership Ecosystem	15
1.11.1 Market Sizing	15
1.11.2 Target Customer Segment	15
1.11.3 Partner Ecosystem	16
1.11.4 Incentive Alignment Through the Blockchain Platform	16

2	Literature Review	18
2.1	Preliminaries	18
2.1.1	Review Methodology (PRISMA-Style Frame)	19
2.2	Review of Existing Research	20
2.2.1	Decentralized Lending and DeFi Protocol Design	21
2.2.2	Machine Learning for Blockchain Security	21
2.2.3	Explainable AI in Financial Decision Systems	22
2.2.4	Blockchain for Financial Inclusion	22
2.2.5	Group Lending and Microfinance Digitization	23
2.2.6	Smart Contract Security and Governance	23
2.2.7	AI Security Features for Blockchain Lending	23
2.2.8	Gas Cost Optimization and Layer 2 Scalability	24
2.2.9	Correspondent Banking and Cross-Border Settlement	24
2.2.10	Monetary Policy Distribution and Financial Inequality	25
2.2.11	Real-World Asset Tokenization and Institutional DeFi	25
2.2.12	Graph Neural Networks for Relational Fraud Detection	25
2.2.13	Federated Learning Across Banking Tiers	25
2.2.14	Stablecoin Regulation: MiCA and GENIUS Act	26
2.2.15	On-Chain Credit Passport and Soulbound Tokens	26
2.2.16	Institutional DeFi Gap and the mBridge/Agora Landscape	26
2.2.17	LLMs in Finance and Hallucination Risk	26
2.3	Literature Review Summary	27
2.4	Comparative Protocol Analysis	30
2.4.1	Literature Synthesis	31
2.5	Summary of Key Findings	31
3	System Architecture and Design	34
3.1	Prototype Scope	34
3.2	High-Level Architecture	34
3.3	Blockchain Platform Selection	37
3.3.1	Transaction Verification and Consensus	38
3.3.2	Oracle Architecture: Off-Chain AI to On-Chain Decision	39
3.4	Data Model and Database Design	39
3.4.1	Entity Summary	43
3.4.2	EER Constructs Applied	44
3.4.3	Normalization	44
3.4.4	Indexing Strategy	44
3.4.5	Functional Dependencies	45
3.4.6	Relational Integrity Constraints	46
3.5	On-Chain and Off-Chain Data Partitioning	46
3.6	Digital Identity System	46
3.6.1	ZKP KYC and zkAML Compliance Architecture	47
3.7	User Taxonomy and Onboarding Flows	48
3.7.1	ERC-4337 Account Abstraction for Retail Onboarding	49
3.7.2	Tiered Risk-Based KYC	50
3.7.3	Five-Stage Conversion Funnel for Non-Crypto Users	50
3.8	Kinked Interest Rate Model	51
3.9	Liquidation Engine	51

3.10	SavingsVault and FixedDeposit Modules	52
3.11	On-Chain Credit Passport (Soulbound Token)	52
3.12	Cross-Chain Bridge Architecture	53
3.13	Multi-Entity and Cross-Tier Capital Operations	54
3.13.1	InterBankLendingPool: Fully Specified Same-Tier Lending	54
3.13.2	Upward Surplus Repatriation	55
3.13.3	Syndicated and Club Lending	56
3.13.4	Senior–Junior Tranching Lending Pools	57
3.13.5	Cross-Tier Treasury FX Swap	57
3.13.6	Multilateral Settlement Netting Engine	58
3.13.7	Database, Sprint, and Contract Consistency for Multi-Entity Operations	58
3.14	Reentrancy and Security Analysis	59
3.15	System Modeling	60
3.15.1	Use Case Diagram	60
3.15.2	Activity Diagrams	62
3.15.3	Data Flow Diagrams	64
3.15.4	Sequence Diagrams	65
3.15.5	Four-Tier Capital Flow	68
3.16	Auxiliary Dual-Currency Facility	69
3.17	Banking Product Suite	69
3.17.1	Savings Products	69
3.17.2	Checking and Transactional Accounts	70
3.17.3	Group / Solidarity Lending	70
3.17.4	Foreign Exchange and Multi-Currency Operations	72
3.17.5	Trade Finance Facilitation (Planned)	72
3.17.6	Privacy-Preserving Identity Compliance (Planned)	72
3.18	Governance Framework	73
3.18.1	Network Membership Governance	73
3.18.2	Business Network Governance	73
3.18.3	Technology Infrastructure Governance	74
3.18.4	Regulatory Compliance Considerations	74
3.18.5	Asset Tokenization	74
3.19	Five-Layer Defense-in-Depth Security Architecture	76
3.20	Threat Model and Security Controls	79
4	Methodology	82
4.1	Development Methodology	82
4.2	Planned AI/ML Support and Risk-Score Wiring	83
4.3	Graph Neural Network Extension	84
4.4	Federated Learning Across Banking Tiers	84
4.5	Formal Verification of Reserve Invariants (Certora)	85
4.6	Foundry Invariant and Fuzz Test Suite	85
4.7	Agent-Based Economic Feasibility Simulation	85
4.8	Real-Time Dashboard Pipeline and Runtime Monitoring	86
4.9	Transaction-State Machine for Frontend UX	87
4.10	EIP-712 Authentication and API Hardening	88
4.11	Evaluation Methodology	88

4.11.1	LLM Assistant Evaluation Protocol	89
4.12	Sprint Plan and Deliverables	90
4.12.1	Sprint 1: Foundation and Core Banking (Weeks 1–3)	90
4.12.2	Sprint 2: Lending Features and Communication (Weeks 4–6)	90
4.12.3	Sprint 3: AI/ML Security and Finalization (Weeks 7–8)	91
4.13	SDLC Stage Mapping	92
4.14	Design Decisions and Alternatives	92
4.14.1	Justification of Selected Technologies	95
4.15	Design Patterns	97
4.16	Software Testing Strategy	97
5	Market Analysis and Feasibility	99
5.1	Market Sizing	99
5.2	Target Customer Segment	99
5.3	Partner Ecosystem	100
5.4	Competitive Landscape	100
5.5	Risk Taxonomy	103
5.6	Technical Feasibility	104
5.7	Economic Feasibility	105
5.8	Revenue Projection	105
5.8.1	Transaction Economics: Interest Rates	108
5.8.2	Global Economic Impact	109
5.8.3	Value Proposition and Go-to-Market	110
5.9	Currency Risk and the Stablecoin Imperative	110
5.10	MiCA and GENIUS Act Compliance Mapping	111
5.11	Bangladesh Regulatory Reality	112
5.12	Bootstrap Funding and the Tier 1 Capitalization Problem	113
5.13	Prototype Scope and Limitations	113
5.14	Accessibility Assessment: A Borrower in Rural Sylhet	114
6	Conclusion	116
A	Technology Stack	119
A.1	In-product assistant: local large language model (LLM) integration (prototype)	119
B	Smart Contract Capabilities	123
Appendix C: Deployed Testnet Contract Addresses		125
Appendix D: WorldBankReserve Contract Interface		126
References		128

List of Tables

1.1	Borrower tier access rules.	11
1.2	Tiered Borrower Access Rules by Borrower Type and Loan Size	11
1.3	Market segments.	15
1.4	Market Sizing: TAM, SAM, and SOM for the Crypto World Bank	15
1.5	Target customer segment profile.	15
1.6	Target Retail Customer Segment Profile and Constraints	15
1.7	Partner categories and roles.	16
1.8	Partner Ecosystem: External Dependencies and Integration Points	16
2.1	Top-10 ranked literature subset under the PRISMA-style frame. Ranking is by direct architectural impact on the CWB design; each entry maps to a v15 section that operationalizes the finding.	20
2.2	Literature review summary (Part A): DeFi lending, fraud detection, and XAI.	27
2.3	Literature Review Summary (Part 1): DeFi Lending and Hierarchical Architecture	27
2.4	Literature review summary (Part B): DeFi protocol systematisation and adaptive lending.	28
2.5	Literature Review Summary (Part 2): Governance, Settlement, and Institutional Adoption	28
2.6	Literature review summary (Part C): blockchain P2P lending, privacy-preserving ML, and smart contract auditing.	28
2.7	Literature Review Summary (Part 3): Cross-Border Settlement and Security Tooling	28
2.8	Literature review summary (Part D): CBDC design, cross-border settlement, and multi-chain DeFi.	29
2.9	Literature Review Summary (Part 4): Risk Monitoring and AI/ML in Finance	29
2.10	Literature review summary (Part E): multi-chain DeFi, smart microfinance, gas optimisation, and SHAP-based credit models.	30
2.11	Literature Review Summary (Part 5): Institutional Adoption and Financial Inclusion	30
2.12	Comparative protocol analysis: existing DeFi lending protocols vs. the Crypto World Bank (CWB). ✓ = implemented/present; ⚡ = designed/partial; ● = planned; ✕ = absent.	31
3.1	Prototype scope: feature implementation status as of pre-thesis submission. ✓ = Implemented and testnet-verified; ⚡ = Designed or partially scaffolded; ● = Planned for final thesis phase.	34
3.2	Blockchain platform selection criteria and justification.	38
3.3	Blockchain Platform Selection: Comparison of EVM-Compatible Networks	38
3.4	Blockchain platform selection (continued): operational and deployment factors.	38
3.5	Blockchain Platform Selection (Continued): Operational and Deployment Factors	38
3.6	Database entity summary (15 entities).	43

3.7	Database Entity Summary: 15 Normalized Entities in the Relational Schema	43
3.8	EER constructs applied: specialisation, hierarchy, and constraints.	44
3.9	EER Constructs Applied: Specialization, Hierarchy, and Constraints	44
3.10	Indexing strategy: B-tree indexes for time-window and high-frequency queries.	45
3.11	Indexing Strategy: B-tree Indexes for Time-Window and High-Frequency Queries	45
3.12	Representative functional dependencies.	45
3.13	Functional Dependencies in the Core Lending and Identity Schema	45
3.14	Relational integrity constraints.	46
3.15	Relational Integrity Constraints: Referential and Domain Rules	46
3.16	Data partitioning between on-chain and off-chain storage.	46
3.17	On-Chain vs. Off-Chain Data Partitioning: State, Analytics, and Privacy Boundaries	46
3.18	Seven-user taxonomy with KYC level, wallet type, and data-residence assumption. The taxonomy expands the actor list referenced in the use-case diagram of Section 3.15.1.	49
3.19	Tiered, risk-based KYC ladder. Higher KYC level unlocks larger loans but requires proportionally more verification.	50
3.20	Network membership governance.	73
3.21	Banking Product Suite: Deposit, Credit, and Ancillary Contract Specifications	73
3.22	Business network governance.	73
3.23	Security Threat Model: Vulnerability Classes, Attack Vectors, and Mitigations	73
3.24	Technology infrastructure governance.	74
3.25	Governance Framework: Operational, Business, and Technology Governance Layers	74
3.26	Five-layer defense-in-depth model for the Crypto World Bank platform. Each layer is necessary; breaching the contract layer requires defeating every layer above it.	78
3.27	Threat model and security controls mapping, expanded to span all five layers of the defense-in-depth stack (Section 3.19).	80
4.1	Sprint 1 backlog: smart contract development (21 pts).	90
4.2	Agile Sprint Plan: Three-Sprint Development Timeline and Deliverables .	90
4.3	Sprint 1 backlog: frontend foundation (13 pts) and database schema (8 pts).	90
4.4	Technology Stack: Primary Choices and Justifications	90
4.5	Sprint 2 backlog: lending and communication features (50 pts).	91
4.6	Technology Stack (Continued): Framework and Tooling Selection	91
4.7	Sprint 3 backlog: AI/ML security and polish (38 pts).	91
4.8	Technology Stack: Second-Choice Alternatives and Trade-offs	91
4.9	SDLC stage mapping.	92
4.10	Technology Stack: Additional Alternative Evaluations	92
4.11	Design decisions and alternatives considered.	93
4.12	Software Testing Strategy: Acceptance Criteria Across Four Test Layers . .	93
5.1	Market segments with supporting data.	99
5.2	Market Sizing: DeFi Lending TVL, Remittances, and MSME Financing Gap	99
5.3	Target customer segment profile.	100
5.4	Target Customer Segment: Retail Borrower Profile and Product Fit	100

5.5	Partner categories and roles.	100
5.6	Partner Ecosystem: Integration Partners and External Dependencies (Chapter 5)	100
5.7	Detailed competitive landscape analysis (Part 1).	101
5.8	Competitive Landscape (Part 1): DeFi Lending and Payment Rail Protocols	101
5.9	Detailed competitive landscape analysis (Part 2).	102
5.10	Competitive Landscape (Part 2): Inclusion Wallets and Institutional Blockchain Systems	102
5.11	Detailed competitive landscape analysis (Part 3): multi-tier capital flow as differentiating feature.	102
5.12	Competitive Landscape (Part 3): Multi-Tier Capital Flow as Differentiating Feature	102
5.13	Risk taxonomy and mitigation.	103
5.14	Risk Taxonomy: Technical, Financial, Regulatory, and Operational Risk Categories	103
5.15	Technical feasibility assessment.	104
5.16	Technical Feasibility Assessment: Infrastructure Readiness and Ecosystem Maturity	104
5.17	Economic feasibility — zero-cost prototype.	105
5.18	Economic Feasibility: Cost Drivers vs. Revenue Potential on Layer-2 Networks	105
5.19	Revenue projection assumptions.	106
5.20	Revenue Projection by Tier (Base Case, USD Millions)	106
5.21	System-wide annual revenue summary.	106
5.22	Revenue Projection Assumptions and Sensitivity Inputs	106
5.23	Default rate sensitivity scenarios with economic basis.	107
5.24	Interest rate parameters.	108
5.25	Tiered Interest Rate Parameters: APR Spreads Across the Four-Tier Hierarchy	108
5.26	Go-to-market phases.	110
5.27	Value Proposition and Go-to-Market: User Benefits Mapped to Platform Features	110
5.28	MiCA and GENIUS Act compliance mapping. Each row identifies a regulatory article relevant to a stablecoin-denominated retail lending platform and the corresponding CWB design control.	111
A.1	Technology stack summary.	122
A.2	Local LLM Assistant Integration: Components, Configuration, and Prototype Stance	122
B.1	Deployed smart contract addresses, Polygon Amoy testnet (as of pre-thesis submission).	125

List of Figures

3.1	Three-layer decentralized application architecture: presentation (React 18, TypeScript, Wagmi/Viem), smart-contract layer on the EVM (World Bank Reserve, National Bank, Local Bank), and off-chain services (Express REST API, PostgreSQL, FastAPI ML service, event listener, Redis).	35
3.2	Component diagram showing interactions between the presentation layer, smart contract layer, off-chain backend services, and external systems. . .	36
3.3	Layered blockchain and application stack: L1 settlement (Polygon PoS for retail, Ethereum Sepolia for institutional, plus Chainlink CCIP bridge, The Graph indexer, and Tenderly monitor); L2 smart-contract platform (Solidity 0.8.20, OpenZeppelin v5, UUPS, TimeLock, RBAC); L3 data (PostgreSQL 16 in 3NF, Redis, IPFS); L4 API and services (Express, FastAPI, WebSocket, EIP-712); L5 presentation (React + MetaMask).	37
3.4	Core system graph showing entity relationships across the four-tier banking hierarchy (WORLD_BANK → NATIONAL_BANK → LOCAL_BANK → BANK_USER), the central BORROWER entity, and the lending-lifecycle sub-graph (LOAN_REQUEST, LOAN, INSTALLMENT, TRANSACTION, INCOME_PROOF, CHAT_MESSAGE, AI_ML_LOG, CREDIT_PASSPORT).	40
3.6	Extended ERD entities introduced in v15: the banking-product entities (SAVINGS_ACCOUNT, FIXED_DEPOSIT, CURRENT_ACCOUNT, LOAN_GROUP, GROUP_MEMBER, INSURANCE_FUND) and the multi-entity / cross-tier operational entities (INTERBANK_LOAN, UPWARD_DEPOSIT, SYNDICATE, SYNDICATE_MEMBER, TRANCHED_POOL, TREASURY_SWAP, NETTING_BATCH, NETTING_ENTRY). Foreign-key relationships into the core entities of Figure 3.5 are preserved.	41
3.7	Enhanced Entity-Relationship (EER) model showing generalization / specialization (BANK_USER → National-/Local-bank-admin / Approver subtypes), the weak entity INSTALLMENT identified by its parent LOAN, the multi-valued attribute INCOME_PROOF, the loan-centric aggregation, and participation constraints (total / partial).	41
3.5	Entity-Relationship Diagram (ERD) of the Crypto World Bank database: core lending and governance entities in Third Normal Form (3NF), with primary keys, foreign keys, attribute types, and crow's-foot cardinality. Multi-entity / cross-tier and extended-product entities are shown in Figure 3.6. .	42
3.8	Compliance and identity stack: (a) zkKYC issuance via licensed identity provider with W3C Verifiable Credential, (b) zkAML continuous monitoring with sanction-list non-membership proofs, (c) the tiered KYC ladder (L1 zkKYC → L4 entity onboarding) gated by loan size, and (d) ERC-4337 Account-Abstraction onboarding for non-crypto users with gas sponsored by a Paymaster.	47
3.9	Tiered borrower access model. KYC level is monotone non-increasing down the tier ladder; the loan-band ceiling is enforced on-chain via borrowerLevel[wallet]; stablecoin denomination is enforced at Tier 4 (Section 1.10.3); and the SBT (Section 3.11) carries the simultaneous-loan cap that the GroupLendingPool reads at every application.	50

3.10	Multi-entity and cross-tier capital operations: (a) InterBankLendingPool with utilization-kinked rate model and default cascade, (b) UpwardDepositFacility with asymmetric rate structure, (c) SyndicatedLoan with Lead Arranger and Co-Lenders, (d) TranchPool with senior–junior waterfall, (e) TreasurySwap for cross-tier asset exchange, and (f) NettingEngine for multilateral settlement.	54
3.11	Use-case diagram with the seven-user taxonomy of Section 3.7: <i>Visitor</i> , <i>Retail User</i> , <i>Borrower</i> , <i>Approver</i> , <i>Local-Bank Operator</i> , <i>National-Bank Admin</i> , and <i>World-Bank Admin</i> , mapped to 20 representative use cases covering KYC, deposits, lending, multi-entity operations, and regulator oversight. .	61
3.12	Lending activity flows: (a) loan request to repayment, (b) hierarchical capital flow from World Bank Reserve to Local Bank, (c) borrowing-limit enforcement via the credit-passport SBT.	62
3.13	Onboarding and identity flows: (a) income verification (Open-Banking pull or officer review), (b) profile management with EIP-712 signed updates, (c) tiered, risk-based KYC ladder from zkKYC (L1) to entity onboarding (L4). .	63
3.14	Auxiliary activity flows: (a) borrower–bank chat over authenticated WebSocket, (b) RAG-augmented AI chatbot with hallucination guard, (c) live market-data retrieval and pre-filled loan sizing.	64
3.15	Data-flow diagrams: Level-0 context view of the Crypto World Bank platform with five external entities; Level-1 decomposition of the core lending subsystem (origination, ML scoring, disbursement, installment engine, and their data stores); and Level-1 decomposition of the deposit-mobilization, InterBankLendingPool, TreasurySwap, and NettingEngine subsystems. . .	64
3.16	Sequence diagram for the loan-approval flow with reject alternative: borrower submits an EIP-712 signed request, ML oracle returns a SHAP-explained score via commit-reveal, and the approver either approves (revealing the commit) or rejects (with reason logged on-chain).	65
3.17	Sequence diagrams: (a) installment payment loop using the CEI pattern with SBT update on the final installment, and (b) income verification with Open-Banking pull or manual officer review.	66
3.18	Sequence diagrams: (a) hierarchical capital flow with reserve-ratio gates at every tier, (b) market-data retrieval via cached Chainlink feeds, (c) borrowing-limit calculation reading the credit-passport SBT before ML scoring.	67
3.19	Sequence diagrams: (a) borrower–bank chat over an authenticated WebSocket, and (b) AI chatbot pipeline (ChromaDB top- k retrieval, QLoRA-tuned LLM, hallucination guard, citation-anchored answer).	68
3.20	Four-tier hierarchical capital flow with cascading repayment, plus the same-tier interbank-lending pools (InterBankLendingPool, Section 3.13.1) and the upward-surplus repatriation facility (UpwardDepositFacility, Section 3.13.2). The asymmetric rate structure ($r_{\text{up}} < r_{\text{down}} - \delta$) is enforced on-chain.	68

3.21	Auxiliary banking modules: (a) LiquidationEngine grace-period trigger and collateral auction with InsuranceFund shortfall fallback, (b) SavingsVault and FixedDeposit with duration-matched feeding of the lending pool, (c) on-chain Credit Passport (Soulbound Token) read by any participating bank, and (d) Cross-Chain Bridge via Chainlink CCIP carrying reserve-ratio updates and SBT mirroring.	69
3.22	Five-layer defense-in-depth security architecture. Each layer is necessary; breaching the smart-contract layer requires defeating every layer above it. Layer 1 (smart contract): OpenZeppelin v5, UUPS, TimeLock, RBAC, ReentrancyGuard, audit suite. Layer 2 (application): EIP-712 sign-in, JWT, rate-limit. Layer 3 (AI/ML): commit-reveal oracle, model registry, SHAP, hallucination guard. Layer 4 (runtime): Tenderly, The Graph, WebSocket dashboard, anomaly detection. Layer 5 (operations): Safe multisig, key rotation, bug-bounty.	76
3.23	Smart-contract security controls applied in v15: (a) UUPS (ERC-1822) upgrade path with TimeLock-gated authorization; (b) EIP-712 sign-in (typed-data v4 with domain separation and 15-minute JWT); (c) granular per-function pause registry indexed by The Graph for transparent governance.	77
4.1	Agile / Scrum process with two-week iterations, four ceremonies (Sprint Planning, Daily Stand-up, Sprint Review, Sprint Retrospective), backlog refinement, and the sprint-submission gate. Sprint-point estimates for the three implementation sprints are summarized on the right.	82
4.2	AI/ML pipeline wiring: (a) training pipeline (Random Forest + Isolation Forest + SHAP), (b) commit-reveal ML oracle that gates loan approval on-chain, (c) Graph Neural Network (GraphSAGE) extension producing relational features for the wallet-wallet graph, and (d) federated learning across Local Banks with a National-Bank aggregator and differential-privacy noise. . .	83
4.3	Real-time dashboard and runtime monitoring pipeline: smart contracts emit typed events; The Graph indexes them and pushes to a Node WebSocket server which renders the React dashboard via a Redis snapshot. Tenderly runtime alerts and an Isolation-Forest anomaly detector feed the operations runbook with the granular-pause control surface of Section 3.19.	86
4.4	Transaction state machine of the loan lifecycle: DRAFT → PENDING_KYC → PENDING_LIMIT → PENDING_SCORE → PENDING_APPROVAL → ACTIVE → {CLOSED, DEFAULTED → {LIQUIDATED, cured-back-to-ACTIVE}}, with every transition guarded by a CEI-ordered contract function and every transition emitting an indexed event.	87
4.5	SDLC stage mapping with Agile sprints. Requirements / Architecture / Design are covered by Pre-thesis 1; Implementation, Verification, Validation, and Maintenance span Sprints 1–3 of the final thesis phase, with Foundry invariants, Certora proofs, Mesa-based agent-based simulation, and Tenderly runtime monitoring layered on top.	89

4.6	Key design decisions and alternatives considered: smart-contract platform (EVM vs. Cosmos / Solana), chain choice (Polygon + Sepolia vs. L1-only vs. single L2), upgradeability (UUPS vs. Transparent Proxy vs. immutable), identity (DID/VC + zkKYC vs. raw on-chain KYC vs. off-chain-only), ML oracle (commit-reveal vs. trusted-backend vs. on-chain ML), and cross-chain bridge (Chainlink CCIP vs. LayerZero vs. Axelar).	94
5.1	Annual revenue projection by platform tier (USD millions, at \$2,500/ETH conservative mid-point).	108
5.2	Hierarchical interest-rate spread (APR) across the four-tier lending structure.	109
A.1	Compact local LLM assistant request path: browser UI → Vite dev proxy → CWB Express API (SSE) → LM Studio streaming endpoint → incremental Markdown rendering in the UI.	120
A.2	Expanded local LLM assistant data flow with component boundaries: the browser UI streams from the CWB Express API, which proxies to a local OpenAI-compatible model server (LM Studio) over Server-Sent Events. Optional user context is attached when the request is authenticated.	121

List of Formulas

Utilization Rate (U): $U = \frac{L}{L+B}$ (lending utilization; L is lent amount, B is available balance/liquidity)

Collateral Ratio (CR): $CR = \frac{C}{D}$ (collateral-to-debt ratio)

Loan-to-Value (LTV): $LTV = \frac{\text{Loan Amount}}{\text{Collateral Value}}$

APR (simple): $APR = r_{\text{period}} \times N$ (annualized simple rate over N periods)

Compound Interest (Savings): $A = P \left(1 + \frac{r}{n}\right)^{nt}$ (deposit growth; P principal, r annual rate, n compounding frequency)

EMI (Installment Payment): $EMI = \frac{P \cdot r \cdot (1+r)^n}{(1+r)^n - 1}$ (installment for principal P , periodic rate r , n installments)

Health Factor (HF): $HF = \frac{C \times LT}{D}$ (liquidation trigger; if $HF < 1.0$ liquidation is triggered, where LT is liquidation threshold)

Reserve Ratio (RR): $RR = \frac{\text{Reserves}}{\text{Total Deposits}}$ (minimum reserve enforcement constraint)

Capital Adequacy Ratio (CAR): $CAR = \frac{\text{Tier 1 Capital}}{\text{Risk-Weighted Assets}}$ (Basel-style prudential ratio)

Money Multiplier (M): $M = \frac{1}{RR}$ (fractional reserve multiplier approximation)

FX Conversion: $\text{Amount}_B = \text{Amount}_A \times \left(\frac{P_A}{P_B}\right)$ (oracle-priced conversion between assets A and B)

Group Loan Individual Share: $\text{Share}_i = \frac{\text{LoanTotal}}{N_{\text{members}}}$

On-Chain Credit Score (conceptual): $CS = \sum_i w_i \cdot \text{feature}_i$ (weighted features learned off-chain and stored as a score on-chain)

Isolation Forest Anomaly Score: $s(x, n) = 2^{-\frac{E(h(x))}{c(n)}}$ (anomaly scoring for instance x ; where $h(x)$ is the path length of instance x from the root of an isolation tree and $c(n)$ is the average path length for n samples)

Random Forest Fraud Probability: $P(\text{fraud} \mid x) = \frac{1}{T} \sum_{t=1}^T f_t(x)$ (ensemble average over T trees; where $f_t(x)$ is the binary fraud prediction of the t -th decision tree)

Platform Net Interest Allocation: $\text{NetInterest} = \text{InterestCollected} - \text{DepositorYield} - \text{InsuranceAllocation}$

SHAP (Shapley value): $\phi_i = \sum_{S \subseteq F \setminus \{i\}} \frac{|S|!(|F|-|S|-1)!}{|F|!} [f(S \cup \{i\}) - f(S)]$

List of Abbreviations

AA: Account Abstraction (ERC-4337)

ALM: Asset-Liability Management

AI: Artificial Intelligence

AI/ML: Artificial Intelligence / Machine Learning

AML: Anti-Money Laundering

APR: Annual Percentage Rate

API: Application Programming Interface

BRAC: Bangladesh Rural Advancement Committee

CAR: Capital Adequacy Ratio

CBDC: Central Bank Digital Currency

CCIP: Cross-Chain Interoperability Protocol (Chainlink)

CCS: ACM Computing Classification System

CR: Collateral Ratio

CVL: Certora Verification Language

DeFi: Decentralized Finance

DID: Decentralized Identifier (W3C)

DSCR: Debt Service Coverage Ratio

EIP: Ethereum Improvement Proposal

EMI: Equated Monthly Installment

ERC: Ethereum Request for Comments (token / interface standard)

EVM: Ethereum Virtual Machine

FL: Federated Learning

FX: Foreign Exchange

GNN: Graph Neural Network

HF: Health Factor

KYC: Know Your Customer

LLM: Large Language Model

L2: Layer 2

LTV: Loan-to-Value Ratio

MFI: Microfinance Institution

MiCA: Markets in Crypto-Assets Regulation (EU)

ML: Machine Learning

NIM: Net Interest Margin

PRISMA: Preferred Reporting Items for Systematic Reviews and Meta-Analyses

PoS: Proof of Stake

QRC: QR code

RBAC: Role-Based Access Control

RR: Reserve Ratio

SBT: Soulbound Token (non-transferable ERC standard)

SHAP: SHapley Additive exPlanations

SOFR: Secured Overnight Financing Rate

SSE: Server-Sent Events

SSI: Self-Sovereign Identity

SoK: Systematization of Knowledge

SWC: Smart Contract Weakness Classification

TVL: Total Value Locked

U: Utilization

UUPS: Universal Upgradeable Proxy Standard (ERC-1822)

VC: Verifiable Credential (W3C)

XAI: Explainable Artificial Intelligence

ZKP: Zero-Knowledge Proof

zkAML: Zero-Knowledge Anti-Money-Laundering proof

zk-SNARK: Zero-Knowledge Succinct Non-Interactive Argument of Knowledge

Chapter 1

Introduction

The Crypto World Bank is not a lending protocol. It is a complete institutional banking architecture implemented on programmable blockchain infrastructure. Where existing decentralized finance protocols address isolated financial functions—Aave handles collateralized lending, Uniswap handles exchange, MakerDAO issues stablecoins—the Crypto World Bank integrates the full spectrum of banking functions within a single, hierarchically governed platform. These functions include deposit mobilization, credit allocation, installment-based repayment, interbank liquidity management, foreign exchange facilitation, group lending, reserve enforcement, and risk-based governance—coordinated across four institutional tiers that mirror the structural logic of real-world development finance.

The system is designed to serve three categories of participants simultaneously. At the global level, it functions as a programmable central reserve institution—managing capital allocation across national jurisdictions with transparent, auditable reserve ratios enforced by smart contract code rather than periodic audit. At the national and regional level, it functions as a commercial and development bank—channeling capital to local lending institutions, managing interbank liquidity, and providing institutional borrowers with structured credit facilities. At the individual level, it functions as a retail bank—offering savings products, group loans, installment-based personal lending, and digital payment services to end customers, including the estimated 1.4 billion adults currently excluded from formal financial systems [14]. The complete source code, smart contracts, and supporting material for this project are available in our GitHub repository.

Blockchain functions here as a coordination technology, providing visibility of shared states, audit trails that are difficult to alter, and programmable enforcement of rules among institutions that may have competing interests. Governance design, security controls, and regulatory considerations are incorporated into the platform according to a staged implementation roadmap that includes academic validation and regulatory sandbox pilots. Analogous hybrid institutional structures in blockchain-based finance include the World Bank’s FundsChain programme, which tracks fund disbursement across projects on a permissioned ledger, and JPMorgan Kinexys, which settles multi-billion-dollar institutional payments daily on a private EVM-compatible chain—demonstrating that open ledger infrastructure and institutional hierarchy are not mutually exclusive.

1.1 Background

Understanding why such a system is necessary requires examining the structural failures of the institutions it is designed to complement.

Capital is distributed to local borrowers through layered institutional arrangements in which supranational development institutions such as the World Bank and IMF, national financial intermediaries, and local lenders collaborate to channel development finance to

end borrowers. Although this model allows risk sharing and penetrating local markets, it is linked to great operational issues. In cross-border correspondent banking, financial institutions are obligated to have pre-funded nostro and vostro accounts at correspondent banks in every currency corridor, which maroon huge amounts of capital in idle balances unable to be invested or lent out [24]. Crossborder transactions are regularly settled in two to five business days with an average cost of about \$42 per transaction by using the correspondent banking network [25]. The remittance market, which is valued at about \$860 billion annually, loses between \$48–56 billion per annum to transfer charges, with the average cost to transfer in the world being 6.49% as of 2024, more than the United Nations Sustainable Development Goal limit of 3% [26].

At the same time, it is estimated that there are 1.4 billion unbanked adults in the world as defined by the World Bank Global Findex [14], and most of them are found in the developing economies where documentation, geographic location, and minimum balance requirements leave out high populations of people in the formal financial sector. The International Finance Corporation estimates that there is a financing gap in the world of micro, small and medium enterprises at \$4.5 trillion per year [20], which is unmet credit demand that cannot be effectively met by the existing institutions through the conventional intermediation chains.

The DeFi industry has already shown that lending, collateral management, and interest calculation can be automated on a smart contract platform and that all transactions are transparent, on-chain. The largest DeFi lending protocol, Aave v3, has amassed \$26.3 billion in total value locked across ten blockchain networks [27], and the DeFi lending market has more than \$55 billion in TVL [13]. Nevertheless, these protocols utilize flat, pool based designs where all the lenders feed into one pool and all borrowers withdraw off the same source—irrespective of institutional status, creditworthiness and geographies. There is no current DeFi scheme that reflects the multi-tier institutional structure, inter-tier capitals movement and tiered borrower access provision that is typical of real-world development finance.

1.2 Rationale of the Study

The combination of three forces, namely: (1) the long-standing inefficiencies in the traditional development finance, which restrict the transparency and speed of settlement; (2) the absence of multi-tier banking functions in existing DeFi designs, particularly deposit mobilization and solidarity group lending alongside hierarchical lending; (3) the possibility of integrating blockchain auditability with lightweight analytics and monitoring support to enhance operational control, drives this study. With an architecture that will bridge these areas, we hope to show a plausible way forward toward transparent, programmable, and institutionally significant banking services for underserved populations.

1.3 Problem Statement

The international system of the development finance works based on a stratified system in which funds are directed down by large supranational organizations such as the World Bank and IMF to national institutions, which transmit them to regional and local banks, and they are ultimately delivered to individual borrowers. This intermediary chain brings into being a number of significant inefficiencies:

1. **Lack of transparency.** Financial records at each tier of the intermediation chain are not openly accessible to participants below that tier. Lower-level institutions and individual borrowers cannot independently verify how capital is being managed, what reserve levels are maintained, or on what basis lending decisions are made at higher tiers. Reserve adequacy is predominantly self-reported and audited at most quarterly, providing no mechanism for real-time verification of institutional financial health.
2. **Sluggish settlements and unproductive capital.** Cross-border money transmission requires multiple intermediaries, compliance verification steps, and correspondent bank confirmations that routinely delay settlement by two to five business days. Banks must also maintain pre-funded accounts at partner institutions in every currency corridor, locking up capital that would otherwise be available for lending [24]. An average transaction via the correspondent banking system costs approximately \$42, a burden that falls disproportionately on small-value transactions and participants in developing economies [25].
3. **Inconsistent risk evaluation.** Fraud screening and credit assessment rely heavily on individual human judgment, introducing bias, inconsistency, and limited scalability. Because no unified and verifiable credit history exists across institutional tiers, borrowers must be re-evaluated from scratch at each stage of the intermediation chain, adding time and cost without improving accuracy.
4. **Obstacles to access and institutional trust.** Establishing inter-institutional trust through legal agreements, compliance processes, and third-party audits is both expensive and time-consuming, placing smaller institutions and borrowers in developing countries at a particular disadvantage. According to the World Inequality Report 2026, the global monetary system continues to redirect resources from developing to wealthier economies, with developing-country investors receiving returns approximately three percentage points lower than those in developed economies [28]—a structural gap that further impedes access to international credit markets.
5. **Absence of programmable savings instruments.** Traditional savings accounts in developing economies offer near-zero real yields due to inflation, currency risk, and institutional opacity. Depositors have no real-time visibility into whether their savings are being deployed responsibly or sitting idle in reserve accounts.
6. **Inaccessibility of group credit mechanisms.** Over 1.4 billion unbanked individuals lack individual collateral but could access credit through solidarity group structures, a model demonstrated by BRAC, Grameen Bank, and ASA in Bangladesh. No existing DeFi protocol models programmable mutual liability or on-chain group repayment enforcement.

At the same time, the DeFi ecosystem has shown that the lending, collateral management, and interest calculation can be completely automated through smart contract platforms with complete transparency. The TVL of Aave v3 is \$26.3 billion and active borrows are \$17.7 billion [27]; the TVL of Compound v3 is \$1.4 billion [29]; and the combined DAI and USDS stablecoin issuance of MakerDAO (renamed Sky Protocol) is approximately \$10.5 billion with projected 2026 revenue of \$611 million [30]. Despite this scale, these protocols exhibit fundamental limitations in the context of institutional development finance:

- They employ **flat**, **peer-to-peer**, or **over-collateralized** lending models, where everyone and large institutional borrowers, along with individual retail clients, are dealing with the undifferentiated liquidity pool. No existing protocol models institutional

hierarchy, tiered capital allocation or cross-tier lending flows in as similar a manner as the World Bank → National Bank → Local Bank model of development finance.

- They are not connected with **AI-based risk analytics** and **explainable decision support** systems. DeFi lending decisions are determined solely using collateral ratio and utilization curve and no behavioral fraud detection, anomaly detection or explainable credit judgement is present.
- They do not model **role based, tiered governance** of the development finance institutions. It is a form of leadership, in which a majority of the large token holders will wield the power of decision-making, without the institutional role distinction (regulator, wholesale lender, retail lender, borrower) of hierarchical finance.
- The credit-based and emerging-market lending has also been introduced with other new institutional DeFi projects such as Maple Finance (\$2.6–3.8 billion TVL, undercollateralized institutional lending) [31] and Goldfinch (\$680 million loan originations in 18+ developing countries) [32], however, are single-level projects with no hierarchical capital flows and no interbank lending facilities.

1.4 Objectives

The objectives of this project are:

1. To **design, formally specify, and partially implement** a four-tier lending architecture (World Bank → National Bank → Local Bank → Borrower) on an EVM-compatible blockchain that maintains institutional hierarchy and provides shared access to the ledger. The current prototype fully implements the Tier 1 World Bank Reserve contract and the lending request/approval workflow; Tier 2 and Tier 3 contracts are specified and partially scaffolded, with full implementation planned for the final thesis phase.
2. To specify and justify an extended banking product suite—including deposit mobilization, savings products, and solidarity group lending—that can be integrated on top of the hierarchical lending foundation.
3. To investigate a lightweight off-chain analytics support layer, including fraud detection, anomaly detection, and explainable review support, to supplement human credit governance.
4. To demonstrate transparent and programmable lending processes with configurable borrowing limits, installment payments, and role-based access control implemented through smart contracts.
5. To test the prototype on public testnets (e.g., Polygon, Ethereum Sepolia) and verify hierarchical controls, transparency properties, and operational workflows.
6. To document the design of governance mechanisms, security controls, regulatory considerations, and a controlled rollout pathway toward academic validation and potential regulatory sandbox piloting.

1.5 Research Questions

The research questions used in the study are as follows:

1. **RQ1:** Does a hierarchical blockchain-based lending architecture reflect more faithfully the real-world capital flow mechanism of development finance than present decentralized lending?

2. **RQ2:** Can on-chain transparency, programmable reserve controls and role-based governance system make settlements less opaque and less frictional across institutional tiers?
3. **RQ3:** Does a lightweight off-chain analytics layer offer practical and auditable support to monitor fraud and lending in such a system?
4. **RQ4:** Does the proposed architecture work technically and operationally on current test networks in the public and yet retain their relevance to the real-life institutional and financial-use cases?
5. **RQ5:** Can deposit-funded lending and solidarity group lending be represented as programmable, auditable mechanisms on-chain while preserving practical feasibility for retail users in developing economies?

1.6 Research Contribution

This work makes four original and precisely scoped research contributions to the field of decentralized finance and blockchain-based development banking:

Contribution 1 — Four-Tier Hierarchical DeFi Architecture. We are the first to design and partially implement a four-tier smart-contract lending hierarchy (World Bank Reserve → National Bank → Local Bank → Borrower) on an EVM-compatible blockchain that mirrors the capital-flow model of multilateral development finance. No existing DeFi protocol—including Aave, Compound, MakerDAO, Maple Finance, or Goldfinch—models institutional hierarchy or cross-tier capital allocation; all employ flat, single-tier pool architectures with undifferentiated liquidity.

Contribution 2 — On-Chain Solidarity Group Lending. We are the first to specify and prototype a programmable solidarity group lending mechanism on-chain, encoding mutual liability enforcement, group formation consent, and installment splitting into smart contract logic. This model was previously realised only in analogue microfinance institution (MFI) operations, such as BRAC’s 30–40 member groups and Grameen Bank’s groups of five, where enforcement relies on social pressure rather than programmable contract rules.

Contribution 3 — Oracle-Mediated AI/ML Integration with Explainability. We propose an architectural pattern for integrating off-chain Random Forest fraud detection and SHAP-based explanations into an on-chain lending decision workflow via a trusted oracle relay, providing a blueprint for auditable, AI-assisted credit governance in DeFi. The design addresses the oracle problem—the challenge of securely bridging on-chain logic with off-chain data—and specifies three concrete integration options (centralized relay, Chainlink Functions, and commit-reveal scheme) with explicit trust trade-offs.

Contribution 4 — Compliance-Aware ZKP / zkAML Identity Pathway. We design a two-circuit privacy-preserving compliance architecture combining a zk-SNARK KYC verifier (Groth16 proofs via Circom 2.0 + snarkjs) with a zkAML circuit [R19] that proves a wallet has not transacted with sanctioned addresses and remains within velocity limits, without revealing the underlying transaction graph. Both circuits compose with a W3C Decentralized Identifier (DID) and Verifiable Credential (VC) layer [R20] anchoring user identity off-chain. This positions the platform’s compliance gateway against the broader Self-Sovereign Identity (SSI) framework and is grounded in the Piper et al. (2025) TU Berlin permissioning paper [R8].

Positioning against institutional DeFi. The Sygnum Bank *Institutional DeFi in 2025* report (February 2026) documents a specific gap: protocols work technically, but no large institutional allocator participates until legal and regulatory risks are resolved [R21]. The Crypto World Bank is positioned not as a technically novel curiosity but as an attempt to resolve precisely the governance, compliance, and audit gaps that report identifies as the barriers to institutional DeFi adoption. Combined with the mBridge / Agora positioning in Section 1.10.3, this reframes the contribution from “an alternative to traditional banking” to “a composable lending protocol compatible with emerging institutional CBDC and tokenized-deposit settlement infrastructure”—a substantially more defensible and publishable claim.

The remainder of this paper is organized as follows. Chapter 2 surveys the academic and industry literature on DeFi lending, hierarchical financial architecture, AI-assisted credit assessment, and blockchain-based financial inclusion, identifying the research gaps that motivate this work and presenting the synthesis under a PRISMA-style methodological frame. Chapter 3 presents the system architecture and design, including the smart contract hierarchy, banking product specifications, data model, identity and onboarding flows, security architecture, and governance framework. Chapter 4 describes the development methodology, sprint plan, AI/ML and federated-learning pipelines, formal-verification strategy, and technology stack justification. Chapter 5 evaluates the technical, economic, regulatory, and sustainability feasibility of the proposed system. Chapter 6 presents conclusions and future research directions.

1.7 Blockchain Justification

The multi-party coordination and trust problem of incompatible institutions with potential conflicting interests is the root cause of the above problem. We argue that blockchain technology is the solution to this kind of problems than its traditional counterparts due to the following reasons:

1. **Minimizing trust based consensus.** Distributed consensus algorithms make sure that no one can change the ledger state unilaterally and a trusted central operator is not required.
2. **Programmable enforcement.** The rules of lending, such as limits on borrowing, installment payments and workflows are stored in smart contracts as deterministic self-executable programs [58], reducing the usage of manual processes and bilateral contracts.
3. **Cryptographic auditability.** Each change of state is recorded in an immutable and publicly verifiable record of transactions and will be subject to real-time auditing by regulators, partners, and borrowers without expensive third-party interactions.
4. **Composable incentive structures.** On-chain reputation systems, governance tokens, and programmable fee distributions can align incentives across all network participants. In the Crypto World Bank’s architecture, this takes the form of on-chain repayment history contributing to a borrower’s credit score, interest spread revenue being programmatically split between depositor yield and an insurance fund, and governance-controlled parameter adjustments that require multi-tier approval rather than unilateral decision-making.

A conventional cloud-based data architecture would delegate ledger integrity, access control, and audit assurances to a trusted central operator, re-establishing the same trust de-

pendency that this project is designed to eliminate. Blockchain eliminates this single point of trust failure.

1.7.1 Blockchain Fundamentals and the EVM Execution Model

A. What blockchain is (technically precise). A blockchain is an append-only distributed ledger in which each block contains a cryptographic hash of the previous block, a Merkle root of all transactions in the block, a timestamp, and nonce or validator signature data. Immutability is not absolute—it is probabilistic and consensus-enforced: it holds as long as no coalition controlling more than 50% of stake (in Proof-of-Stake) cooperates to rewrite history [R17]. On Polygon PoS specifically, finality is achieved through the Heimdall validator layer (a Tendermint-based BFT consensus layer), which provides stronger finality guarantees than pure longest-chain PoW, but still assumes that fewer than one-third of validators are Byzantine. This trust assumption is a design constraint, not a deficiency, and must be stated explicitly in any security analysis.

B. The EVM Execution Model (Ethereum Yellow Paper [58]). The Ethereum Virtual Machine is a stack-based, deterministic, Turing-complete virtual machine that executes smart contract bytecode. Every operation (opcode) has a fixed gas cost specified in the Ethereum Yellow Paper [R7]. Key opcodes relevant to this system:

- SSTORE (write to storage): 20,000 gas for a new slot, 2,900 for modification—the most expensive operation class.
- SLOAD (read from storage): 2,100 gas (warm access), 100 gas (cold).
- CALL (external contract call): 700 gas base plus additional costs.
- Events (LOG2, LOG3): $\approx 375 + 8$ per data byte—inexpensive, used for off-chain indexing.

Each installment repayment involves at minimum one SSTORE for updating the loan balance, one SSTORE for recording payment history, and one event emission. A 12-installment loan therefore involves at minimum 24 SSTORE operations. A complete retail loan lifecycle involves approximately 27–32 individual on-chain state changes, as detailed in Section 5.8.

C. Smart contracts as state machines. A lending smart contract is best modeled as a finite state machine: PENDING → APPROVED → ACTIVE → REPAYING → COMPLETED | DEFAULTED. State transitions are triggered by role-authorized function calls, and the nonReentrant guard ensures atomicity of each transition. This state machine framing strengthens the argument that smart contracts are more auditable than off-chain systems, because every state transition is permanent, timestamped, and publicly verifiable.

D. The Oracle Problem. Smart contracts are deterministic and isolated from external data by design. This is a feature—consensus would break if different nodes observed different external states—but it creates a fundamental architectural gap for any system that requires off-chain data. The oracle problem, formally identified by Beniiche (2020) [R1] and Pasdar et al. (2023) [R2], is the challenge of securely bridging on-chain logic with off-chain information. For the Crypto World Bank, the AI/ML risk scores produced by the FastAPI Random Forest service are off-chain data that must influence on-chain lending decisions. Three architectural options exist: (1) a centralized relay (trusted backend calls `updateRiskScore` on-chain—simple but re-introduces a central trust point); (2) a decentralized oracle network such as Chainlink Functions (removes the trust assumption at the cost of 30–60 seconds latency and \$0.10–\$1.00 per call); and (3) a commit-reveal scheme (the ML service commits

a hash of the risk score before the loan decision window, reveals it after). The current prototype uses a centralized relay, acceptable in the testnet context; Chainlink Functions integration is planned for the final thesis phase.

E. Polygon PoS Trust Model. Polygon PoS operates a delegated proof-of-stake consensus with a validator set that, as of 2025, comprises approximately 100 active validators. Check-points are periodically submitted to Ethereum mainnet, providing a finality anchor against which Polygon state can be independently verified. The trust assumption is: the platform is secure as long as no coalition of validators controlling more than one-third of staked MATIC colludes to finalize invalid state. This assumption is weaker than Ethereum's own security guarantee, but substantially stronger than the central-operator model of a traditional banking platform.

1.8 Proposed Solution

The Crypto World Bank is based on a **four tier on-chain lending model**:

- **Tier 1 — World Bank:** Custodian of the global crypto reserve. Allocates capital to registered National Banks via lending mediated by smart contracts.
- **Tier 2 — National Banks:** Borrow from the World Bank reserve. Lend to Local Banks incorporated in their jurisdiction. Aggregate the risk exposure at the national level.
- **Tier 3 — Local Banks:** Borrowing of National Banks. Loan applications by process originators from individual borrowers. Administer loan lifecycle using particular approvers.
- **Tier 4 — Borrowers:** Loan requests to Local Banks. Repay through configurable installment schedules. Construct on-chain payoff history which determines borrowing in the future.

The platform also incorporates:

- **Risk-sensitive borrowing limits** computed from rolling 6-month and 1-year transaction windows.
- **Automatic installment generation** for loan amounts exceeding a configurable threshold (e.g., 100 ETH equivalent).
- **Off-chain AI/ML security analytics** (planned) for fraud detection (Random Forest), anomaly identification (Isolation Forest), and explainable risk assessment (SHAP).

1.8.1 Banking Functions of the Platform

A functionally complete bank performs six core activities: deposit mobilization, credit allocation, payment and settlement, risk intermediation, liquidity management, and ancillary financial services. The Crypto World Bank is designed to implement each of these functions across its four-tier hierarchy.

Deposit Mobilization. The process by which a bank accepts funds from savers and transforms them into productive capital. On the platform, depositors at any tier can place funds into savings products—standard savings accounts with variable yield, fixed-term deposits with locked periods and agreed APY, and institutional yield accounts for large participants. Deposits and accrual are recorded on-chain, and interest accrues automatically via deterministic rules rather than discretionary accounting.

Credit Allocation. Loans flow downward through the four-tier hierarchy—from the World Bank reserve to national banks, from national banks to local banks, and from local banks to end borrowers. Each tier applies its own interest rate spread, collateral requirement, and borrowing limit, enforced by smart contract rules and supported by data-informed monitoring.

Payment and Settlement. Transfers between registered accounts settle atomically in a smart contract transaction, avoiding the intermediate “funds-in-transit” state that commonly produces disputes in traditional systems.

Risk Intermediation. Implemented through reserve-ratio enforcement constraints, role-based approval workflows, and an AI/ML monitoring layer (Random Forest for fraud detection, Isolation Forest for anomaly detection, SHAP for explainability) designed for transparent, auditable decision support.

Liquidity Management. Enforced via minimum reserve ratios at each tier, with same-tier interbank lending pools for short-term liquidity balancing and planned asset-liability monitoring to detect unsafe duration gaps.

Ancillary Financial Services. Includes foreign exchange, group lending, trade finance facilitation, and digital identity management. FX is designed around decentralized price oracles; group lending enables pooled collateral and mutual liability; trade finance instruments can be added as planned extensions.

1.8.2 Service Delivery Across Scale

The platform is designed to operate at three scales of engagement: the global institutional level, the national and regional commercial level, and the individual retail level.

Global institutional level (World Bank Reserve tier). The World Bank Reserve functions as a programmable reserve authority by managing system-level capital allocation and reserve constraints across jurisdictions. This tier is designed to support transparent reporting of reserves and enforce reserve-ratio constraints as code, reducing informational asymmetry relative to periodic self-reporting.

National and regional level (National Bank and Local Bank tiers). National Banks borrow wholesale capital from the reserve tier and allocate it to registered Local Banks under jurisdiction-aware governance and tiered approval workflows. The intent is to mirror real-world development finance intermediation while improving settlement speed and auditability through on-chain execution.

Retail level (borrowers and depositors). End customers interact through Local Bank interfaces to open accounts, deposit savings, apply for loans (including group loans), repay installments, and build an on-chain credit history. In Bangladesh, where roughly 40% of adults lack access to formal banking services [54], a mobile-accessible interface with low transaction costs can reach users that branch-based systems do not, while keeping loan terms and reserve behavior verifiable through on-chain transparency.

1.8.3 Cross-Tier Lending System

Capital does not necessarily flow down in the old banking system. Banks at the same level deposit with one another regularly in the interbank lending market (e.g., the federal funds

market) and less significant banks that have surplus cash inject the extra cash upwards with their home institutions or central banks. The Crypto World Bank is the extension of its hierarchical architecture to accommodate such **multi-directional lending flows**, developing a more realistic and robust financial ecosystem.

Same-Tier Lending

Banks at the same hierarchical level can lend to one another to manage short-term liquidity:

- **National Bank ↔ National Bank:** A national bank, which has excess reserves, can lend to a second party that is in a liquidity crunch, similar to the traditional interbank lending market. The lending rate is adjusted according to the supply and demand through national banks of the system, like the Secured Overnight Financing Rate (SOFR).
- **Local Bank ↔ Local Bank:** Local banks in same or different national jurisdictions can share liquidity through a peer lending pool. This prevents local liquidity crunches when one bank possesses surplus deposits and the other is experiencing elevated loan demand.

Same-tier lending is done by an on-chain **InterBankLendingPool** contract at every tier level in which surplus banks provide short-term liquidity and those banks that need it are allowed to borrow at utilization floating rates. v15 specifies this pool fully in Section 3.13.1 (rate model, settlement, default cascade, operations); the present chapter introduces only the role it plays in the four-tier hierarchy. The current pre-thesis 1 prototype implements the downward flow (World Bank → National Bank → Local Bank → Borrower); the InterBankLendingPool is built in Sprint 2 of the final-thesis phase.

Upward Lending

Banks with excess capital of lower tier are then able to lend up the ladder:

- **Local Banks → National Banks:** When local banks accumulate reserves beyond the minimum reserve ratio, they are allowed to deposit excess capital in their parent national bank's liquidity pool with earned deposit yield. This mirrors how commercial banks save the surplus with central banks by means of standing deposit facilities.
- **National Banks → World Bank:** National banks may provide excess to the international deposit, enhancing the total capital base of the system. This provides a back route to well capitalized national banks and raises the reserve available for downward distribution to banks with higher demand.

The downward lending rates are higher than the upward lending rates (the low risk of lending to a higher-level institution), developing a natural interest rate term structure across the hierarchy.

Tiered Borrower Access

Different borrower classes access different tiers based on loan size and organizational type:

Table 1.1: Borrower tier access rules.

Borrower Type	Accessible Tiers	Loan Range	Use Case
Individual / End User	Local Bank only	0.01–10 ETH	Personal, micro-enterprise
Small Business / SME	Local Bank, National Bank	1–100 ETH	Working capital, equipment
Large Corporate	National Bank, World Bank	50–10,000 ETH	Infrastructure, large projects
Institutional / Sovereign	World Bank only	1,000+ ETH	Development programs

This table summarizes tiered borrower access rules by mapping borrower type and loan size to the appropriate institutional tier. The intent is to preserve real-world banking intermediation: retail borrowers primarily access Local Banks, while larger institutional borrowers can be routed to higher tiers under stricter verification. This design supports risk segmentation and prevents large actors from consuming retail liquidity.

Local Banks are used by the end users and retail borrowers to gain access to the system, maintaining the hierarchical intermediation model. Greater levels can be accessed directly by large corporate accounts and institutional borrowers, but are subject to rigorous verification (on-chain credit history with off-chain documentation) and a borrower type designation in the smart contract that determines tier access permissions.

This multi-way lending scheme—downward capital allocation, same-tier interbank lending, upward repatriation of surpluses, and tiered access by borrowers—generates a more detailed model of the banking flows of the real world in the decentralized architecture. The current prototype fully implements the downward capital distribution and tiered borrower access components; same-tier interbank lending and upward surplus repatriation are specified in full in Section 3.13 (InterBankLendingPool, UpwardDepositFacility) and scheduled for implementation in Sprint 2 of the final-thesis phase. The genuinely multi-entity operations that build on top of these flows—syndicated lending, tranching pools, cross-tier treasury FX, and multilateral settlement netting—are likewise specified at full depth in Section 3.13 so that the paper describes a complete banking system rather than only retail-facing lending.

1.9 Methodology in Brief

Our approach combines a lightweight Agile/Scrum delivery process with an explicit Software Development Life Cycle (SDLC) frame. Pre-thesis 1 covers the requirements, architecture, and design phases of the SDLC; the implementation, verification, and validation phases run across three subsequent sprints during the final thesis stage. Sprint 1 builds the four-tier smart contract scaffold (World Bank Reserve, National Bank, Local Bank), the wallet authentication flow, the frontend skeleton, and the PostgreSQL schema. Sprint 2 implements lending services (loan application, approval, installment generation, borrowing limit enforcement), the deposit-mobilization modules (SavingsVault, FixedDeposit), the group-lending pool, and the same-tier / upward and treasury-swap modules (InterBankLendingPool per Section 3.13.1, UpwardDepositFacility per Section 3.13.2, TreasurySwap per Section 3.13.5). Sprint 3 wires the off-chain AI/ML risk-scoring pipeline (Random Forest, Iso-

lation Forest, SHAP) into the on-chain loan decision via the commit-reveal oracle, layers the Graph-Neural-Network and federated-learning extensions on top, adds the syndicated-lending, tranching-pool, and netting-engine contracts (SyndicatedLoan per Section 3.13.3, TranchingPool per Section 3.13.4, NettingEngine per Section 3.13.6, plus the off-chain Settlement Coordinator service), and adds the Foundry invariant suite, Certora reserve-invariant specifications, and Tenderly runtime monitoring. The implementation stack is Solidity 0.8.20 (smart contracts), React 18 with TypeScript and ERC-4337 account abstraction (frontend), Express.js + FastAPI (backend), scikit-learn + PyTorch Geometric (ML), Circom 2.0 + snarkjs (ZKP), and ChromaDB + a locally hosted 9B-parameter LLM (RAG assistant). All deployments target Polygon Amoy and Ethereum Sepolia testnets so that the prototype carries zero real-cryptocurrency cost; stablecoin pools use testnet USDC.

1.10 Scopes and Challenges

The complete banking architecture described in this report defines both what is implemented in the current prototype and what constitutes the full system design, clearly distinguishing prototype scope from architectural intent throughout.

Scope. This prototype is restricted to publicly available testnets, i.e. no actual cryptocurrency is involved in the system. It encompasses basic features including four-tier lending architecture, loan request and approval process, installment based repayment, calculation of borrowing limits, communication between borrowers and banks and verification of income. The system also uses Random Forest with SHAP to explain explainability in fraud detection and Isolation Forest to detect anomalies. An additional feature of dual-currency is also introduced to enhance the current banking infrastructure. The prototype, however, does not cover mainnet deployment, fiat on/off-ramp integration, automated KYC/AML compliance or production-scale stress testing.

Challenges. There are some issues that come with this system. The unlabeled nature of DeFi lending fraud data is one of those problems, as it limits the successful training of the model and might necessitate the application of synthetic data or transfer learning methods. The other issue is regulatory uncertainty, since crypto lending may have different jurisdictional restrictions; and this risk can be partially reduced by working the testnets only. The sensitivity of gas costs is an issue as well, as intricate on-chain programs can be costly; to solve this, AI/ML work requiring computation is performed off-chain. Additionally, model interpretability is essential to regulatory compliance, especially in describing the process of lending, which is obtained by SHAP-based feature attribution. Last but not the least, economic sustainability should be taken into account, because the platform should be able to earn enough income, like interest revenues, to cover gas expenses at all four levels. The following Chapter 5 discusses this balance in greater depth.

1.10.1 Economic Sustainability: Interest Revenue Versus Gas Costs

The most important problem facing any blockchain-based lending system is whether the revenue earned by the lending activities will be in a sustainable position to cover the total gas expenses of the transactions on-chain. Every level in the hierarchy will have deposits, loan applications, approvals, disbursements, repayment and installment processing gas fees. One complicated smart contract interaction can cost between \$5 and \$50 on Ethereum mainnet, based on network congestion, but around \$0.001 to \$0.01 in Polygon PoS. At an average retail loan of 10 ETH (\$20,000 at \$2,000/ETH) with an 8% APR, the borrower pays \$1,600

in annual interest. With 10–15 on-chain transactions per loan lifecycle (request, approval, disbursement, 12 monthly installments), the overall gas costs on Polygon are under \$0.15—less than 0.01% of the interest obtained. The platform thus provides a sustainable margin on Layer 2 networks, but mainnet implementation would need to consider gas optimisation or batched transaction processing to ensure it would be viable to serve micro-loans.

1.10.2 Resistance to Institutional Capture

As the Crypto World Bank grows in adoption, a critical question is whether large capital holders and established financial institutions—which already dominate the traditional banking system—could migrate to the new architecture and replicate the existing patterns of financial concentration. This risk is mitigated by the platform in a variety of architectural ways:

- **Visible reserve ratios:** In contrast to traditional banks, where reserve adequacy is reported and audited quarterly, on-chain reserves are publicly verifiable in real time. There is no way a given institution can hide its real financial status to have unfair advantage.
- **Algorithmic interest rates:** Interest rates are determined by the smart contract parameters, not by opaque internal pricing committees, and thus no single participant can capture the entire pool of capital.
- **Immutable audit trails:** Every transaction, approval, and governance decision is permanently recorded on-chain, making regulatory capture and corruption detectable by any participant.
- **Open-source governance:** The smart contract code is publicly auditable, preventing hidden backdoors or preferential treatment encoded at the protocol level.
- **Tiered access with caps:** Borrowing limits and tier access rules are enforced programmatically, preventing any single entity from monopolizing the capital pool.

Although no system can entirely deter wealth concentration in a free market, the detectability and programmability characteristics of blockchain infrastructure make exploitative behavior prohibitively expensive and highly observable by comparison to opaque intermediaries.

1.10.3 Stablecoin-First Lending and Supply Scalability

Denomination matters more than any other design choice for a retail lending platform that targets developing economies. A borrower in Bangladesh who takes a 0.5 ETH micro-loan during a period of price stability and is later required to repay the same 0.5 ETH after a 50% drawdown effectively faces a $2\times$ increase in the real value of their debt. ETH has experienced exactly this drawdown twice in the past five years (May 2021 and 2022), and the stablecoin literature has formalized the resulting liability-side volatility risk for crypto-asset borrowers [R12, R13]. The Crypto World Bank therefore elevates **stablecoin-denominated lending** from a future-work footnote to an explicit design requirement at every retail tier.

The current prototype continues to use ETH as the testnet denomination for simplicity, but every loan-bearing contract has been designed so that the loan currency is a per-pool parameter rather than a hard-coded asset. In the final thesis phase, the retail Local Bank tier will deploy with USDC as the default loan currency; ETH remains acceptable only for institutional Tier 1 / Tier 2 operations where the borrower has the balance-sheet capacity

to hedge crypto-asset exposure.

Three regulatory anchors shape this choice. First, the EU Markets in Crypto-Assets (MiCA) Regulation, fully in effect since December 2024, classifies the kind of fiat-pegged tokens used here as electronic-money tokens (EMTs) and requires 1:1 reserve backing, authorization before public offering, and regular audits [R36, R37]. Second, the United States GENIUS Act, signed into law in July 2025, introduces the first federal framework for payment stablecoins and codifies a similar reserve and disclosure regime [R38]. Third, the BIS *mBridge* platform reached Minimum Viable Product status in mid-2024 and now settles real-value cross-border CBDC transactions among China, Hong Kong, Thailand, the UAE, and Saudi Arabia, while the BIS / central-bank *Project Agora* explores tokenized bank deposits on a unified ledger [R39, R40]. Neither initiative implements a retail lending hierarchy: they are settlement rails. The Crypto World Bank is positioned as the lending layer that can compose on top of such institutional settlement infrastructure rather than as a replacement for it.

The remaining supply-side concerns are addressed by three complementary measures:

1. **Stablecoin pool support.** Each Local Bank deploys an ERC-20 stablecoin loan pool (USDC primary, USDT and DAI as fallbacks). The InsuranceFund and SavingsVault contracts are denominated in the same unit so depositors, borrowers, and the reserve share a single numeraire.
2. **Multi-chain and Layer 2 deployment.** The platform deploys on Polygon PoS for low-cost retail traffic and on Ethereum Sepolia for high-value institutional flows, with the documented cross-chain bridge architecture in Section 3.12 preserving the hierarchy invariant across chains.
3. **Fractional reserve via the four-tier hierarchy.** The reserve ratio enforced at every tier (Formula RR) provides a multiplicative lending capacity: stablecoin deposits at Tier 1 fund proportionally larger downstream lending volumes, recreating the multiplier effect of traditional banking while keeping the multiplier transparent and on-chain.

The supply concern that originally motivated this subsection is therefore reframed: scarcity is a property of base-layer crypto-assets, not of the lending denomination. Fiat-backed stablecoins issued under MiCA-/GENIUS-compliant reserves provide the elastic, audit-friendly supply a retail banking platform requires, while the hierarchical reserve architecture provides the credit-multiplier behavior. Borrower-side currency risk is removed, deposit-side regulatory compliance is preserved, and the protocol does not depend on the monetary discretion of any single issuer.

1.11 Market Analysis and Partnership Ecosystem

1.11.1 Market Sizing

Table 1.3: Market segments.

Segment	Description	Estimated Scale
Total Addressable Market (TAM)	Global DeFi lending (\$55B+)	\$55B – 5T+
Serviceable Addressable Market (SAM)	Institutional and semi-institutional lending requiring hierarchical structures	\$5B – 15B
Serviceable Obtainable Market (SOM)	Pilot deployments in regulatory sandboxes, academic prototypes, NGO-backed microfinance	\$50 – 200M

This market sizing table quantifies the total addressable opportunity by combining DeFi lending TVL, global remittance volume, and the MSME financing gap. The key implication is that even a small market share yields meaningful lending volume and social impact, justifying the platform’s focus on low-cost settlement and credit access.

Sources: (a) TAM: DefiLlama [13] reports DeFi lending TVL exceeding \$55B; World Bank [26] values global remittances at \$860B; IFC [20] estimates the MSME financing gap at \$4.5T; (b) SAM and SOM: project-derived estimates based on industry segmentation [18].

The market for **transparent, hierarchy-preserving decentralized lending** is presently unaddressed by existing DeFi protocols, which universally adopt flat, pool-based architectures. This represents a significant whitespace opportunity.

1.11.2 Target Customer Segment

The Crypto World Bank targets the **retail customer segment** — individual borrowers and small businesses seeking transparent, accessible crypto-based lending services.

Table 1.5: Target customer segment profile.

Characteristic	Description
Primary Users	Individual retail borrowers seeking personal or small business loans
Geographic Focus	Developing economies with limited traditional banking access (e.g., Bangladesh, Southeast Asia, Sub-Saharan Africa)
Loan Size Range	Micro to mid-range: 0.1 ETH – 500 ETH equivalent (~\$200 – \$1,000,000 at current rates)
User Profile	Digitally literate individuals with cryptocurrency wallet access; small business owners; gig-economy freelancers
Key Pain Points	High interest rates from informal lenders; lack of credit history in traditional systems; exclusion from banking due to documentation barriers

This table profiles the target retail segment and highlights constraints that motivate the design (mobile-first UX, low fees, and progressive credit building). It supports the thesis claim that retail adoption is both the largest inclusion lever and the volume driver required for sustainable platform economics.

Why Retail: The retail segment represents the largest underserved population in developing economies. According to the World Bank’s Global Findex Database, approximately 1.4 billion adults remain unbanked, with the majority concentrated in developing countries. By targeting retail customers, the platform maximizes financial inclusion impact while generating the transaction volume necessary to sustain the lending ecosystem. The hierarchical banking model (World Bank → National → Local → Borrower) mirrors how traditional microfinance institutions reach retail customers in underserved regions, but with blockchain-enforced transparency and AI-enhanced risk assessment.

1.11.3 Partner Ecosystem

Table 1.7: Partner categories and roles.

Partner Category	Functional Role	Blockchain-Mediated Incentive
Financial Regulators	Regulatory approval; oversight	Reduced enforcement cost through on-chain transparency and audit trails
Banking Institutions	Network membership as National/Local Banks	Access to diversified global reserve; reduced inter-bank settlement friction
Payment Gateway Providers	Fiat-to-crypto on-ramp and off-ramp services	Volume-based transaction fees; expanded market reach
Academic & Research Institutions	Validation of AI/ML models; publication of research findings	Access to anonymised datasets; collaborative research opportunities
Non-Governmental Organizations	Pilot deployment; field testing with underserved borrower populations	Transparent, low-friction credit access for beneficiaries

This partner ecosystem table identifies the external entities required for a deployable banking workflow, such as wallet infrastructure, RPC providers, oracle networks, and regulatory sandbox pathways. The mapping clarifies which components are on-chain versus off-chain and where integration risk exists.

1.11.4 Incentive Alignment Through the Blockchain Platform

The system manages and enforces partner incentives directly on the blockchain, making the process more transparent and reliable:

- **Immutable repayment records** serve as an on-chain reputation system through permanent repayment records, allowing credit decisions to be based on actual data without relying on external credit bureaus.
- **Transparent reserve verification** enables on-chain reserve verification, so all participants can independently confirm that allocated funds are being used as intended.
- **Programmable fee structures** (with potential for future expansion) allow transaction fees to be distributed fairly among network participants according to their roles

and contributions.

Chapter 2

Literature Review

2.1 Preliminaries

This part defines important terminology and concepts which are employed in the literature review and the project design.

Decentralized Finance (DeFi). DeFi means financial services that are based on blockchain places that do not use conventional intermediaries. Lending protocols allow borrowing and lending of assets via smart contracts, at interest rates that are usually algorithmically calculated or market utilized.

Smart contracts. Self-sovereign programs run on a blockchain that execute when certain preset conditions are fulfilled. Smart contracts are used to manage deposits in lending, loan applications, loans, loan repayments, and interest calculations without human intermediaries.

Over-collateralization. Borrowing model in which borrowers are required to give security at a higher amount than the loan amount. This is the model that prevails in DeFi (e.g., Aave, Compound) because it lacks credit scoring; it does not include under-collateralized lending of the conventional finance.

Explainable AI (XAI). Methods that cause machine learning model predictions interpretable to humans. SHAP and Local Interpretable Model-agnostic Explanations (LIME) are leading techniques of assigning importance of features to individual forecasts, which are important in the regulatory compliance of loans.

Proof of Stake (PoS). A network security mechanism of validators through postage of tokens; applied by Ethereum (after the merge) and Polygon. PoS is more efficient than Proof of Work and allows the block finality to be faster.

Central Bank Digital Currency (CBDC). A computerized variant of fiat currency of a nation, centrally issued. CBDCs incorporate leveled distribution models that are parallel to the multi-tier design that was discussed in this project.

Solidarity/Group Lending. A credit model pioneered by the Grameen Bank in Bangladesh in which a group of borrowers (typically 3 to 20) jointly guarantee each other's loans. Peer social pressure and shared liability significantly reduce default rates, with BRAC and Grameen Bank reporting repayment rates above 95% in solidarity group portfolios, a figure consistently documented in the microfinance literature [54].

Asset-Liability Management (ALM). The practice of managing mismatches between the duration and liquidity of a bank's assets (loans) and its liabilities (deposits). Duration

mismatch, where long-term loans are funded by short-term deposits, is a primary cause of bank runs and liquidity crises.

Flash Loans. Uncollateralized loans in DeFi that must be borrowed and repaid within a single blockchain transaction. If repayment fails, the entire transaction reverts. Flash loans are used for arbitrage and liquidation but are also exploited in price oracle manipulation attacks against lending protocols.

Zero-Knowledge Proof (ZKP). A cryptographic method by which one party proves to another that a statement is true without revealing any information beyond the truth of that statement. In the context of financial compliance, ZKPs allow a user to prove KYC verification to a smart contract without exposing personal data on-chain.

Health Factor (HF). A numerical measure of the safety of a collateralized loan position, defined as the ratio of collateral value adjusted by the liquidation threshold to outstanding debt. When HF falls below 1.0, the position is eligible for automated liquidation.

2.1.1 Review Methodology (PRISMA-Style Frame)

The literature underlying this thesis is synthesized under a PRISMA-style evidence-synthesis frame [R22] rather than as a narrative summary alone. Inclusion criteria were: (i) peer-reviewed publications or formally published institutional reports issued between 2017 and 2026; (ii) direct relevance to at least one of seven scoped CWB modules—DeFi lending mechanics, ML-based fraud / anomaly detection, ZKP / DID-based identity, group lending and microfinance, smart-contract security, financial-inclusion economics, or institutional DeFi compliance; and (iii) sufficient methodological transparency to allow re-use of design parameters (e.g., reserve invariants, gas accounting, F1 scores). Sources were identified through ACM Digital Library, IEEE Xplore, Elsevier ScienceDirect, MDPI, arXiv (with venue check), and the BIS / World Bank publication portals; queries combined the keywords blockchain, DeFi, lending, fraud detection, ZKP, microcredit, and financial inclusion with regional anchors (Bangladesh, MFI). The full evidence synthesis (114 entries) is held in `references.bib`; the top-10 ranked subset is summarized below and in Table 2.1. Recent systematic reviews of blockchain in banking using PRISMA on 38 studies confirm that this methodological frame is the current expected standard for journal-quality synthesis [R23].

Table 2.1: Top-10 ranked literature subset under the PRISMA-style frame. Ranking is by direct architectural impact on the CWB design; each entry maps to a v15 section that operationalizes the finding.

#	Author / Source	Module	Headline Finding	Operationalized In
1	Werner et al. 2022 [1]	DeFi mechanics	DeFi lending is uniformly flat, pool-based, overcollateralized	Four-tier architecture (Ch. 3)
2	Tan 2023 (IMF) [5]	CBDC & inclusion	Hierarchical distribution model for CBDC in developing economies	Section 5.11
3	Palaiokrassas et al. 2023 [3]	Fraud detection	Multi-chain ML on 54M tx, F1 0.76–0.85 with features	Section 4.2
4	Adom et al. 2022 [4]	Explainable AI	SHAP outperforms LIME on lending datasets	Section 4.2
5	Weber et al. 2019 / El-iptic [R24]	Graph fraud	Graph-structured fraud signal in BTC transactions	Section 4.3
6	McMahan et al. 2017 / FedAvg [R25]	Federated learning	Privacy-preserving model averaging across institutions	Section 4.4
7	BIS WP 905 [R13]	Stablecoin risk	Algorithmic stablecoins unsuitable for EM DeFi; USDC/USDT recommended	Sections 1.10.3, 5.10
8	Atzei et al. 2017 [7]	Smart contract security	SoK of EVM attack vectors	Section 3.19
9	EU MiCA + GENIUS Act [R36, R38]	Regulation	Stablecoin issuer authorization, reserve disclosure	Section 5.10
10	Bangladesh Bank SP2025-02 [R44]	Inclusion data	Female-to-male account parity 49%/49% in 2025; 89% BRAC loans to women	Section 5.11

2.2 Review of Existing Research

The idea of the Crypto World Bank is designed based on the examination of peer-reviewed materials, institutional reports, and industry figures covering more than one sphere of direct interest to our architecture: blockchain machine learning architecture, decentralized lending protocol architecture, security, explainable AI in finance, blockchain for financial inclusion, smart contract security, correspondent banking economics, monetary policy distribution effects, and real world asset tokenization.

2.2.1 Decentralized Lending and DeFi Protocol Design

In their SoK paper, Werner et al. [1] systematize DeFi protocol design and found that current lending markets are homogenous, pool based, and over-collateralized, and they do not exist institutional hierarchy—a gap which is directly addressed by this project. Their taxonomy of DeFi protocols indicates that there is no current system modeling that can represent multi-tier capital flow in a similar manner to development finance, ensuring the innovativeness of our four-level approach.

Bastankhah et al. [2] present a data-driven DeFi lending protocol that is adaptive dual fast/slow control architecture which proves the dynamic interest rate adjustment compares very well with the static utilization curves employed by Aave and Compound [59]. Their experience confirms that it is a viable method to optimize the algorithmic lending parameters, which is justified by work we go through our adjustable levels of interest rates, through the four levels of the hierarchy.

Li et al. [46] introduce a multi-chain lending model, which allocates lending activities on various blockchain networks in order to enhance throughput and lower single-chain congestion. Their cross-chain settlement system certifies technical feasibility of executing lending protocols in heterogeneous blockchain settings, directly promoting our intended multi-chain implementation approach on Ethereum and Polygon.

Xu et al. [47] come up with a DeFi lending protocol evaluation system that quantifies protocol risk by measuring such metrics as the stability of utilization ratio, liquidation efficiency, and interest rate responsiveness. Their assessment system gives them a flexible methodology, which can be implemented to evaluate the performance of our hierarchical lending layers versus the current flat-pool protocols.

Sharma et al. [48] introduce a peer-to-peer lending framework that is powered by blockchain that removes intermediary overhead by the use of smart contract-mediated loan origination and repayment. Although they are still single-tier in architecture, the cost of their gas analysis on-chain credentials and management of identity through optimization strategies and borrowers guides our strategy in reducing transaction costs in the four level hierarchy. Dao et al. [60] demonstrate that credit scoring models can be adapted for DeFi lending environments, validating the feasibility of data-informed borrowing limits within decentralized protocols.

2.2.2 Machine Learning for Blockchain Security

Palaiokrassas et al. [3] use machine learning on multichain DeFi fraud over 54 million transactions in 23 protocols which show a demonstration that the behavioral characteristics of DeFi enhance Neural Network and XGBoost classifiers to F1-scores of 0.76–0.85 versus 0.08 using transactional features only. Their finding that the behavioral features are superior to raw transaction data directly informed our feature engineering method for the random forest fraud detection model.

The algorithm of unsupervised anomaly detection presented by Liu et al. [6] is the Isolation Forest algorithm. The algorithm identifies anomalies using recursive partitioning of data, which assigns shorter distances to outliers. We used Isolation Forest as the second detection model when there are limited labeled fraud data to analyze wallet behavior, it is possible to detect new patterns of attacks not pre-labeled.

Hassan et al. [49] introduce blockchain and machine learning to detect fraud with the help of a privacy-protecting and adjustment-flexible incentive-based method. Their framework trains ML classifiers on encrypted transaction characteristics via federated learning, whereby the operators of the detection model never get to see the individual transaction data. Our future extensions of our AI/ML layer involve the privacy-preserving paradigm, in which the histories of borrower transactions should be confidential and at the same time allow system-wide fraud pattern recognition.

2.2.3 Explainable AI in Financial Decision Systems

Direct comparison of LIME and SHAP on loan approval is given by Adom et al. [4], SHAP offers more consistent and in-depth feature attributions, which are found to be deeper via Shapley values whereas LIME provides quicker running time but less sure descriptions across repeated evaluations. We have used their comparison to make the decision to implement SHAP since the main explainability technique of loan risk measurements, trading off interpretability depth with regulatory compliance requirements.

Gupta et al. [56] use interpretable models that are based on SHAP on credit default assessment proving that SHAP feature attributions on Gradient Boosting and Random Forest classifiers are able to score over 0.92 with complete interpretability of individual predictions. Their approach to causing per-borrower risk explanations tells our intended credit risk dashboard, where the approvers of loans look at SHAP waterfall plots before lending decisions are made.

2.2.4 Blockchain for Financial Inclusion

Tan [5] constructs a model of the International Monetary Fund (IMF) according to which CBDCs in developing countries can bank large unbanked populations using two-tier distribution model (central bank $\rightarrow$ commercial banks $\rightarrow$ users) that directly parallels our four-tier hierarchy. The paper shows that the distributed infrastructure of digital currency can prove to be effective by use of available institutional means, can access those populations not covered by traditional banking—facilitating the mission of the Crypto World Bank, which is transparent and programmable lending to underserved populations.

The article by Alam et al. [54] examines how blockchain technology can be used in the microcredit industry particularly in Bangladesh, it can be proven that smart contract-based loan management is capable of minimizing the administrative overhead by up to 60% as compared to manual microfinance processes. They have analyzed the Bangladeshi financial environment where about 40% of adult population do not have access to formal banking—directly confirms the geographic and demographic attractiveness of the target market of the Crypto World Bank.

The article by Islam et al. [53] examines the design requirements and challenges of Central Bank Digital Currencies, there are two levels of retail CBDC model (central bank to commercial banks to end users) as the distribution architecture of choice. Their analysis of interoperability challenges between CBDC systems and existing payment infrastructure informs our design of the dual-currency facility that bridges fiat and cryptocurrency within participating banks.

2.2.5 Group Lending and Microfinance Digitization

The solidarity group lending model, pioneered by Grameen Bank in Bangladesh and scaled by BRAC (Bangladesh Rural Advancement Committee) into one of the world's largest microfinance institutions, demonstrates that peer-monitored mutual liability can achieve repayment rates exceeding 95% even among borrowers with no individual collateral or formal credit history. The blockchain extension of this model replaces social enforcement with programmable contract enforcement: multi-signature consent, on-chain group formation, and automatic collateral pool claims in the event of member default. Alam et al. [54] show that smart contract-based loan management reduces administrative overhead in Bangladeshi microfinance by up to 60%, directly validating the economic case for the platform's group lending module. Güçük et al. [61] provide a broader literature review of privacy and trust considerations in blockchain-based microlending, reinforcing the importance of privacy-preserving mechanisms in the group lending architecture. The convergence of solidarity lending's social model and blockchain's programmable mutual liability represents a research gap that the Crypto World Bank is designed to partially address.

2.2.6 Smart Contract Security and Governance

Atzei et al. [7] list Ethereum smart contract attack vectors such as reentrancy, access control vulnerabilities, integer overflow, and access control vulnerabilities. They were directly informed by their taxonomy our implementation of the ReentrancyGuard by OpenZeppelin, Solidity 0.8.20 inbuilt overflow protection, and role-based access control modifiers. On their advice we took up their suggested defense-in-depth strategy of combining security primitives with formal verification with planning using static analysis (Slither) and symbolic execution (Mythril).

The article by Wang et al. [50] introduces ContractWard which is an automated vulnerability detection system that uses machine learning classifiers (Random Forest, SVM, k-NN) to classify six types of smart contract vulnerabilities based on the features of bytecodes. ContractWard scores F1-reentrancy and F1-access control with 0.96 and 0.93 respectively, proving that security auditing using ML can be used to supplement traditional static analysis tools. Our proposed security verification pipeline will utilize the use of machine learning-based static analysis tools.

Liao et al. [51] present SoliAudit, that is an integration of machine learning classification and vulnerability assessment of smart contracts fuzz testing. Their hybrid approach—using ML to first prioritize probable vulnerable code paths then target fuzz testing—reduces audit time by an estimated 70% of that spent on manual review. SoliAudit's methodology informs our intended security audit plan on the current three-contract prototype and its planned extensions toward the full nine-contract architecture.

So et al. [52] come up with VERISMART, an extremely accurate Ethereum safety verifier which employs constraint-based abstract interpretation to detect arithmetic overflow, division-by-zero and array out-of-bound errors. VERISMART achieves 98.4% accuracy on benchmark data, but has zero false positives, so it is a candidate tool to formally verify our smart contract invariants (e.g., reserve ratio maintenance, cascading interest rate limits).

2.2.7 AI Security Features for Blockchain Lending

In addition to the currently existing fraud detection and anomaly identification models in place in the platform, there are other security features that are based on AI that can enhance

the Crypto World Bank's resilience to threats of change:

- **Graph Neural Network (GNN) transaction analysis:** GNN-based models are able to analyze the graph structure of transaction to identify coordinated rings of fraud—groups of wallets which conspire to rig the borrowing limits or make circular lending patterns. A study by Palaiokrassas et al. [3] proves that graph-based features are much superior to flat transaction features as an indicator of DeFi fraud detection.
- **Federated learning for cross-bank fraud intelligence:** Hassan et al. [49] demonstrate that federated learning allows the collaboration between several institutions to detect and train fraud models without distributing raw transaction information. Deploying federated learning of the Crypto World Bank's National and Local Banks would allow detecting threats system wide without compromising the privacy of the per-bank data.
- **Reinforcement learning for adaptive interest rates:** As a reinforcement-based learning algorithm, adaptive interest rates can be learned dynamically; RL agents can dynamically modify interest rate parameters to match the varying market conditions, utilization rates, and risk profiles, decreasing the lag of manual parameter governance.
- **Real-time smart contract monitoring:** ML-based monitoring agents (informed by ContractWard [50] and SoliAudit [51]) can continuously analyze incoming transactions for patterns matching known exploit signatures, triggering the platform's pause mechanism before significant losses occur.
- **Natural language processing for income verification:** NLP models are able to extract and validate income information in uploaded documents, saving the bank officers the manual review load but not eliminating verification accuracy.

2.2.8 Gas Cost Optimization and Layer 2 Scalability

In DeFi, Tolmach et al. [55] examine the optimal approaches to minimization of gas fees, demonstrating that transaction batching, calldata compression, and tactical time of on-chain operations can save 30–50% of gas costs on Ethereum mainnet. Their findings test our architectural choice of deploying to Polygon PoS (where fees already are sub-cent) and inform optimization strategies of the contemplated Ethereum mainnet deployment. The research also finds out that complicated multi-step operations of DeFi (analogous to our four-tier loan lifecycle) benefit disproportionately from Layer 2 deployment due to the multiplicative savings in gas of sequential calls of contracts.

2.2.9 Correspondent Banking and Cross-Border Settlement

The Bank of International Settlement [21] and Financial Stability Board [33] have extensively documented the structural inefficiencies of the correspondent banking network. In the traditional model, the banks are required to have pre-funded nostro accounts in every currency corridor, establishing a world pool of idle capital which can not be put into effective productive lending. SWIFT settlements and correspondent banks involves routing of messages using MT103 payment directions, screening of compliance at all intermediates, and settlement time extensions to two–five business days. According to the World Bank Migration and Development Brief [26] it is stated that the cost of transferring \$200 on cross-border is 6.49% on average, with Sub-Saharan African corridors averaging more than 8%. These results encourage our development of on-chain settlement with close to instant finality and sub-cent transaction costs on Layer 2 networks.

2.2.10 Monetary Policy Distribution and Financial Inequality

Recent empirical studies have reported the distributional impact of monetary policy on wealth inequality. A 2025 cross-country study spanning 49 nations (1999–2019) discovered that the relationship between central bank asset purchases programs and greater wealth inequality, whose outcomes are stronger in the distribution of wealth and long lasting [34] than income inequality effects. The Federal Reserve’s own 2025 contractionary monetary analysis based on the U.S. metropolitan tax data confirmed that policies harm the income of low income employees [35]. These findings deliver economic incentive to the design principles of Crypto World Bank: transparent and monetary parameters (interest rates, reserve ratios, supply rules) that have been algorithmically determined represented in smart contracts as opposed to being established discretionally by central bank decisions. On-chain transparency of the platform makes sure that all participants follow the same interest rates and reserve conditions, which removes the informational asymmetry permitting the Cantillon Effect—the effect that early receivers of new money created benefit at the expense of subsequent recipients who will suffer greater prices [36].

2.2.11 Real-World Asset Tokenization and Institutional DeFi

The tokenization of real-world assets is an emerging convergence of traditional and decentralized finance. The Corda platform of R3 boasts of more than \$17 billion in tokenized real-world assets on its permissioned ledger as of September 2025, with institutional participants including HSBC and Bank of America [37]. Centrifuge has deployed in eight blockchain networks of more than \$1 billion in tokenized institutional fund products [38]. The World Bank is using blockchain on its own public chain side to track the disbursement of funds using its FundsChain program (based on Hyperledger Besu) which is flown on 13 projects in 10 countries and increases to 250 projects by mid-2026 [39]. These developments confirm the institutional interest of blockchain-based financial infrastructure and the viability of hierarchical fund distribution design models on distributed ledgers. This institutional adoption is continued in the Crypto World Bank’s trajectory through carrying out not only fund tracking but total hierarchical lending system that has on-chain interest rates, reserves and credit assessment.

2.2.12 Graph Neural Networks for Relational Fraud Detection

Recent work has demonstrated that DeFi fraud is fundamentally a relational phenomenon: coordinated borrowing rings, sybil attacks, and flash-loan manipulation appear in the wallet-interaction graph, not in individual transaction features. Cheng et al. (2024) [R24] and the DeFiGuard graph-neural-network architecture [R25] consistently outperform flat ML baselines on blockchain fraud, with reported detection above 70% of illicit transactions at sub-1% false-positive rates on the Elliptic dataset; Wang and Wang (2025) report similar advantages on cross-chain laundering [R26]. This evidence motivates a Graph Neural Network (GNN) extension to the platform’s Random-Forest fraud-detection layer, treated as an ablation that demonstrates the marginal value of relational features over the flat behavioral features already used by Palaiokrassas et al. [3]. The GNN extension is described in Section 4.3.

2.2.13 Federated Learning Across Banking Tiers

Hassan et al. (2022) [49] introduced privacy-preserving fraud detection via federated learning on the blockchain. The PrivChain-AI study [R27] in *Nature Scientific Reports* (Decem-

ber 2025) demonstrates the same pattern with 94.7% fraud-detection accuracy across multiple banks, while Abbassi et al. (2025) [R28] and FED-SPFD [R29] document the regulatory benefits of FL for cross-border fraud intelligence. The Crypto World Bank’s four-tier structure—where National and Local Banks are independent legal entities that cannot share raw transaction data—makes federated learning architecturally appropriate rather than merely fashionable: each Local Bank trains a local fraud model on its own borrower data, parameter updates are aggregated at the National Bank tier via FedAvg, and no raw records leave the bank. The detailed FL pipeline is given in Section 4.4.

2.2.14 Stablecoin Regulation: MiCA and GENIUS Act

Two recent statutes reshape the legal context of any stablecoin-denominated lending platform. The EU Markets in Crypto-Assets (MiCA) Regulation, fully in effect from December 2024, requires authorized issuers, 1:1 reserve backing, and regular audits for electronic-money tokens—categories that directly cover USDC and USDT used at retail tiers of this platform [R36, R37]. The United States GENIUS Act of July 2025 creates the first federal payment-stablecoin framework with similar reserve and disclosure obligations [R38]. These statutes are read in this thesis not as obstacles but as the regulatory anchor that makes stablecoin-first design defensible (Section 1.10.3); the corresponding compliance mapping is presented in Section 5.10.

2.2.15 On-Chain Credit Passport and Soulbound Tokens

Buterin, Hitzig, and Weyl (2022) [R30] introduce the concept of Soulbound Tokens (SBTs)—non-transferable ERC-style credentials that bind reputational data to an identity rather than to a tradable asset. For a multi-tier lending platform, SBTs provide a natural primitive for a portable on-chain credit passport: a borrower’s successful repayment history at one Local Bank becomes a credential they can present to any other CWB Local Bank, and (with the schema fixed) to any external lending protocol that adopts the standard. MicroSave Consulting (December 2025) [R31] documents the Bangladesh microfinance sector’s active shift toward performance-based credit identity, providing direct empirical grounding for this design choice. The full credit-passport schema is in Section 3.11.

2.2.16 Institutional DeFi Gap and the mBridge/Agora Landscape

Sygnium Bank’s February 2026 report *Institutional DeFi in 2025* [R21] documents the central gap that this thesis attempts to fill: DeFi protocols work technically, but no large institutional allocator will participate until legal and regulatory risks are resolved. The BIS mBridge platform [R39] (China, Hong Kong, Thailand, UAE, Saudi Arabia) reached Minimum Viable Product status in mid-2024 and now settles real-value cross-border CBDC transactions, and the BIS / central-bank *Project Agora* [R40] explores tokenized bank deposits on a unified ledger. The IMF CBDC survey 2025 reports that 94% of central banks are now engaged in CBDC research [R32]. None of these initiatives implements a retail lending hierarchy: they are settlement rails. This positions the Crypto World Bank explicitly as the lending layer compatible with such institutional CBDC infrastructure rather than as a competitor to it.

2.2.17 LLMs in Finance and Hallucination Risk

The CWB AI assistant uses a 9B-parameter open-weight LLM with QLoRA fine-tuning, RAG over the platform’s policy and contract documents, and an explicit refusal layer for

regulated questions. The LLM-in-finance literature [R33, R34, R35] gives both the design rationale and the necessary cautions: Morgan Stanley’s GPT-based equity-analysis assistant achieved a reported 50% reduction in analyst time, McKinsey estimates LLMs can cut back-office costs by 40%, but the 2025 LLM-vulnerability-in-finance corpus [R35] documents specific regulatory-hallucination failure modes when LLMs are asked compliance questions without RLHF from domain experts. The evaluation methodology in Section 4.11.1 includes a red-teaming protocol for adversarial financial prompts directly motivated by this work.

2.3 Literature Review Summary

The most important reviewed works with their methodologies, key findings, and relevance to our project are presented in Table 2.3, in the form of literature table suggested by Michigan State University Libraries [23].

Table 2.2: Literature review summary (Part A): DeFi lending, fraud detection, and XAI.

Author Year	Research Focus	Methodology	Key Findings	Relevance to Our Project
Palaiokrassas et al. (2023) [3]	DeFi fraud detection	XGBoost, NN classifiers on 54M+ multi-chain transactions	F1: 0.76–0.85 vs. 0.08 with features alone	Informs Random Forest fraud modeling
Adom et al. (2022) [4]	XAI in loan approval	LIME / SHAP comparison on lending datasets	SHAP provides deeper, more consistent explainability; LIME is faster	Justifies SHAP as our primary explainability method
Tan (2023) [5]	CBDC and IMF financial inclusion	Model for developing nations	TBD	Informs dual-currency facility design
Liu et al. (2008) [6]	Anomaly detection	Isolation Forest on synthetic and real datasets	Isolation via recursive partitioning; shorter paths	Adopted as secondary unsupervised detection for wallet behaviour
Atzei et al. (2017) [7]	Smart contract security	SoK of Ethereum attack vectors	Over-reliance, access control vulnerabilities	Directly informs our security primitives and planned formal verification

Of the works reviewed, none model a four-tier hierarchical lending structure, confirming the architectural novelty of this project. The synthesis shows that DeFi and institutional finance have developed in parallel without converging on a governance-aware, multi-tier design.

Table 2.4: Literature review summary (Part B): DeFi protocol systematisation and adaptive lending.

Author Year	Research Focus	Methodology	Key Findings	Relevance to Our Project
Werner et al. (2022) [1]	DeFi protocol systematisation	SoK survey of 12+ DeFi protocol categories	Lending platforms are uniformly pool-based and overcollateralised; no institutional hierarchy exists	Identifies the core gap our four-tier architecture addresses
Bastankhah et al. (2023) [2]	Adaptive DeFi lending	Dual fast/slow control; simulation on historical data	Dynamic rate adjustment outperforms static utilisation curves	Validates algorithmic lending parameter optimisation

This continuation extends coverage to governance, settlement, and institutional blockchain adoption. The key observation is that enabling infrastructure (Layer 2 networks, oracle pricing, EVM tooling) is mature, while the specific hierarchical banking architecture remains architecturally unexplored.

Table 2.6: Literature review summary (Part C): blockchain P2P lending, privacy-preserving ML, and smart contract auditing.

Author Year	Research Focus	Methodology	Key Findings	Relevance to Our Project
Sharma et al. (2021) [48]	Blockchain P2P lending	Smart contract-mediated loan origination framework	Eliminates intermediary overhead; gas cost minimisation analysis	Informs four-tier architecture for peer-to-peer lending
Hassan et al. (2022) [49]	Privacy-preserving fraud detection	Federated learning on encrypted blockchain features	ML achieves comparable accuracy to centralised models	Informs future privacy-preserving fraud detection extensions
Wang et al. (2020) [50]	Smart contract vulnerability detection	ML classifiers on bytecode features	F1: 0.96 for access control; 0.93 for data leakage	Supports our planned security verification models
Liao et al. (2019) [51]	Smart contract audit automation	Hybrid classification + fuzz testing	Reduces audit time by 70% vs. manual review	Informs our security audit strategy for three-contract architecture

This continuation captures empirical and systems findings on cross-border settlement and security tooling, supporting the claim that Layer 2 deployment and hybrid ML-static analysis pipelines substantially

reduce feasibility risk for an academic prototype.

Table 2.8: Literature review summary (Part D): CBDC design, cross-border settlement, and multi-chain DeFi.

Author	Year	Research Focus	Methodology	Key Findings	Relevance to Our Project
Islam et al.	(2024) [53]	CBDC design requirements	Analysis of retail CBDC distribution architecture	Two-tier model preferred; interoperability challenges identified	Informs dual-currency facility design
BIS/FSB	[21]	Cross-border settlement infrastructure	Analysis of cross-border payment costs and capital trapped in nostro accounts	2–5 days avg. settlement; \$42/transfer avg. cost	Motivates four-tier settlement with near-instant finality
Beyer et al.	(2025) [34]	Monetary policy and inequality	Cross-country analysis (1999–2019)	QE increases wealth inequality; effects more persistent than monetary parameters	Motivates transparent, algorithmic monetary parameters
World Bank	(2025) [39]	Blockchain for development finance	Pilot on 13 projects, 10 countries	Blockchain tracking reduces reporting burden for fund distribution	Supports our planned multi-chain deployment strategy

This block summarizes works informing the platform’s risk and monitoring approach. The synthesis justifies the choice of lightweight, auditable ML components rather than opaque deep learning models, prioritizing regulatory interpretability over raw detection performance.

Table 2.10: Literature review summary (Part E): multi-chain DeFi, smart microfinance, gas optimisation, and SHAP-based credit models.

Author Year	Research Focus	Methodology	Key Findings	Relevance to Our Project
Li et al. (2024) [46]	Multi-chain DeFi lending	Cross-chain model design and implementation	Multi-chain distribution improves throughput and reduces congestion	Supports our planned multi-chain deployment strategy
Xu et al. (2023) [47]	DeFi lending protocol evaluation	Quantitative risk metrics for lending protocol assessment	Utilisation ratio, liquidation efficiency, and reserve responsiveness as key metrics	Provides benchmarks for evaluating our hierarchical tiers
Alam et al. (2021) [54]	Smart micro-credit in Bangladesh	Smart contract-based microfinance	Reduces admin overhead by 60%; validates geographic and demographic markets	Directly validates our target geographic and demographic market
Tolmach et al. (2024) [55]	DeFi gas fee optimisation	Transaction batching and calldata compression analysis	Gas savings of 30–50% through optimisation strategies	Validates Layer 2 deployment and mainnet optimisation
Gupta et al. (2024) [56]	SHAP-based credit default models	SHAP, Random Forest, and Gradient Boosting classifiers	AUC above 0.92 with full interpretability	Informs per-borrower risk explanation methodology

This final block closes the synthesis with institutional and inclusion-related sources. The combined evidence supports the thesis that blockchain adoption is increasing at the institutional level while inclusion gaps persist at the retail level, matching the platform’s intended multi-scale scope.

2.4 Comparative Protocol Analysis

The literature survey above reveals that no existing DeFi protocol combines multi-tier institutional hierarchy with AI-assisted governance and compliance-aware identity. Table 2.12 provides a structured comparison of the Crypto World Bank against representative existing protocols on eleven architectural dimensions.

Table 2.12: Comparative protocol analysis: existing DeFi lending protocols vs. the Crypto World Bank (CWB). ✓ = implemented/present; ⌚ = designed/partial; ● = planned; ✕ = absent.

Feature	Aave v3	Compound v3	MakerDAO	Maple	Goldfinch	CWB
Institutional hierarchy	✕	✕	✕	✕	✕	✓ 4-tier
Cross-tier capital flow	✕	✕	✕	✕	✕	⌚ Designed
Same-tier interbank lending	✕	✕	✕	✕	✕	⌚ Designed
Solidarity group lending	✕	✕	✕	✕	Partial	● Planned
AI/ML fraud detection	✕	✕	✕	Manual	Manual	⌚ Built, not inte
SHAP explainability	✕	✕	✕	✕	✕	⌚ Built, not inte
ZKP KYC compliance	✕	✕	✕	✕	✕	● Planned
Kinked interest rate curve	✓	✓	Via gov.	✓	✕	⌚ Designed
Role-based access control	Partial	Partial	Via gov.	✓	Partial	✓ Implemen
Developing-economy focus	✕	✕	✕	✕	✓	✓ Banglade
TVL / Status (2026)	\$26.3B	\$1.4B	\$10.5B	\$2.6B	\$680M	Testnet

This comparison directly answers RQ1: no existing protocol models institutional hierarchy, cross-tier capital flow, or solidarity group lending in one decentralized architecture. The Crypto World Bank is architecturally distinct across every dimension that characterizes development banking.

2.4.1 Literature Synthesis

The literature synthesis above directly informs four architectural decisions in this work: (1) Gudgeon et al.'s (2020) [R3] empirical finding that utilization above 90% causes liquidity crises motivates the kinked interest rate model in Section 3.8; (2) Piper et al.'s (2025) [R8] ZKP permissioning framework provides the technical foundation for the compliance pathway in Section 3.6.1; (3) the empirical finding of Howlader & Halder (2025) [R11] that mobile financial inclusion in Bangladesh grew by 99% between 2004 and 2021 validates the retail-tier accessibility argument in Section 1.6; and (4) Alam et al.'s (2021) [54] IEEE TENSYP paper on blockchain microcredit in Bangladesh provides direct prior art that this thesis advances.

2.5 Summary of Key Findings

The following summarized findings are obtained in the literature review and are directly informative to our design:

- **DeFi lending is structurally flat.** Available protocols (Aave, Compound, MakerDAO) operate pool-based or over-collateralized models with total value locked exceeding \$55 billion [13]. Institutional hierarchy has no comparable protocol models to development finance. The research by CBDC [5] ascertains that tiered distribution is workable and promotes financial inclusion.
- **Correspondent banking is inefficient in structure.** Cross-border settlement spends two to five days at average costs of \$42 per transaction [25]. The system traps

important capital lying idle in nostro/vostro accounts [24], and remittance fees are estimated to consume between \$48–56 billion a year [26]. On-chain settlement on Layer 2 networks is known to cut the latency (to seconds) and the cost (to sub-cent levels) down.

- **ML enhances fraud detection in blockchain.** DeFi-specific behavioral features significantly work well as compared to transactional features (F1: 0.76–0.85 vs. 0.08 [3]). The Isolation Forest [6] allows unsupervised detection in which labeled data is scarce.
- **SHAP is explainable by regulators.** SHAP offers greater consistency and is better than the LIME in loan approval systems [4] in meeting prudential demands on open-minded automated decision-making.
- **Layered defense is needed in smart contract security.** Reentrancy, overflow, and vulnerabilities to access control are well documented [7]. ML-based vulnerability detection achieves F1-scores of above 0.93 [50], and hybrid ML-fuzz methods reduce audit time by 70% [51]. Verification tools are formal, i.e., VERISMART achieves 98.4% precision [52]. Automated primitives used with OpenZeppelin are recommended for production deployments to be verified.
- **Monetary policy leads to distributional inequality.** Cross-country evidence [34] and Federal Reserve research [35] confirm that quantitative easing disproportionately favors asset holders, and workers of lower income cover the expenses of both inflation and consequent tightening. Transparent, algorithmic monetary parameters can decrease this informational asymmetry.
- **The use of blockchains in institutions is gaining momentum.** The World Bank's FundsChain [39], tokenized assets of R3 Corda worth \$17 billion [37], and JPMorgan Kinexys settling multi-billion dollars per day [40] reflect the fact that financial institutions are actively implementing blockchain on settlement, funds tracking, and asset tokenization.
- **Blockchain allows financial inclusion.** Approximately 1.4 billion adults were not banked in any country around the world [14], and DeFi was found by the World Economic Forum as a leaping frog technology that allows peoples to avoid the banking systems infrastructure [41]. The MiniPay wallet of Celo has onboarded 14 million users in over 60 countries with less than cent transaction fees [42]. Blockchain-based microcredit in Bangladesh shows 60% decrease on administrative overhead [54].
- **Multi-chain lending enhances scalability.** Cross-chain lending models [46] and Layer 2 gas optimization strategies [55] prove the fact that the lending is distributed when operating on several networks, decreasing congestion and transaction costs by 30–50%.
- **ML can be used to detect cross-institutional fraud using privacy preserving ML.** Federated learning methods [49] are such that enable several lending institutions to cooperate and learn fraud detection models in a non-intrusive manner, i.e. no individual transaction data is exposed, resolving the conflict between institution-wide security and data privacy.
- **Group lending reduces default risk in underserved markets.** Microfinance research from BRAC, Grameen Bank, and ASA consistently reports repayment rates above 95% in solidarity group structures [54]. On-chain enforcement of mutual liability, replacing social pressure with programmable contract enforcement, has not been studied in the academic literature, representing an open research gap that this project begins to address.

- **Sustainable microfinance requires deposit-funded lending.** Literature on sustainable microfinance institutions consistently shows that deposit-funded lending rather than donor-funded or externally capitalized models is the only economically viable approach at scale. The platform's banking product suite is designed to close this loop by mobilizing retail savings at the Local Bank tier to partially fund the lending pool.

The results explain why we have a four-tier architecture, AI/ML integration (Random Forest, Isolation Forest, SHAP), cross-tier lending structure, governance structure, and planned security verification pipeline.

Chapter 3

System Architecture and Design

3.1 Prototype Scope

Evaluators should distinguish between what has been **built and tested**, what has been **designed and partially scaffolded**, and what is **planned** for the final thesis phase. Table 3.1 provides this mapping for every major platform feature.

Table 3.1: Prototype scope: feature implementation status as of pre-thesis submission. ✓ = Implemented and testnet-verified; ⌘ = Designed or partially scaffolded; ● = Planned for final thesis phase.

Feature	Status
Four-tier role system (RBAC)	✓ Implemented
World Bank Reserve contract (Tier 1)	✓ Implemented
Tier 2 National Bank contracts	⌘ Designed, partial
Tier 3 Local Bank contracts	⌘ Designed, partial
Cross-tier fund transfer	⌘ Designed, unimplemented
Loan request / approval workflow	✓ Implemented
Installment EMI auto-generation	⌘ Designed
SavingsVault contract	● Planned (final thesis)
FixedDeposit contract	● Planned (final thesis)
GroupLendingPool contract	● Planned (final thesis)
InterBankLendingPool	● Planned (final thesis)
AI/ML fraud detection (Random Forest)	⌘ Built, not integrated
SHAP explainability output	⌘ Built, not integrated
Oracle integration (Chainlink Functions)	● Planned (final thesis)
ZKP KYC compliance layer	● Planned (final thesis)
Testnet deployment evidence	✓ Addresses in Appendix C

This table eliminates any ambiguity between present-tense architectural description and empirical implementation evidence throughout the following sections.

3.2 High-Level Architecture

The system employs a **three-layer decentralized application architecture** (Figure 3.1): a presentation layer (React + wallet integration), a smart-contract layer (the World Bank, National Bank, and Local Bank contracts on the EVM), and an off-chain services layer (REST

API, PostgreSQL, FastAPI ML service, event listener, and Redis cache). Figure 3.2 elaborates this view with the fifteen-contract target architecture and external integrations.

Three-Layer Decentralised Application Architecture

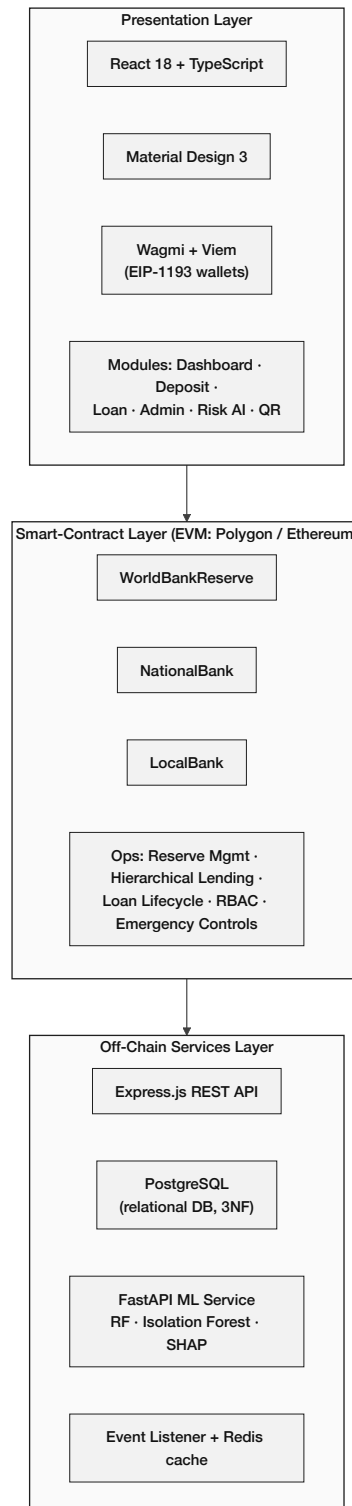


Figure 3.1: Three-layer decentralized application architecture: presentation (React 18, TypeScript, Wagmi/Viem), smart-contract layer on the EVM (World Bank Reserve, National Bank, Local Bank), and off-chain services (Express REST API, PostgreSQL, FastAPI ML service, event listener, Redis).

The current prototype implements three core contracts (World Bank Reserve, National Bank, Local Bank). The complete architecture extends to fifteen modular contracts covering the full banking product suite described in Section 3.17 and the multi-entity / cross-tier operations of Section 3.13. The remaining twelve contracts are: SavingsVault, FixedDeposit, GroupLendingPool, FXModule, InsuranceFund, CurrentAccount (banking product suite); InterBankLendingPool, UpwardDepositFacility, SyndicatedLoan, TranchPool, TreasurySwap, NettingEngine (multi-entity / cross-tier operations). All are planned for implementation across Sprints 2 and 3 of the final thesis phase.

Figure 3.2 shows the component interactions across these three layers. The diagram reflects the current three-contract prototype view; the nine-contract target architecture is specified in Appendix B and will be reflected in an updated diagram in the final thesis phase.

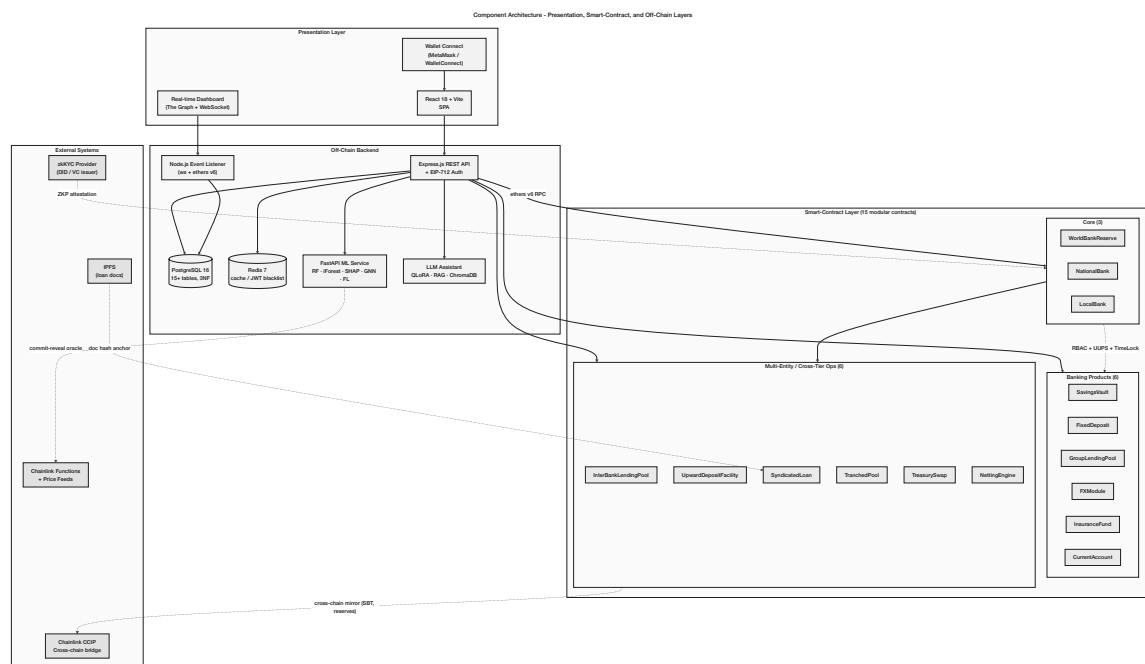


Figure 3.2: Component diagram showing interactions between the presentation layer, smart contract layer, off-chain backend services, and external systems.

3.3 Blockchain Platform Selection

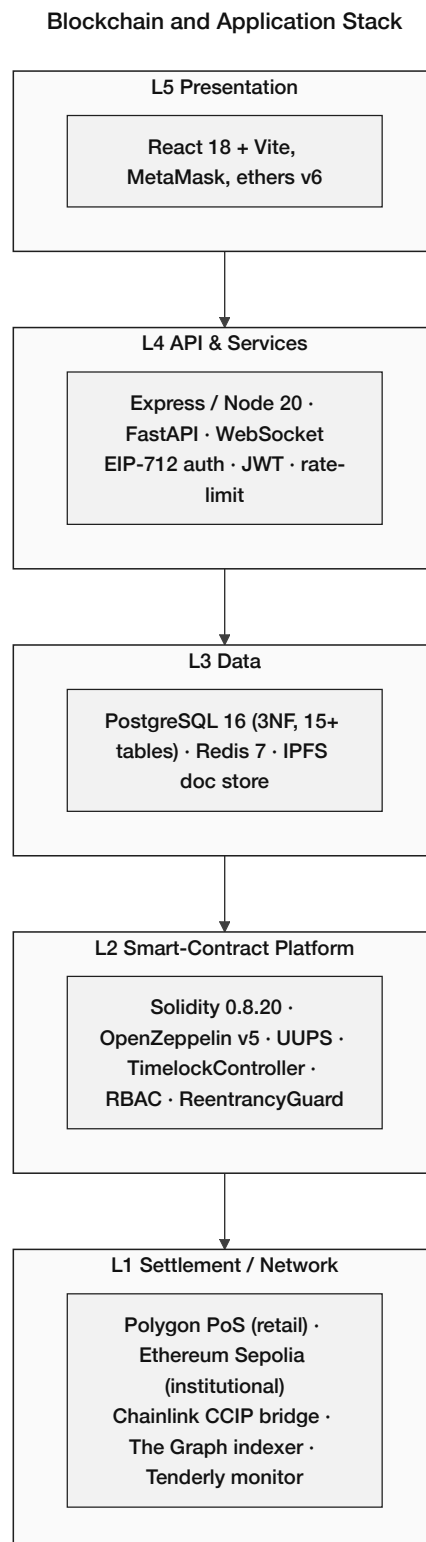


Figure 3.3: Layered blockchain and application stack: L1 settlement (Polygon PoS for retail, Ethereum Sepolia for institutional, plus Chainlink CCIP bridge, The Graph indexer, and Tenderly monitor); L2 smart-contract platform (Solidity 0.8.20, OpenZeppelin v5, UUPS, TimeLock, RBAC); L3 data (PostgreSQL 16 in 3NF, Redis, IPFS); L4 API and services (Express, FastAPI, WebSocket, EIP-712); L5 presentation (React + MetaMask).

Table 3.2: Blockchain platform selection criteria and justification.

Criterion	Selection	Justification
Platform	Ethereum Virtual Machine (EVM)	Largest developer ecosystem; battle-tested security model; extensive tooling.
Network	Polygon Amoy / Ethereum Sepolia	Zero-cost deployment; production-equivalent behaviour; free faucet access.
Consensus	Proof-of-Stake via Polygon validators	Energy-efficient; sub-2-second block finality; decentralised validator set.
Smart contract language	Solidity 0.8.20	Industry standard; mature compiler with overflow protection; rich library ecosystem.

This platform selection table compares candidate networks using criteria relevant to banking workflows, including cost, finality, throughput, and ecosystem maturity. It motivates the selection of an EVM-compatible Layer 2 deployment for low-fee retail operations while retaining portability to other EVM chains.

Table 3.4: Blockchain platform selection (continued): operational and deployment factors.

Criterion	Selection	Justification
Gas cost	\$0.001–\$0.01 per transaction	Orders of magnitude cheaper than Ethereum mainnet (\$5–\$50); enables micro-loan economics.
Block finality	≈2 seconds	Sub-second practical finality for retail UX; checkpointed to Ethereum for security.
Developer tooling	Hardhat + OpenZeppelin	Automated test suite, deployment scripts, and audited security primitives.
Testnet availability	Amoy (Polygon), Sepolia (Ethereum)	Free faucets; EVM-identical behaviour; no real cryptocurrency required for prototype.
Migration path	EVM-compatible	Same Solidity contracts deploy to any EVM chain; reduces future L2 migration cost.

This supplementary comparison expands the platform evaluation to additional operational and deployment factors. The conclusion is that Polygon PoS provides a practical balance of stability and low transaction costs for an academic prototype, with a clear migration path if future requirements demand alternative L2s.

3.3.1 Transaction Verification and Consensus

- **Polygon PoS:** An independent set of PoS validators checks transactions, reaching block finality in approximately two seconds. Checkpoints are periodically committed to the Ethereum mainnet for additional security.

- **On prototype testnets:** Amoy and Sepolia use equivalent consensus models at no financial cost, enabling incremental development and testing without exposure to real-asset risk.

3.3.2 Oracle Architecture: Off-Chain AI to On-Chain Decision

The Crypto World Bank requires a mechanism to convey off-chain AI/ML risk assessments into the on-chain loan approval workflow. This is an instance of the oracle problem [R1,R2]. The prototype implements a **commit-reveal relay** pattern, illustrated conceptually as follows:

1. The off-chain FastAPI ML service evaluates a loan application and produces a risk score $s \in [0, 1]$.
2. Before the loan decision window opens, the service submits a commitment: $h = \text{keccak256}(s \parallel \text{nonce})$ to the LoanController contract via the `commitRiskScore(loanId, h)` function.
3. Within the decision window, the service reveals s and the nonce: `revealRiskScore(loanId, s, nonce)`.
4. The contract verifies $h = \text{keccak256}(s \parallel \text{nonce})$, stores s immutably, and applies the governance-defined threshold: loans with $s > 0.75$ are flagged for manual review regardless of collateral status.

This pattern prevents score manipulation between commitment and decision, keeps ML computation off-chain (saving gas), and creates an immutable on-chain audit trail of every risk score used in a lending decision. In the final thesis phase, this centralized relay will be replaced by Chainlink Functions, removing the remaining trust assumption at the cost of additional latency and per-call fees.

The diagram placeholder for this architecture (Figure pending): `oracle_architecture.png`—
off-chain FastAPI ML service → commit → reveal → on-chain LoanController.

3.4 Data Model and Database Design

The relational database schema comprises 15 normalized entities in Third Normal Form (3NF). Figure 3.4 presents the core system graph showing entity relationships, Figure 3.5 presents the full Entity-Relationship Diagram (ERD), and Figure 3.7 presents the Enhanced Entity-Relationship (EER) diagram.

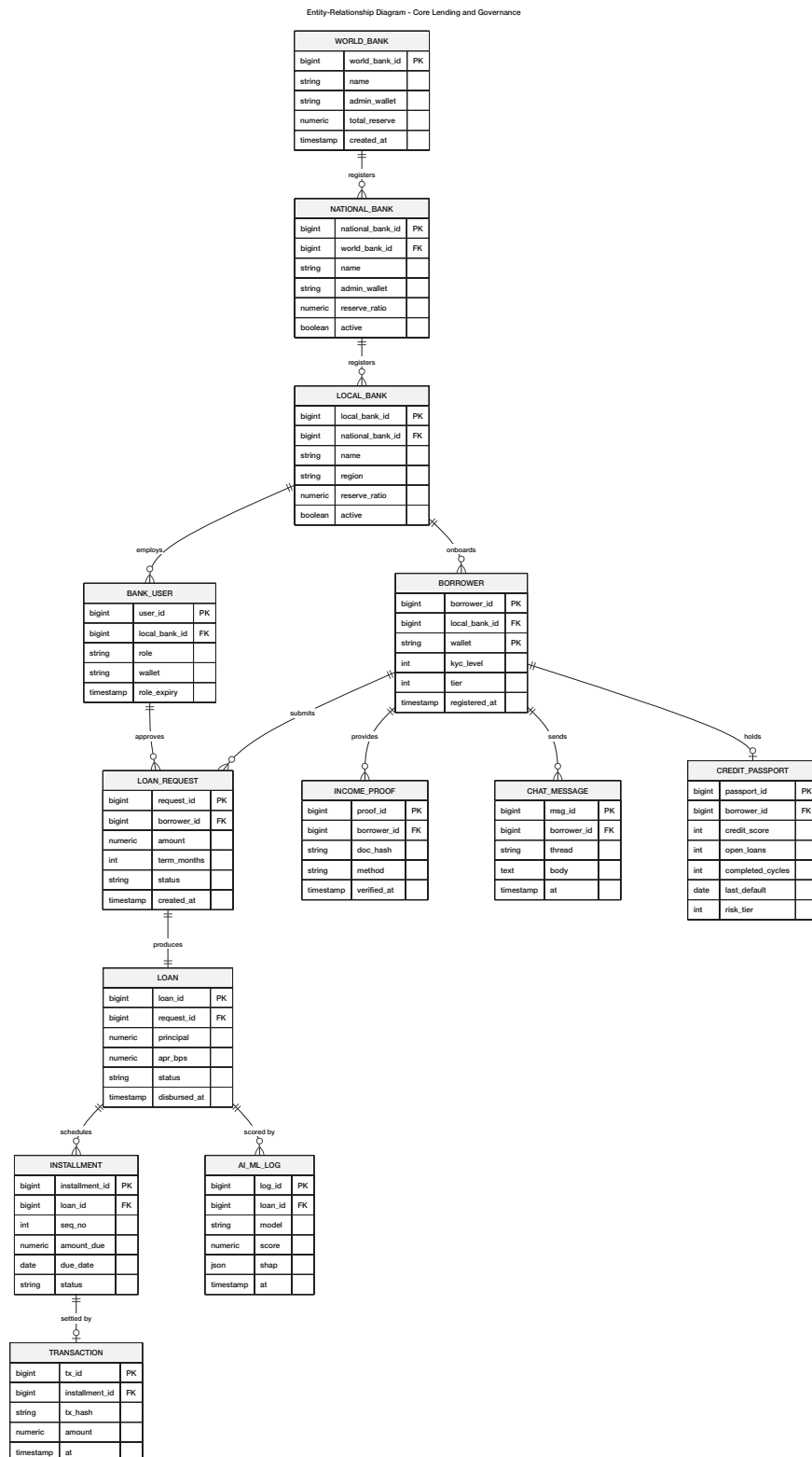


Figure 3.4: Core system graph showing entity relationships across the four-tier banking hierarchy (WORLD_BANK → NATIONAL_BANK → LOCAL_BANK → BANK_USER), the central BORROWER entity, and the lending-lifecycle sub-graph (LOAN_REQUEST, LOAN, INSTALLMENT, TRANSACTION, INCOME_PROOF, CHAT_MESSAGE, AI_ML_LOG, CREDIT_PASSPORT).

Reading guide. Arrows denote *one-to-many (1:N)* relationships flowing downward through the institutional hierarchy. **BANK_USER** specialises into *National* and *Local* variants (disjoint generalisation—see Figure 3.7). **BORROWER** is the pivot entity: it aggregates identity, credit history, and all transactional records. **LOAN_REQUEST** is the *aggregation hub* for the lending lifecycle: **INSTALLMENT** (weak entity), **CHAT_MESSAGE** (audit trail), and **AI_ML_LOG** (risk scoring) all depend on it existentially. The complete attribute sets for every entity are shown in the ERD (Figure 3.5) and the full EER model (Figure 3.7).

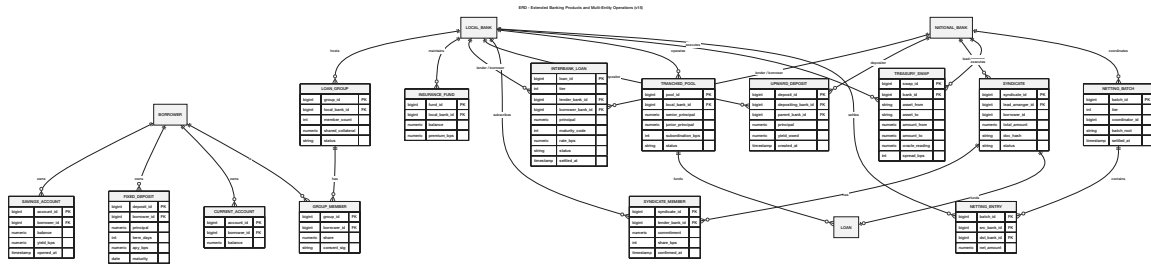


Figure 3.6: Extended ERD entities introduced in v15: the banking-product entities (SAVINGS_ACCOUNT, FIXED_DEPOSIT, CURRENT_ACCOUNT, LOAN_GROUP, GROUP_MEMBER, INSURANCE_FUND) and the multi-entity / cross-tier operational entities (INTERBANK_LOAN, UPWARD_DEPOSIT, SYNDICATE, SYNDICATE_MEMBER, TRANCED_POOL, TREASURY_SWAP, NETTING_BATCH, NETTING_ENTRY). Foreign-key relationships into the core entities of Figure 3.5 are preserved.

The current ERD covers the core lending and governance entities. Extended banking entities including SavingsAccount, FixedDeposit, LoanGroup, GroupMember, CurrentAccount, and InsuranceFund are designed for the full system and will be incorporated into the updated data model in the final thesis. v15 additionally introduces the multi-entity / cross-tier operational entities described in Section 3.13.7: INTERBANK_LOAN, UPWARD_DEPOSIT, SYNDICATE and SYNDICATE_MEMBER, TRANCED_POOL, TREASURY_SWAP, and NETTING_BATCH with NETTING_ENTRY. Their foreign-key relationships into the core lending entities (LOAN, BANK_USER, REPAYMENT) are detailed in Section 3.13 and will be drawn into the updated ERD figure in the diagram-audit session. The data model therefore remains in one-to-one consistency with the contract architecture and Sprint plan.

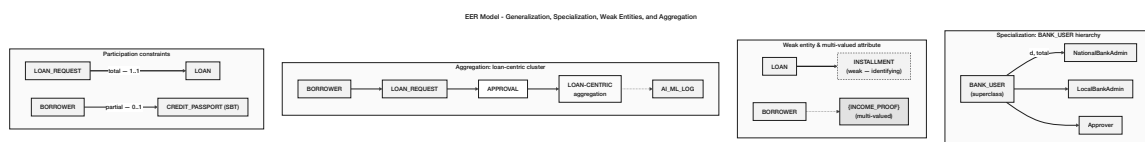


Figure 3.7: Enhanced Entity-Relationship (EER) model showing generalization / specialization (BANK_USER → National/Local-bank-admin / Approver subtypes), the weak entity **INSTALLMENT** identified by its parent **LOAN**, the multi-valued attribute **INCOME_PROOF**, the loan-centric aggregation, and participation constraints (total / partial).

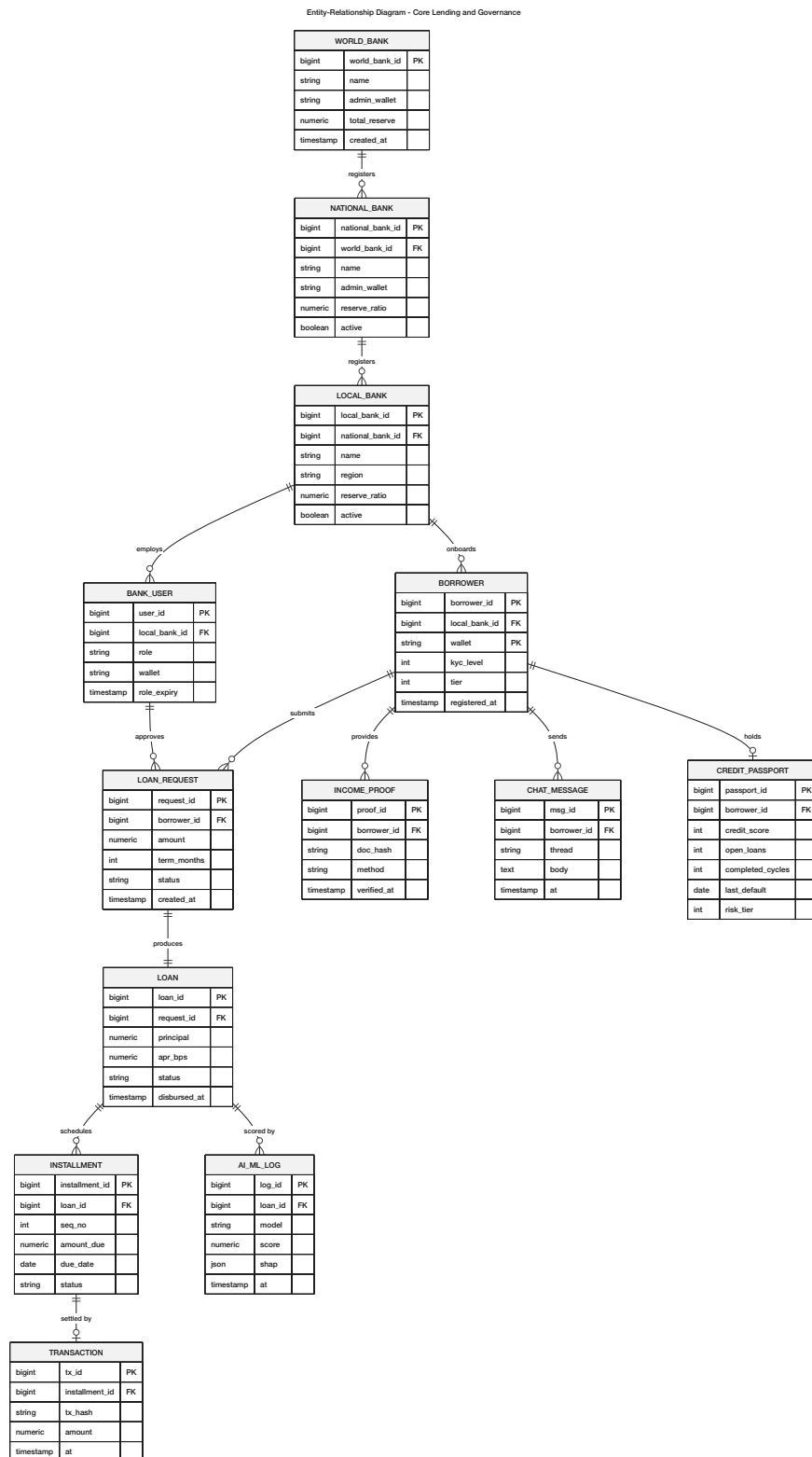


Figure 3.5: Entity-Relationship Diagram (ERD) of the Crypto World Bank database: core lending and governance entities in Third Normal Form (3NF), with primary keys, foreign keys, attribute types, and crow's-foot cardinality. Multi-entity / cross-tier and extended-product entities are shown in Figure 3.6.

3.4.1 Entity Summary

Table 3.6: Database entity summary (15 entities).

Entity	Role / Description
WORLD_BANK	Top-level reserve holder; global lending parameters.
NATIONAL_BANK	Country-level banks; borrow from World Bank.
LOCAL_BANK	City-level banks; borrow from National Bank and lend to users.
BANK_USER	Bank staff with role-based permissions (approve/reject loans).
BORROWER	End-users requesting and repaying loans.
LOAN_REQUEST	Loan applications and full lifecycle tracking.
INSTALLMENT	Repayment schedule records (weak entity dependent on LOAN_REQUEST).
TRANSACTION	Financial transaction records.
BORROWING_LIMIT	Per-borrower limits with 6-month and 1-year rolling windows.
INCOME_PROOF	Income verification documents (multi-valued per borrower).
CHAT_MESSAGE	Borrower-bank communication records.
AI_CHATBOT_LOG	AI chatbot interaction records.
AI_ML_SECURITY	Security and ML monitoring events.
MARKET_DATA	Cryptocurrency price feeds.
PROFILE_SETTING	User profile and platform preferences.

This entity summary table enumerates the relational entities used to model hierarchical institutions, lending lifecycle, verification, and communication. The on-chain/off-chain split implied by the entities supports the architectural principle that high-integrity state transitions occur on-chain while analytics and document workflows remain off-chain.

3.4.2 EER Constructs Applied

Table 3.8: EER constructs applied: specialisation, hierarchy, and constraints.

EER Construct	Applied To	Notes
Specialisation (dis-joint)	BANK_USER $\rightarrow$ NationalBankUser, LocalBankUser	A bank user belongs to exactly one bank type; enforced by CHECK constraint.
Generalisation	WORLD_BANK, NATIONAL_BANK, LOCAL_BANK share institution attributes	Avoids attribute duplication across institution types.
Weak entity	INSTALLMENT depends on LOAN_REQUEST	Existence and identity determined by parent loan.
Multi-valued attribute	INCOME_PROOF per BORROWER	Modelled as separate entity to maintain 1NF.
Aggregation	LOAN_REQUEST aggregates BORROWER + LOCAL_BANK	Captures the ternary relationship between borrower, bank, and loan.
Participation constraint	BORROWER must have ≥ 1 LOAN_REQUEST to access services	Enforced at application layer; architecturally planned for on-chain.

This table captures the EER constructs applied to represent specialization, hierarchy, and constraints in the data model. It demonstrates how institutional roles and banking participants are modeled without duplicating attributes across tables, improving data integrity and query clarity.

3.4.3 Normalization

The schema has been normalized to Third Normal Form (3NF) and checked against Boyce-Codd Normal Form (BCNF):

- **1NF:** There are no repeating groups in any of the attributes. Separate entities store multi-valued attributes, such as proof of income.
- **2NF:** There are no partial dependencies. The full composite key (loan_id, installment_number) determines the non-key attributes of INSTALLMENT.
- **3NF:** No dependencies that go through other dependencies. Instead of storing redundant values like total_borrowed in bank entities, they are calculated at query time.
- **BCNF:** All determinants are candidate keys. A CHECK constraint on BANK_USER enforces that the generalization hierarchy's specializations are not the same.

3.4.4 Indexing Strategy

We use B-tree indexes (the default in PostgreSQL) for efficient retrieval on frequently queried columns:

Table 3.10: Indexing strategy: B-tree indexes for time-window and high-frequency queries.

Index	Table / Column(s)	Rationale
idx_loan_borrower	LOAN_REQUEST(borrower_id)	High-frequency lookup for borrower loan history.
idx_loan_status	LOAN_REQUEST(status)	Efficient filtering of active vs. closed loans.
idx_installment_due	INSTALLMENT(due_date)	Range query for overdue installment detection.
idx_txn_created	TRANSACTION(created_at)	Time window aggregation for 6-month and 1-year borrowing limits.
idx_txn_borrower	TRANSACTION(borrower_id)	Per-borrower transaction history retrieval.
idx_market_symbol	MARKET_DATA(symbol, recorded_at)	Composite index for price feed time-series queries.

This indexing table lists the primary indexes used to support time-window queries (e.g., rolling 6-month/1-year transaction windows) and high-frequency lookups. The selection emphasizes B-tree suitability for range filtering common in risk analytics and reporting, improving performance without complicating the schema.

B-tree indexes are chosen for their efficiency with range queries (e.g., transactions within the last 6 months); hash indexes would be suitable only for exact-match lookups.

3.4.5 Functional Dependencies

Table 3.12: Representative functional dependencies.

Relation	Functional Dependency	Notes
LOAN_REQUEST	loan_id $\rightarrow$ all attributes	Primary key determines row.
LOAN_REQUEST	borrower_id, local_bank_id $\rightarrow$ status, amount, ...	One active request per borrower per bank.
BORROWING_LIMIT	borrower_id $\rightarrow$ six_month_limit, one_year_limit, ...	1:1 with BORROWER.
BORROWER	wallet_address $\rightarrow$ borrower_id, all attributes	Wallet address is a unique candidate key.
INSTALLMENT	loan_id, installment_number $\rightarrow$ amount, due_date, status	Composite primary key; fully determined by both columns.

This functional dependency table documents the data-level rules that prevent inconsistent state, such as improper loan status transitions or duplicated relationships. Explicit FDs reinforce that key banking invariants are enforced structurally, complementing smart contract enforcement on the on-chain side.

3.4.6 Relational Integrity Constraints

Table 3.14: Relational integrity constraints.

Constraint Type	Examples
Primary Key	world_bank_id, loan_id, borrower_id, etc.
Foreign Key	local_bank_id in LOAN_REQUEST references LOCAL_BANK.
UNIQUE	wallet_address in BORROWER; blockchain_tx_hash in LOAN_REQUEST.
CHECK	BANK_USER: (bank_type='national' AND national_bank_id IS NOT NULL) OR (bank_type='local' AND local_bank_id IS NOT NULL).
NOT NULL	Core attributes: name, wallet_address, status.

This integrity constraints table summarizes referential and domain constraints that keep loan, repayment, and identity records consistent. These constraints reduce operational risk by preventing orphaned records and invalid lifecycle states, which is critical for auditability in financial systems.

3.5 On-Chain and Off-Chain Data Partitioning

Table 3.16: Data partitioning between on-chain and off-chain storage.

Data Category	Storage	Rationale
Reserve balances, loan requests, approval/rejection events, repayment transactions	On-chain	Immutability, public auditability, trustless verification.
User profiles, income verification documents, chat messages, AI/ML inference logs	Off-chain (database)	Data privacy, query flexibility, storage cost optimisation.
Borrowing limit computations	Off-chain with on-chain enforcement	Complex temporal aggregation; results committed as on-chain constraints.
Cryptocurrency market data	Off-chain (cached)	High-frequency updates; external API dependency.

This partitioning table clarifies which data must live on-chain (role bindings, reserves, and critical state transitions) versus off-chain (documents, analytics features, and operational metadata). The partitioning choice balances transparency and tamper-resistance with practical storage and privacy constraints.

3.6 Digital Identity System

The platform's identity model operates across two layers, combining wallet-based authentication with off-chain document verification, and is designed to accommodate compliance requirements in future phases.

- **Identity based on wallets:** Users authenticate via Ethereum wallet signatures (MetaMask, WalletConnect), eliminating the need for centralised credential storage.
- **Role binding:** In the smart contract permission system, wallet addresses are linked to hierarchical roles like Owner, National Bank, Local Bank, Approver, and Borrower.
- **Limits of wallet-based identity:** Wallet ownership proves transaction history and cryptographic control but cannot independently prove legal identity, jurisdiction, or age. This distinction is critical for regulated banking operations. A compromised or lost wallet requires an off-chain recovery process and on-chain role revocation followed by re-binding to a replacement address, a governance workflow that must be carefully designed to prevent unauthorized role escalation.
- **On-chain versus off-chain identity layers:** The system maintains two identity layers. On-chain identity consists of the wallet address, assigned role, and transaction history recorded permanently on the blockchain. Off-chain identity consists of document-verified income, KYC credentials, and AML screening results stored in the PostgreSQL database and linked to the wallet address by hash reference. The planned ZKP-based compliance extension would allow users to prove off-chain KYC status to the smart contract without exposing personal documents on-chain.

3.6.1 ZKP KYC and zkAML Compliance Architecture

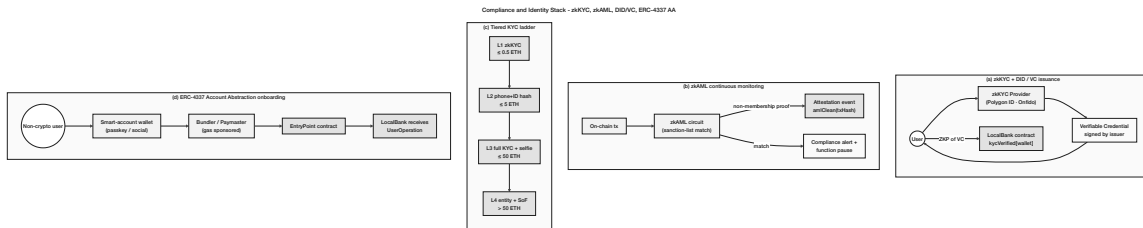


Figure 3.8: Compliance and identity stack: (a) zkKYC issuance via licensed identity provider with W3C Verifiable Credential, (b) zkAML continuous monitoring with sanction-list non-membership proofs, (c) the tiered KYC ladder (L1 zkKYC → L4 entity onboarding) gated by loan size, and (d) ERC-4337 Account-Abstraction onboarding for non-crypto users with gas sponsored by a Paymaster.

The compliance gateway uses two cooperating zero-knowledge circuits rather than a single KYC verifier: a **KYC circuit** and an **AML circuit**, both expressed as zk-SNARKs (Groth16 proofs via Circom 2.0 + snarkjs). KYC proves that the user owns a valid identity credential; AML proves that the user’s wallet has not interacted with sanctioned addresses and stays within velocity limits. Decoupling the two circuits makes it possible to revoke AML certification without invalidating the KYC credential, and matches the actual structure of regulator-mandated compliance.

KYC circuit (identity). An off-chain KYC provider validates the user’s NID and issues a signed credential $credential = \text{Sign}_{\text{KYC}}(wallet_address, country, age_over_18, kyc_passed)$. The user generates a zk-SNARK proof that (a) they possess a valid signed credential from an approved KYC provider, (b) their age is over 18, and (c) their country is in the permitted jurisdiction list—without revealing the underlying credential, NID, or personal data. The smart contract verifies the proof: `KYCVerifier.verify(proof, [wallet, country, age_over_18])` returns true. Piper et al. (2025) [R8] report proof generation in 1–4 s

on consumer hardware; on-chain verification costs $\approx 200\text{--}300\text{ k}$ gas, within a one-time registration budget.

zkAML circuit (anti-money-laundering). The zkAML circuit [R19] (IACR ePrint 2025/465) proves three properties of a wallet without revealing its transaction graph: (1) the wallet has not transacted with any address on a public sanctions list (Chainalysis / TRM Labs); (2) the wallet's 30-day inflow does not exceed a governance-set velocity threshold; and (3) the wallet has not been flagged by the platform's behavioral biometrics layer. The published benchmark for zkAML is 55 TPS on public networks with 226 ms proof-generation latency, both compatible with retail-tier flows on Polygon. The on-chain verifier is a sibling of KYCVerifier that exposes `AMLVerifier.verify(proof, [wallet, blockHeight])`. Both verifiers must return true before a wallet's BORROWER role is activated.

W3C DID / VC anchoring. The KYC credential issued off-chain is shaped as a W3C Verifiable Credential (VC) bound to a W3C Decentralized Identifier (DID) [R20] rather than a free-form signed blob. The DID is anchored on-chain (`did:ethr:0x...` per the EIP-1056 standard), and the smart contract checks possession of the VC by means of the zk-SNARK proof rather than by storing the VC itself. This places the platform's identity layer inside the Self-Sovereign Identity (SSI) framework and aligns it with the European Blockchain Services Infrastructure (EBSI) reference architecture.

Privacy posture. A 2025 ResearchGate study reports a 97% reduction in exposed user data versus conventional on-chain KYC; Decker (2025) [R9] gives the same finding under a different threat model. Implementation for the final thesis phase will deploy both Circom 2.0 circuits to Polygon Amoy testnet, tested with synthetic credential data drawn from the platform's mock KYC provider.

3.7 User Taxonomy and Onboarding Flows

The thesis previously identified five primary actors (Borrower, Local Bank, National Bank, World Bank Admin, Approver). A complete banking platform has seven user types when the distinction between visitor, registered-but-unverified, and KYC-verified retail users is made explicit, and when the institutional roles are split into operator and approver. Table 3.18 gives the complete taxonomy with per-user-type KYC level, wallet type, and database-residence assumption.

Table 3.18: Seven-user taxonomy with KYC level, wallet type, and data-residence assumption. The taxonomy expands the actor list referenced in the use-case diagram of Section 3.15.1.

User Type		Tier	KYC Level	Wallet Type	Data Residence
Anonymous Visitor		—	None	None	Session only (Redis)
Registered User	Retail	Tier 4	Level 1 (NID + selfie)	ERC-4337 abstracted (email / Google login)	Off-chain primary; on-chain role only
Group Borrower		Tier 4	Level 1 + group verification	ERC-4337 abstracted	Off-chain + on-chain group contract
Local Bank Operator		Tier 3	Full institutional KYC	MetaMask (institutional)	Off-chain + on-chain role binding
Local Bank prover	Ap-	Tier 3	Full institutional + background check	MetaMask + Safe co-signer	Off-chain + on-chain role binding
National Bank Admin	Ad-	Tier 2	Full institutional + regulatory license	Safe multisig mandatory	Off-chain + on-chain role binding
World Bank Admin (Governance)		Tier 1	Founding team + supervisor attestation	Safe 3-of-5 multisig	On-chain governance only

3.7.1 ERC-4337 Account Abstraction for Retail Onboarding

The single largest accessibility contradiction in the original thesis was that retail borrowers in developing economies were assumed to already understand MetaMask, seed phrases, gas fees, and Ethereum transactions. This is incompatible with the financial-inclusion mission. The platform resolves the contradiction by adopting **ERC-4337 Account Abstraction (AA)** for all retail-tier (Tier 4) borrowers, with the EIP-7702 Pectra upgrade (May 2025) providing the gas-sponsorship layer.

Under ERC-4337 each retail borrower is issued a smart-contract account that is created on first sign-in. The user can register with email or a Google account via Firebase Auth, never sees a seed phrase, and never needs to acquire ETH for gas: a Paymaster contract sponsors the gas, funded by the World Bank Reserve as a financial-inclusion subsidy line. Account recovery uses a social-recovery guardian (a trusted email / phone) rather than a 24-word mnemonic.

For the demo and final thesis prototype, the Biconomy SDK and Alchemy Account Kit will provide the bundler / Paymaster infrastructure on Polygon Amoy. From the smart-contract perspective, an account-abstracted wallet is indistinguishable from an externally owned account, so none of the four-tier role logic needs to change. From the borrower's perspective, the cryptocurrency stack is invisible: they see a Bengali- or English-language banking interface, sign in with email, and apply for a loan. This re-aligns the implementation with the financial-inclusion contribution claimed in Chapter 1.

3.7.2 Tiered Risk-Based KYC

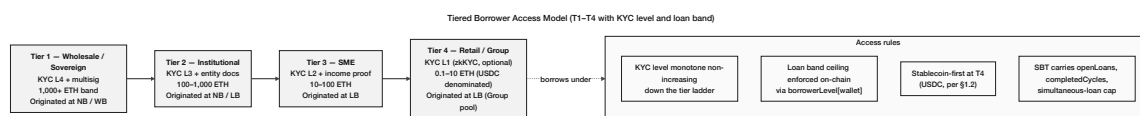


Figure 3.9: Tiered borrower access model. KYC level is monotone non-increasing down the tier ladder; the loan-band ceiling is enforced on-chain via `borrowerLevel[wallet]`; stablecoin denomination is enforced at Tier 4 (Section 1.10.3); and the SBT (Section 3.11) carries the simultaneous-loan cap that the `GroupLendingPool` reads at every application.

Identity verification is gated by a risk-based KYC ladder rather than an all-or-nothing flag. Table 3.19 gives the four levels and the corresponding borrowing limits and verification times. The design is consistent with current Persona / Veriff guidance for retail-fintech KYC and the EU AI Act Article 86 transparency requirements.

Table 3.19: Tiered, risk-based KYC ladder. Higher KYC level unlocks larger loans but requires proportionally more verification.

Risk Level	Required Documents	Borrow Limit	Verify Time
Level 0 (Un-verified)	None	View only	Instant
Level 1 (Basic)	NID photo + selfie liveness check	Up to 0.1 ETH-equivalent (USDC)	2–5 min (automated)
Level 2 (Standard)	NID + proof of income + bank statement	Up to 2 ETH-equivalent	24 h (human review)
Level 3 (Enhanced)	Full document set + video interview	Up to 10 ETH-equivalent	48–72 h

After KYC approval, the smart contract sets `kycVerified[wallet] = true, kycLevel[wallet] = level`, and a one-year expiry `kycExpiry[wallet] = block.timestamp + 365 days`. Re-verification is enforced via the role-expiry mechanism (Section 3.6.1). Personal data—NID number, biometric template, document files—is never stored on-chain; only a SHA-256 hash of the document and the zk-SNARK proof are written, in compliance with GDPR data-minimization principles and Bangladesh’s Personal Data Protection regime.

3.7.3 Five-Stage Conversion Funnel for Non-Crypto Users

The platform’s retail conversion strategy is structured as a five-stage funnel that progressively reveals technical complexity rather than demanding it up front. Stage 1 is information browsing with zero friction (no account required, Bengali / English chatbot answers in plain language). Stage 2 is account creation with email or phone—an ERC-4337 smart account is created in the background; the user never sees a seed phrase. Stage 3 is KYC verification via phone-camera document capture with automated AI verification in 2–5 minutes. Stage 4 is the first loan, with the platform explaining the workflow in plain language and showing the repayment schedule in local-currency equivalent through the Chainlink ETH-USD feed converted to BDT. Stage 5 is the optional power-user mode in which the borrower can

connect a self-custodied MetaMask wallet, view their transactions on a block explorer, or participate in solidarity-group lending. The funnel hides crypto until the user is ready for it and re-frames the platform as “a bank that happens to settle on a public ledger” rather than “a DeFi protocol that happens to call itself a bank”.

3.8 Kinked Interest Rate Model

The Crypto World Bank adopts a **kinked utilization-based interest rate model**, following the design established by Compound Finance and Aave v2/v3 and motivated by empirical findings from Gudgeon et al. (2020) [R3]. Below an optimal utilization rate U^* (set at 80% for retail lending pools), the borrowing rate increases gently:

$$r_b(U) = r_0 + \frac{U}{U^*} \cdot r_1, \quad U \leq U^* \quad (\text{Formula 18})$$

Above U^* , the rate increases steeply to incentivize rapid repayment and new deposits:

$$r_b(U) = r_0 + r_1 + \frac{U - U^*}{1 - U^*} \cdot r_2, \quad U > U^* \quad (\text{Formula 19})$$

where r_0 is the base rate, r_1 is the slope below the kink, and r_2 is the jump multiplier above the kink. This piecewise model prevents liquidity crises in which utilization approaches 100% and depositors are unable to withdraw—a failure mode documented empirically in DeFi lending markets by Gudgeon et al. (2020) [R3] and analyzed theoretically by Mackinga et al. (2023) [R4].

3.9 Liquidation Engine

The Health Factor formula (HF in the List of Formulas) is operationally meaningless without a contract that monitors it and an actor incentivized to act on the result. The original v10 thesis listed HF but contained no liquidation mechanism: when HF dropped below 1.0, the system relied on a human approver noticing and reacting, which is a ledger, not a bank. The Crypto World Bank therefore specifies a dedicated **LiquidationEngine** contract that closes this gap.

Liquidation life-cycle.

1. A borrower’s collateral price is updated from the Chainlink price feed at every block; HF is recomputed in view-only fashion (no gas cost to the borrower).
2. Any external participant may call `liquidate(loanId)` when $HF < 1.0$.
3. The contract verifies HF on-chain, transfers a configurable portion of the collateral (typically equal to the outstanding debt plus a 5–8% liquidation bonus, following Aave v3 parameters) to the liquidator, and updates the borrower’s on-chain credit record.
4. A `LoanLiquidated(loanId, liquidator, bonusBps)` event is emitted and indexed by The Graph for the dashboard.

Incentive design. The 5–8% liquidation bonus is what makes liquidation self-actuating: any wallet (including a public arbitrage bot) can call `liquidate` when HF dips, captures

the bonus, and protects the lending pool from accumulating bad debt. This is a standard DeFi primitive in production protocols since 2020 (Aave, Compound, MakerDAO) and its absence from a paper that calls itself a banking system is a credibility gap that v15 closes.

Hierarchical liquidation queue. The platform extends the standard single-tier liquidation primitive with a hierarchical priority: when a Local Bank exceeds a portfolio-level reserve-ratio threshold (RR below 15%), liquidation calls from authorized liquidators are routed through a National-Bank-level queue that prioritizes the lowest-HF positions first. This converts the same liquidation primitive into a tier-aware portfolio-health control rather than a per-loan reflex, matching how Tier 2 institutions actually manage credit risk.

3.10 SavingsVault and FixedDeposit Modules

A bank is fundamentally a deposit institution, not a credit-only protocol. The original v10 design described the SavingsVault as “planned” but left the contract unspecified, which left the entire “closed-loop sustainable model” claim resting on capital that did not exist. v15 specifies the two deposit-mobilization contracts directly.

SavingsVault. The SavingsVault contract accepts ERC-20 deposits (USDC primary, ETH/USDT/DAI for institutional flows), credits the depositor’s balance on-chain, accrues variable yield tied to pool utilization (the kinked rate of Section 3.8), and permits withdrawal subject to the local-tier reserve ratio gate. The yield is paid in the same token as the deposit; interest accounting is per-second using a cumulative index rather than per-block accrual, exactly matching the Compound v3 / Aave v3 supply-side pattern. The contract is upgradable via UUPS (Section 3.19) and emits `Deposit`, `Withdraw`, `InterestAccrued`, and `ReserveGateTriggered` events for The Graph indexing.

FixedDeposit. FixedDeposit issues a non-transferable on-chain receipt that locks principal for a fixed term (30 / 90 / 180 / 365 days) at an APY written at deposit time. Early withdrawal forfeits a deterministic portion of accrued interest, transparently. Fixed-term deposits provide the Local Bank tier with **duration-matched liabilities**: a 180-day Fixed-Deposit funds the 6-month installment tranche of a retail borrower at the same tier without creating asset-liability mismatch. The contract integrates with the InsuranceFund so that, under stress scenarios, fixed-term depositors are subordinated to demand depositors only after the insurance buffer is exhausted.

Closed-loop economic argument. With SavingsVault and FixedDeposit in place, the platform’s economic claim becomes verifiable: depositor capital flows in on one side, lending flows out on the other, interest collected is split deterministically between depositor yield, the insurance fund, and protocol revenue under Formula NetInterest (see List of Formulas). The model recycles capital without external subsidy, distinguishing the platform from donor-funded MFIs and matching the empirical evidence on sustainable microfinance that Section 2.1.1 cites.

3.11 On-Chain Credit Passport (Soulbound Token)

The platform issues each KYC-verified borrower a non-transferable on-chain credential—a Soulbound Token (SBT) following the Buterin, Hitzig, and Weyl framework [R30]—that en-

codes their cumulative repayment record in a schema that any compatible lending protocol can read. The credential is updated automatically whenever an installment is repaid in full, downgraded after a defined number of missed installments, and revoked on default. Three concrete benefits follow.

First, the SBT solves the cold-start problem at the bank-switching boundary: a borrower who has repaid three loans at a Local Bank in Sylhet can apply for a larger loan at a Local Bank in Dhaka without re-establishing trust from zero. Second, the SBT is the on-chain primitive that the over-indebtedness control (Section 3.17.3) relies on: a borrower's open-loan count across the platform is publicly readable from their SBT, so simultaneous-loan caps are enforceable without an external credit bureau. Third, the schema is intentionally minimal ('creditScore', 'openLoans', 'completedCycles', 'lastDefault', 'riskTier') so other lending protocols that adopt the standard can read the credential, allowing progressive lending to follow a borrower across protocols. MicroSave Consulting (December 2025) [R31] documents that Bangladesh's microfinance sector is actively shifting toward performance-based credit identity; the SBT is the technical implementation of that shift on the Crypto World Bank platform.

3.12 Cross-Chain Bridge Architecture

A multi-chain deployment (Polygon Amoy + Ethereum Sepolia in the prototype, with main-net successors planned) without cross-chain state consistency is a liability: if a borrower has an active loan on Polygon and attempts to open a second loan by routing through Ethereum, the hierarchical borrowing-limit control could be bypassed. v15 makes the bridge architecture explicit rather than leaving it as "portable."

Bridge protocol selection. The platform uses a single canonical bridge protocol for cross-chain messaging rather than a bespoke contract. The 2025–2026 leaders are LayerZero (largest message-volume share), Axelar (GMP for arbitrary payloads), and Chainlink CCIP (institutional-grade with native MEV protection). The current prototype targets Chainlink CCIP because the platform already uses Chainlink Functions for the oracle layer (Section 3.3.2) and Chainlink Price Feeds for FX, which minimizes new trust assumptions. Bridge hacks were the largest single category of crypto loss between 2022 and 2024; concentrating on one well-audited bridge rather than maintaining multiple is the correct security posture for an academic prototype.

Cross-chain hierarchy invariant. The hierarchical control is preserved across chains by requiring that all loan state for a given borrower live on a single chain (the chain of the originating Local Bank). Cross-chain messages are restricted to (a) reserve-ratio updates from World Bank Reserve to National Banks across chains, and (b) credit-passport SBT mirroring across chains so that a borrower's credential is readable on any chain even though their loans are not. This pattern keeps borrowing-limit enforcement single-chain and avoids the consensus problem of cross-chain debt.

3.13 Multi-Entity and Cross-Tier Capital Operations



Figure 3.10: Multi-entity and cross-tier capital operations: (a) InterBankLendingPool with utilization-kinked rate model and default cascade, (b) UpwardDepositFacility with asymmetric rate structure, (c) SyndicatedLoan with Lead Arranger and Co-Lenders, (d) TranchedList with senior–junior waterfall, (e) TreasurySwap for cross-tier asset exchange, and (f) NettingEngine for multilateral settlement.

The Chapter 1 framing of multi-directional flows (Section 1.6) and Section 1.7 promise downward, same-tier, and upward lending plus tiered borrower access. v10 left same-tier and upward flows at design-level only. A paper that claims a *complete* banking system must specify these flows—and the genuinely multi-entity operations they enable—at the same depth as the downward flow. This section closes that gap with six concrete contract-and-flow specifications: a fully-specified InterBankLendingPool (Section 3.13.1), upward surplus repatriation (Section 3.13.2), syndicated / club lending (Section 3.13.3), senior–junior tranching lending pools (Section 3.13.4), cross-tier treasury FX swap (Section 3.13.5), and a multilateral settlement netting engine (Section 3.13.6). The corresponding ERD entities, contract list, Sprint plan, and Future Work are updated downstream so the database design, software-engineering stages, and architecture remain a single consistent system.

3.13.1 InterBankLendingPool: Fully Specified Same-Tier Lending

Pool placement. One InterBankLendingPool (IBLP) contract is deployed per hierarchical tier: IBLP_NB at the National-Bank tier and IBLP_LB at the Local-Bank tier. Membership in a tier’s IBLP is gated by the on-chain role grant from the parent tier (a Local Bank can only deposit into IBLP_LB if its LOCAL_BANK_ROLE is active and unexpired—Section 3.19). The pool denominates positions in the platform’s stablecoin numeraire (USDC per Section 1.10.3); ETH-denominated positions are accepted at higher tiers but cleared through the TreasurySwap contract (Section 3.13.5) before pool entry.

Rate model. The interbank borrowing rate $r_{IB}(U)$ follows a kinked utilization curve identical in shape to the retail kinked rate (Section 3.8) but with two distinguishing properties. First, the slope coefficients are calibrated to the SOFR-style overnight market: the base rate r_0^{IB} is set to the platform’s deposit-yield benchmark, and the kink point $U_{IB}^* = 0.90$ is higher than the retail kink (because banks have lower default risk and tighter operational reserves than retail borrowers). Second, the formula is upper-bounded by the tier’s *downward* lending rate from the tier above:

$$r_{IB}(U) \leq r_{\text{downward}}(U) - \delta,$$

where δ is the governance-set risk premium spread that prevents a Local Bank from arbitraging the tier above by borrowing from IBLP_LB at a cheaper rate than from its parent National Bank. This bound is enforced on-chain inside the rate-setter function.

Settlement and default cascade. Loans through the IBLP are short-tenor (1-, 7-, 30-day) and are settled atomically: the borrowing bank receives the stablecoin amount in the same

transaction in which its repayment obligation is recorded. If the borrowing bank defaults at maturity, the default cascade is: (1) the bank's reserve buffer above the minimum ratio is auto-debited; (2) if insufficient, the InsuranceFund (Section 3.10, with its claim mechanism extended in Section 3.13.2) is drawn; (3) if still insufficient, the parent tier (the bank's National Bank, or the World Bank for an NB default) absorbs the residual as a regulatory backstop and a governance vote is automatically queued through the TimeLock for the defaulting bank's role suspension. Every step emits a typed event indexed by The Graph (Section 4.8) so the cascade is fully auditable.

Operations. IBLP_* exposes four user functions and three governance functions:

- `deposit(uint256 amount, uint8 maturityCode)` — surplus bank supplies liquidity.
- `borrow(uint256 amount, uint8 maturityCode)` — deficit bank draws against pool, gated by per-bank credit ceiling, role expiry, and the IBLP-vs.-downward-rate bound.
- `repay(uint256 borrowId)` — atomic principal + interest settlement.
- `withdraw(uint256 supplyId)` — surplus bank reclaims principal + accrued interest.
- `setRateParams(...)` — TimeLock-gated (24–48 h) governance update.
- `setMembershipPolicy(...)` — role-expiry / per-bank ceiling adjustment.
- `triggerCascade(uint256 borrowId)` — public function any caller can invoke once a borrow is overdue; pays the caller a small bonus from the defaulter's reserve buffer, mirroring the Aave liquidation incentive (Section 3.9).

3.13.2 Upward Surplus Repatriation

The upward direction is the dual of downward lending: a Local Bank whose reserve buffer exceeds the minimum ratio by more than a governance-set “excess threshold” (default 20 % above the minimum) may deposit the excess into its parent National Bank, earning the parent's *deposit yield*. A National Bank with similarly excess capital may deposit into the World Bank Reserve under the same logic. Three controls keep the upward flow safe:

1. **Asymmetric rate structure.** The upward deposit yield is strictly lower than the downward borrowing rate ($r_{\text{up}} < r_{\text{down}} - \delta$), creating a natural term structure across the hierarchy and removing the round-trip arbitrage incentive of borrowing-down to deposit-up.
2. **Withdrawal gate.** The depositing bank may withdraw its upward deposit on demand, but the withdrawal triggers a reserve-ratio recomputation at the depositing bank's own tier first; if the withdrawal would push the depositor below its minimum reserve ratio, the call reverts. This protects retail depositors at lower tiers from a liquidity-squeeze upstream.
3. **Recursive cap.** A given bank's net upward exposure (sum of upward deposits less own borrowings) is capped at 30 % of its on-chain assets, so a single failing bank cannot drain the entire upstream tier.

The on-chain contract is named `UpwardDepositFacility` (one instance per parent tier); it is a thin extension of the `SavingsVault` contract (Section 3.10) restricted by role, with separate accounting registers (`upwardSupplied[bank]`, `upwardYieldOwed[bank]`) so that upward and retail flows do not commingle in the same reserve pool.

3.13.3 Syndicated and Club Lending

For loan sizes that exceed a single bank's prudent exposure ceiling—typically the 1 000 + ETH bracket at the National-Bank tier (Table 1.1)—multiple banks at the same or adjacent tier co-fund a single loan through a `SyndicatedLoan` contract. This is the on-chain analogue of a real-world syndicated loan or club deal and is the answer to the supervisor's question about “multiple entities investing, funding, lending together.”

Roles. A syndicated loan has three distinct on-chain roles: a **Lead Arranger** (the bank that structures and underwrites the deal), **Co-Lenders** (the participating banks supplying capital), and the **Borrower** (a Local Bank, National Bank, or institutional borrower per Table 1.1). The Lead Arranger holds a small underwriting fee (10–20 basis points) plus a portion of the interest spread; Co-Lenders earn pro-rata interest by capital share.

Lifecycle.

1. **Proposal.** Lead Arranger calls `proposeSyndicate(uint256 totalAmount, uint8 borrowerTier, uint256 minSubscription)` which mints a syndicate identifier and emits a `SyndicateProposed` event. Off-chain, the deal terms (covenants, schedule) are pinned to IPFS and the content hash is stored in `syndicateDocHash[id]` for immutable cross-reference.
2. **Subscription window.** Co-Lenders call `subscribe(uint256 syndicateId, uint256 commitment)` within a governance-set window (default 7 days). Their commitment is escrowed in the contract.
3. **Funding threshold.** If aggregate commitments meet $\geq 90\%$ of `totalAmount` by window close, the deal funds; otherwise commitments are returned and the proposal expires. This avoids the partial-funding race condition.
4. **On-chain consent vote.** Each Co-Lender's wallet must call `confirmTerms(uint256 syndicateId)` after subscription; this is a mandatory unanimous step that gates final disbursement. A Co-Lender that subscribes but does not confirm forfeits its escrow up to a small bond.
5. **Disbursement.** On unanimous confirmation, the contract atomically transfers the aggregate to the Borrower in a single transaction. Each Co-Lender's exposure share is recorded as `shareBps[lender][syndicateId]`.
6. **Interest accrual + repayment.** Borrower installments flow into the `SyndicatedLoan` contract; the contract distributes principal + interest pro-rata to each Co-Lender on every installment via the basis-points share registry. The Lead Arranger receives its underwriting fee on the first repayment.
7. **Default handling.** On default, the `LiquidationEngine` (Section 3.9) seizes collateral; recovery is distributed pro-rata to Co-Lenders by the same basis-points share. A Co-Lender's loss is therefore bounded by its commitment, which is the desired multi-entity risk-sharing property.

Cross-tier syndication. The same contract supports *cross-tier* syndication when permitted by governance: one National Bank and three of its child Local Banks may co-fund a single large project loan where the NB is Lead Arranger. This is the genuine “lending from upper tiers as well as lower tier together” pattern the supervisor asked about; the role-grant logic of the `SyndicatedLoan` contract simply permits any wallet holding

NATIONAL_BANK_ROLE or LOCAL_BANK_ROLE (and any WORLD_BANK_ADMIN-approved institutional borrower) to subscribe.

3.13.4 Senior–Junior Tranched Lending Pools

For risk-segmented co-funding, the platform supports the senior–junior tranche pattern established by Goldfinch [32] and Maple [31]. The TranchedPool contract divides depositor capital into two notional tranches with distinct seniority:

- **Senior tranche:** lower yield, first-loss protection, suitable for risk-averse depositors (Local Banks redepositing surplus, or institutional partners).
- **Junior tranche:** higher yield, takes first loss on default, suitable for risk-tolerant capital (governance-token-aligned depositors, the protocol treasury, or impact-investment partners).

Waterfall payment. On each repayment, the contract pays from the inflow in the order: (i) senior accrued interest, (ii) junior accrued interest, (iii) senior principal pro-rata, (iv) junior principal pro-rata. On default the order reverses for losses: the junior tranche is written down first; only when junior principal reaches zero is senior principal touched. The subordination ratio (junior / senior) is fixed at deposit time and ranges from 10 / 90 (institutional-grade) to 30 / 70 (impact-aligned) per the policy table set by the World Bank Admin.

Stress test. A 30 % default-rate stress test (matching the agent-based simulation parameter of Section 4.7) on a 10 : 90 junior:senior pool would zero out the junior tranche but preserve the senior tranche fully; on a 30 : 70 pool a 30 % default writes the junior down to zero and senior takes a 0 % nominal loss. These dynamics are formalized as Foundry invariants (Section 4.6) to give automatic regression detection.

3.13.5 Cross-Tier Treasury FX Swap

The FXModule mentioned in Section 3.17 handles end-user FX. Cross-tier treasury FX—the case where a National Bank holding an ETH treasury must source USDC stablecoin to fund a Local Bank that has elected stablecoin denomination—is a distinct operation served by the TreasurySwap contract.

Mechanism. TreasurySwap is an oracle-priced atomic-swap contract restricted to wallets holding a banking role. A National Bank calls `swap(asset_from, asset_to, amount)`; the contract reads the Chainlink price feed, applies the governance-set treasury spread (typically 5–10 basis points, tighter than retail FX), and executes the swap atomically against either a counterparty bank’s posted offer or the protocol’s own TreasuryAMM liquidity pool. The Chainlink price reading and the swap execution sit inside the same transaction; there is no intermediate state and no settlement risk.

Hierarchy invariant. Treasury swaps cannot push the swapping bank below its minimum reserve ratio (the swap function recomputes the post-swap ratio before any state change). Swaps above a governance-set threshold (default \$1 million-equivalent) require a two-signature confirmation from the bank’s Safe multisig. Each swap emits a TreasurySwapExecuted event with both legs of the swap, the oracle reading used, and the spread, supporting both internal audit and regulatory reporting.

3.13.6 Multilateral Settlement Netting Engine

A four-tier hierarchical bank with same-tier interbank flows accumulates many small inter-bank obligations during a settlement period. Settling each one as a separate on-chain transaction is wasteful; the `NettingEngine` contract compresses them.

Workflow. At the end of each settlement window (1 hour, 1 day, or governance-set), an off-chain Settlement Coordinator computes a netted obligation matrix from the queued `IBLP` and `TreasurySwap` orders. The matrix is committed on-chain as a Merkle root, and a single `settleBatch(bytes32 batchRoot, bytes calldata proofs)` call executes the entire batch in one transaction. Each bank's net debit/credit is applied to its on-chain balance, and the constituent transactions are marked settled. This pattern follows the Hyperledger Cactus and BIS Project Agora cross-ledger netting designs [R39, R40].

Why this matters. For a network with n active banks per tier, naive bilateral settlement requires $O(n^2)$ transactions per period; multilateral netting compresses this to $O(n)$ at the cost of one off-chain coordination step. At full Local-Bank deployment ($n = 50$) this is a $25\times$ reduction in on-chain settlement footprint per settlement window. The trade-off is one trust assumption: the Settlement Coordinator role must be held by a Safe multisig (the same `WORLD_BANK_ADMIN` Safe), and any participating bank may dispute a batch within a challenge window by calling `disputeBatch(uint256 batchId, bytes proof)`.

Database mirror. Each `NettingBatch` is mirrored off-chain to a PostgreSQL `netting_batch` table with one `netting_entry` row per netted obligation, allowing dashboard display, regulator export, and the agent-based simulation (Section 4.7) to read historical settlement-compression ratios.

3.13.7 Database, Sprint, and Contract Consistency for Multi-Entity Operations

To keep the data model, software-engineering stages, and architecture coherent, the following entities, sprint deliverables, and contracts are added downstream of this section:

ERD entities added (data model in Chapter 3 ERD). `INTERBANK_LOAN` {loan_id (PK), tier, lender_bank_id (FK), borrower_bank_id (FK), principal, maturity_code, rate_bps, status, created_at, settled_at}; `UPWARD_DEPOSIT` {deposit_id (PK), depositing_bank_id (FK), parent_bank_id (FK), principal, yield_owed, created_at}; `SYNDICATE` {syndicate_id (PK), lead_arranger_id (FK), borrower_id, total_amount, status, doc_hash, opened_at, funded_at}; `SYNDICATE_MEMBER` {syndicate_id (FK), lender_bank_id (FK), commitment, share_bps, confirmed_at}; `TRANCED_POOL` {pool_id (PK), local_bank_id (FK), borrower_id, senior_principal, junior_principal, subordination_bps, status}; `TREASURY_SWAP` {swap_id (PK), bank_id (FK), asset_from, asset_to, amount_from, amount_to, oracle_reading, spread_bps, executed_at}; `NETTING_BATCH` {batch_id (PK), tier, coordinator_id, batch_root, opened_at, settled_at}; `NETTING_ENTRY` {batch_id (FK), src_bank_id, dst_bank_id, net_amount}. These are appended to the ERD prose at the end of Chapter 3 alongside `SavingsAccount`, `FixedDeposit`, `LoanGroup`, `GroupMember`, `CurrentAccount`, and `InsuranceFund`, and will be drawn into the updated ERD figure in the diagram-audit session.

Smart contracts added (Appendix B). InterBankLendingPool (one per tier), UpwardDepositFacility, SyndicatedLoan, TranchPool, TreasurySwap, NettingEngine. The complete architecture therefore grows from nine to fifteen modular contracts.

Sprint plan update (Chapter 1, Methodology in Brief). Sprint 2 takes ownership of InterBankLendingPool, UpwardDepositFacility, and TreasurySwap; Sprint 3 takes SyndicatedLoan, TranchPool, and NettingEngine (the latter requires the off-chain Settlement Coordinator service, which sits alongside the AI/ML pipeline already scheduled for Sprint 3). The Foundry invariant suite (Section 4.6) is extended in Sprint 3 with three additional invariants: (i) IBLP $r_{IB} < r_{down} - \delta$ at all times; (ii) Syndicate disbursement = sum of confirmed commitments; (iii) NettingBatch sum of net debits = sum of net credits.

3.14 Reentrancy and Security Analysis

Checks-Effects-Interactions (CEI) pattern. The CEI pattern is applied to all state-mutating functions in the lending contracts. The three functions most vulnerable to reentrancy are:

1. `disburseLoan(address borrower, uint256 amount)` — sends ETH to the borrower. Mitigation: update `loanStatus[borrower] = LoanStatus.ACTIVE` *before* calling `payable(borrower).transfer(amount)`, and apply `nonReentrant`.
2. `processInstallment(uint256 loanId)` — updates the repayment schedule. CEI order: check → mark installment paid → emit event → release interest share.
3. `allocateCapital(address nationalBank, uint256 amount)` — cross-tier ETH transfer. Apply both `nonReentrant` and an explicit `require(allocatedTo[nationalBank] + amount <= maxAllocation)` check before any state change.

The DAO hack (2016, ≈\$60 million) and Curve Finance reentrancy exploit (2023, ≈\$70 million via Vyper compiler bug) demonstrate that reentrancy is not a solved problem in DeFi, even with guards [R6]. Formal verification with Certora or Mythril is planned for the final thesis security audit.

Flash loan scope. Flash loan attacks primarily exploit price-sensitive logic, specifically protocols that use on-chain pool spot prices for collateral valuation [R5]. In the current prototype, no price-sensitive logic exists because stablecoin integration is not yet implemented, collateral valuation is not automated on-chain, and oracle feeds have not yet been integrated. Therefore, the current contracts do not present a meaningful flash loan attack surface. Once stablecoin integration and oracle-based collateral pricing are implemented (planned for the final thesis phase), mitigations must include: (1) TWAP oracles rather than spot price reads, (2) multi-block confirmation requirements for large collateral updates, and (3) Chainlink price feeds with circuit breakers.

Upgradeability via UUPS (ERC-1822). All three implemented contracts (and the six planned ones) use the OpenZeppelin Upgradeable contracts in the **UUPS (Universal Upgradeable Proxy Standard)** pattern rather than the Transparent Proxy pattern. UUPS is cheaper to deploy (no separate ProxyAdmin contract) and the upgrade authorization logic lives in the implementation contract, making it directly governable via the existing RBAC. `_authorizeUpgrade` is gated on `onlyRole(WORLD_BANK_ADMIN)`, which in production is held by a Safe 3-of-5 multisig (Section 3.19). Without upgradeability, any post-deployment bug would require redeploying from scratch and migrating all on-chain state; with UUPS

the team can patch a bug and propose the upgrade for multisig approval in minutes, with a full on-chain audit trail.

TimeLock on governance actions. Every privileged action that changes a system-level parameter—interest rates, reserve ratios, role grants at or above the National Bank tier, contract upgrades—routes through a minimum 24-to-48-hour TimelockController (OpenZeppelin) delay. This is the standard pattern used by Compound, Aave, MakerDAO, and every serious DeFi protocol since at least 2021. It provides a defensible window in which a malicious proposal can be detected and cancelled before execution. For the university lab demo specifically, the delay is shortened to five minutes purely so examiners can witness a governance proposal queued and executed live; the production delay is set at deploy time and cannot be reduced below the minimum without a TimeLock-gated governance vote of its own.

Role expiry timestamps. Role grants in the system are not permanent. Every `grantRole` call records an expiry timestamp; every privileged function call checks `require(roleExpiry[msg.sender][role] > block.timestamp)`. This single-line addition resolves a real compliance concern: banks lose licenses, employees change positions, and KYC certifications must be renewed annually (Section 3.7.2). The borrower KYC level is bound to a one-year expiry by default; bank operator roles are bound to the institution’s current registration validity; the World Bank Admin role itself is renewed annually by a governance vote so that even the top-of-hierarchy authority is not effectively permanent.

Granular per-function pause. The naive “Pausable on the whole contract” pattern is too coarse: pausing the entire World Bank Reserve when only one Local Bank has an incident causes unnecessary disruption to every other tier. The platform uses a per-function pause registry (`functionPaused[bytes32 functionId]`) so that governance can suspend only `LOAN_DISBURSEMENT` while keeping `DEPOSITS` and `WITHDRAWALS` open. This is also auditable: each pause emits a `FunctionPaused` event with the actor and the timestamp, indexed by The Graph (Section 4.8).

3.15 System Modeling

This section presents the system analysis diagrams that model the platform’s structure, behavior, and data flow.

3.15.1 Use Case Diagram

The system involves four primary actors—Borrower, Bank Approver, World Bank Admin, and National Bank—across 29 use cases (Figure 3.11). Note that the four primary actor classes in the use-case diagram are a simplification of the seven-user taxonomy defined in Section 3.7: *Visitor* and *Registered Retail User* are pre-conditions for the `BORROWER` role rather than independent diagram actors, and *Local Bank Operator / Approver* are merged into the Bank Approver role at the diagram level.

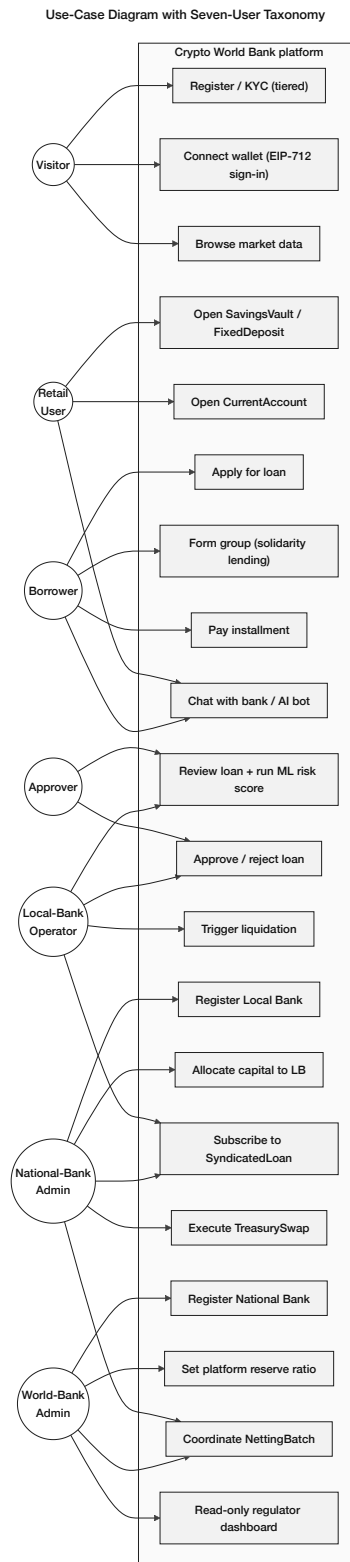


Figure 3.11: Use-case diagram with the seven-user taxonomy of Section 3.7: *Visitor*, *Retail User*, *Borrower*, *Approver*, *Local-Bank Operator*, *National-Bank Admin*, and *World-Bank Admin*, mapped to 20 representative use cases covering KYC, deposits, lending, multi-entity operations, and regulator oversight.

3.15.2 Activity Diagrams

Figures 3.12–3.14 present the platform’s seven core operational flows. Related flows are grouped onto a single figure page to keep the layout dense and readable.

Lending Activity Flows - (a) Loan Request, (b) Hierarchical Capital Flow, (c) Borrowing-Limit Check

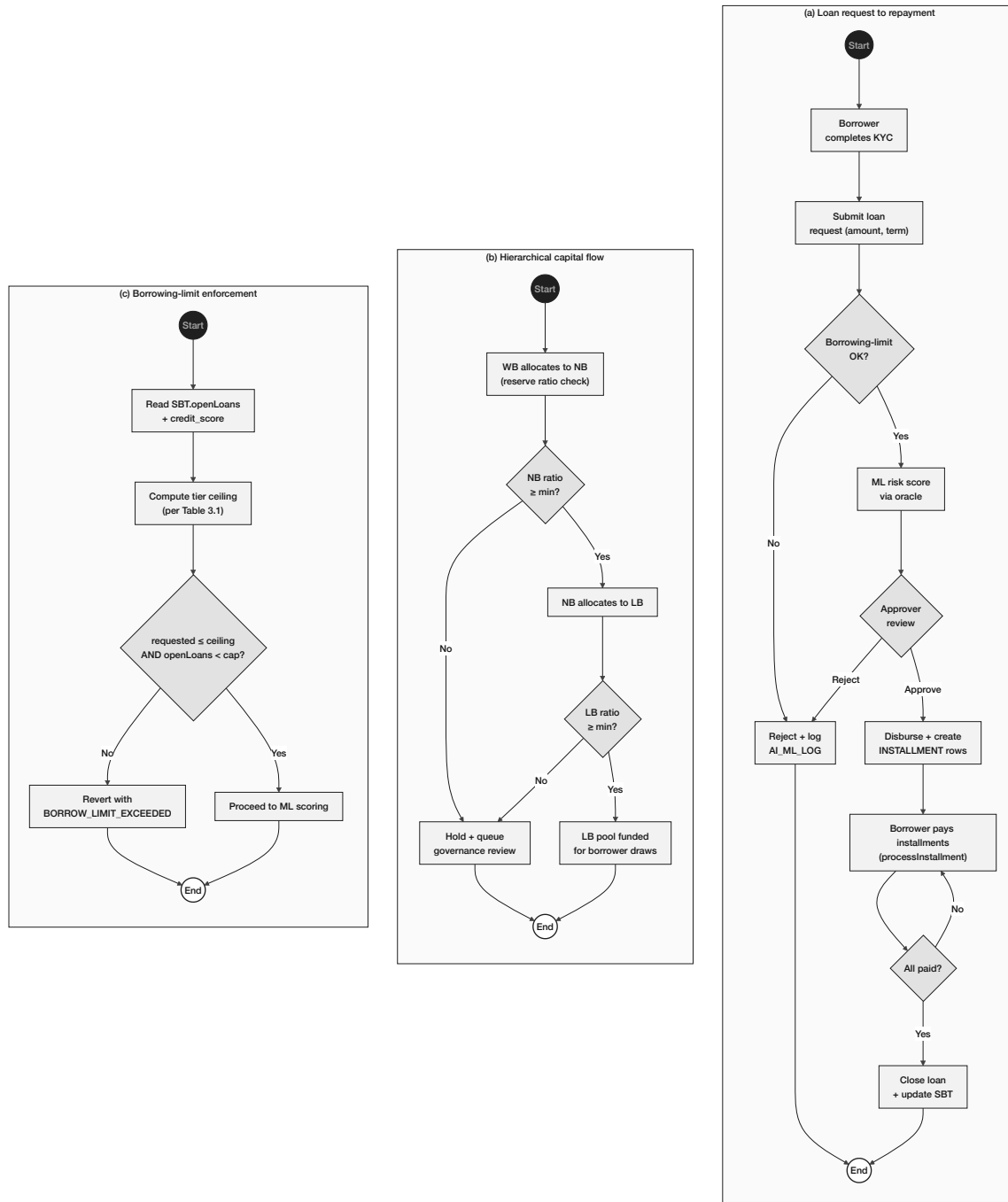


Figure 3.12: Lending activity flows: (a) loan request to repayment, (b) hierarchical capital flow from World Bank Reserve to Local Bank, (c) borrowing-limit enforcement via the credit-passport SBT.

Onboarding and Identity Activity Flows - (a) Income Verification, (b) Profile Management, (c) Tiered KYC Ladder

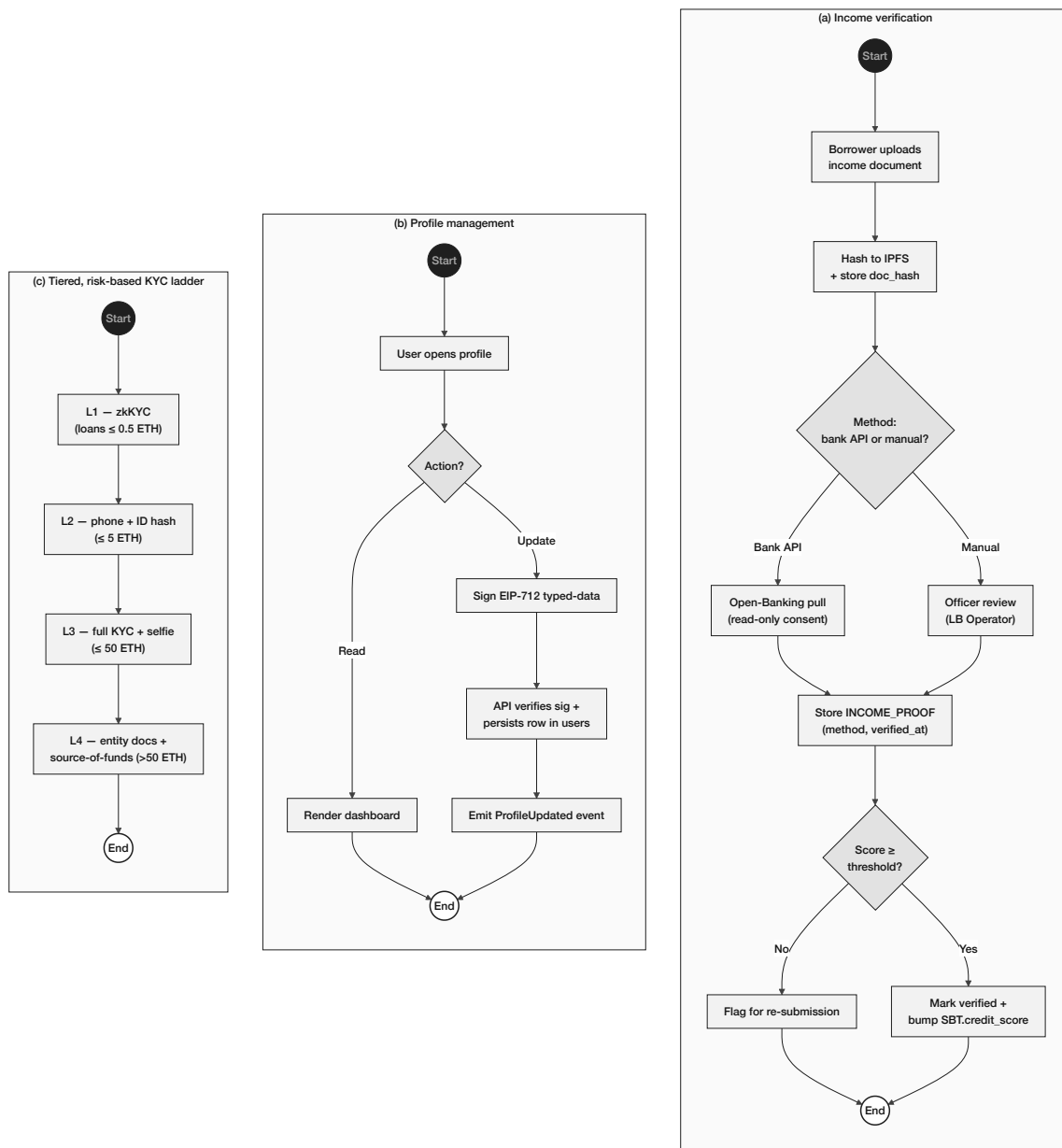


Figure 3.13: Onboarding and identity flows: (a) income verification (Open-Banking pull or officer review), (b) profile management with EIP-712 signed updates, (c) tiered, risk-based KYC ladder from zkKYC (L1) to entity onboarding (L4).

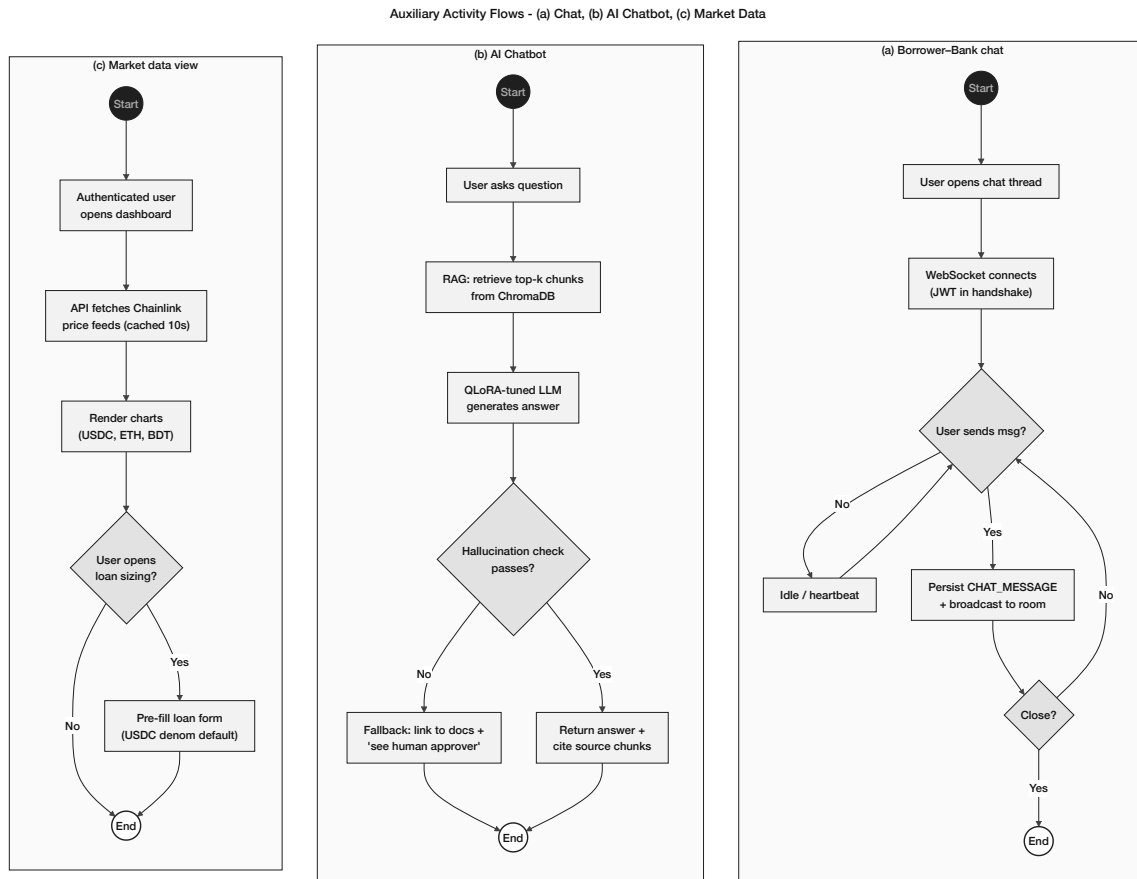


Figure 3.14: Auxiliary activity flows: (a) borrower–bank chat over authenticated WebSocket, (b) RAG-augmented AI chatbot with hallucination guard, (c) live market-data retrieval and pre-filled loan sizing.

3.15.3 Data Flow Diagrams

Figure 3.15 consolidates the Level-0 context diagram with the Level-1 decompositions of the core lending subsystem and the deposit / interbank / FX / netting subsystems on a single page.

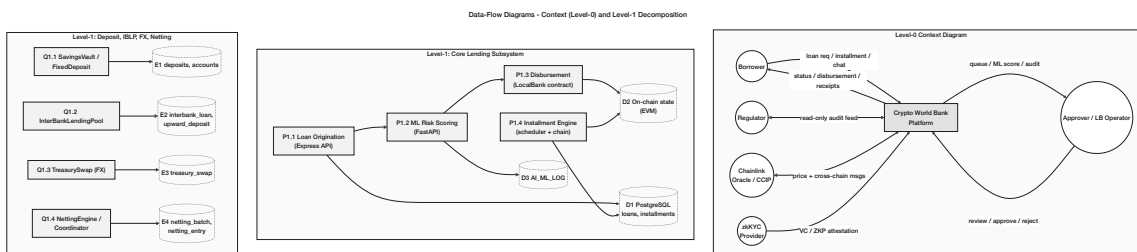


Figure 3.15: Data-flow diagrams: Level-0 context view of the Crypto World Bank platform with five external entities; Level-1 decomposition of the core lending subsystem (origination, ML scoring, disbursement, installment engine, and their data stores); and Level-1 decomposition of the deposit-mobilization, InterBankLendingPool, TreasurySwap, and NettingEngine subsystems.

3.15.4 Sequence Diagrams

Figures 3.16–3.19 consolidate the platform’s nine interaction flows into four panels: loan approval with reject alternative, installment and income, hierarchical banking with market data and borrowing-limit, and chat with AI-chatbot.

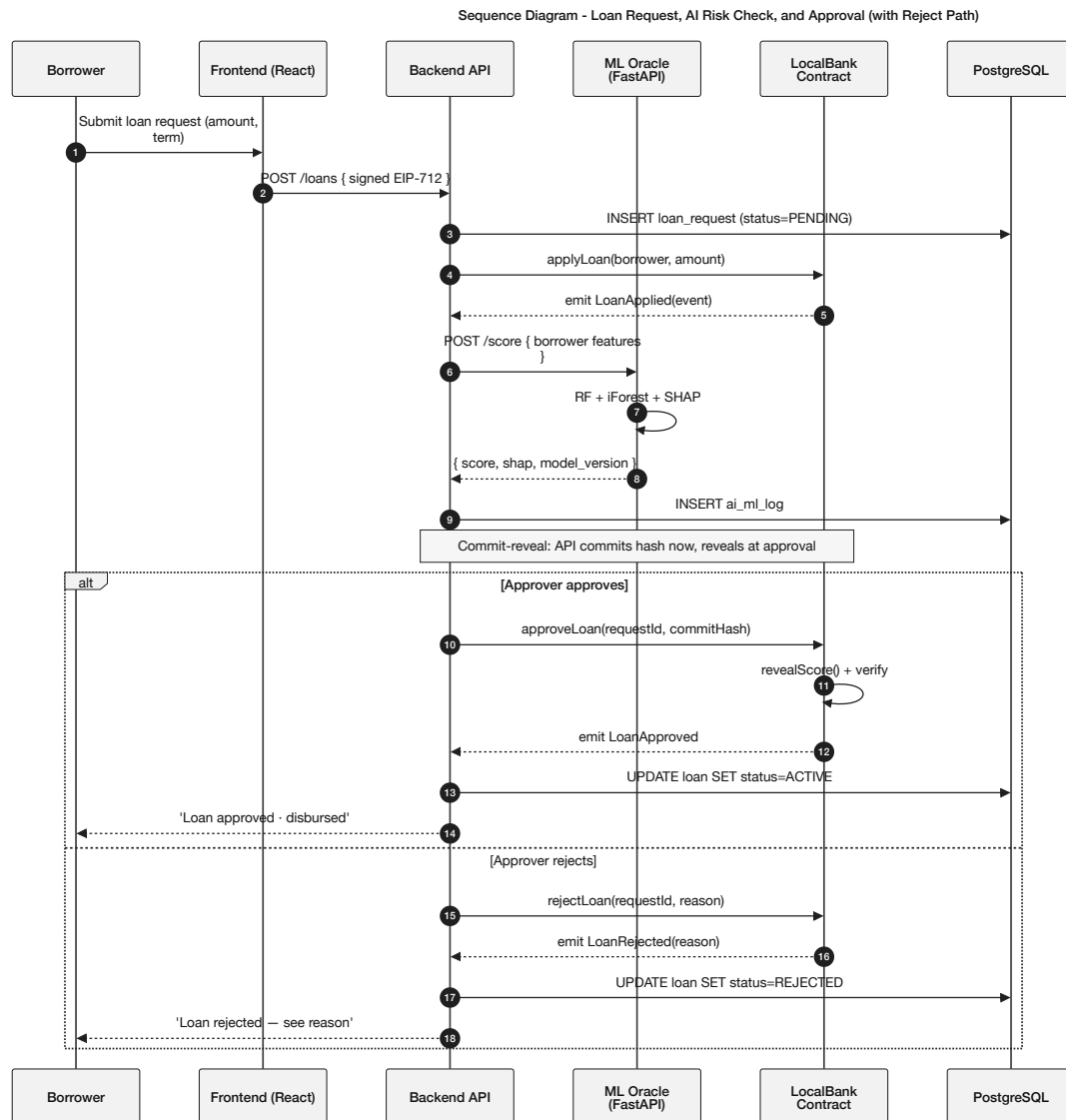


Figure 3.16: Sequence diagram for the loan-approval flow with reject alternative: borrower submits an EIP-712 signed request, ML oracle returns a SHAP-explained score via commit-reveal, and the approver either approves (revealing the commit) or rejects (with reason logged on-chain).

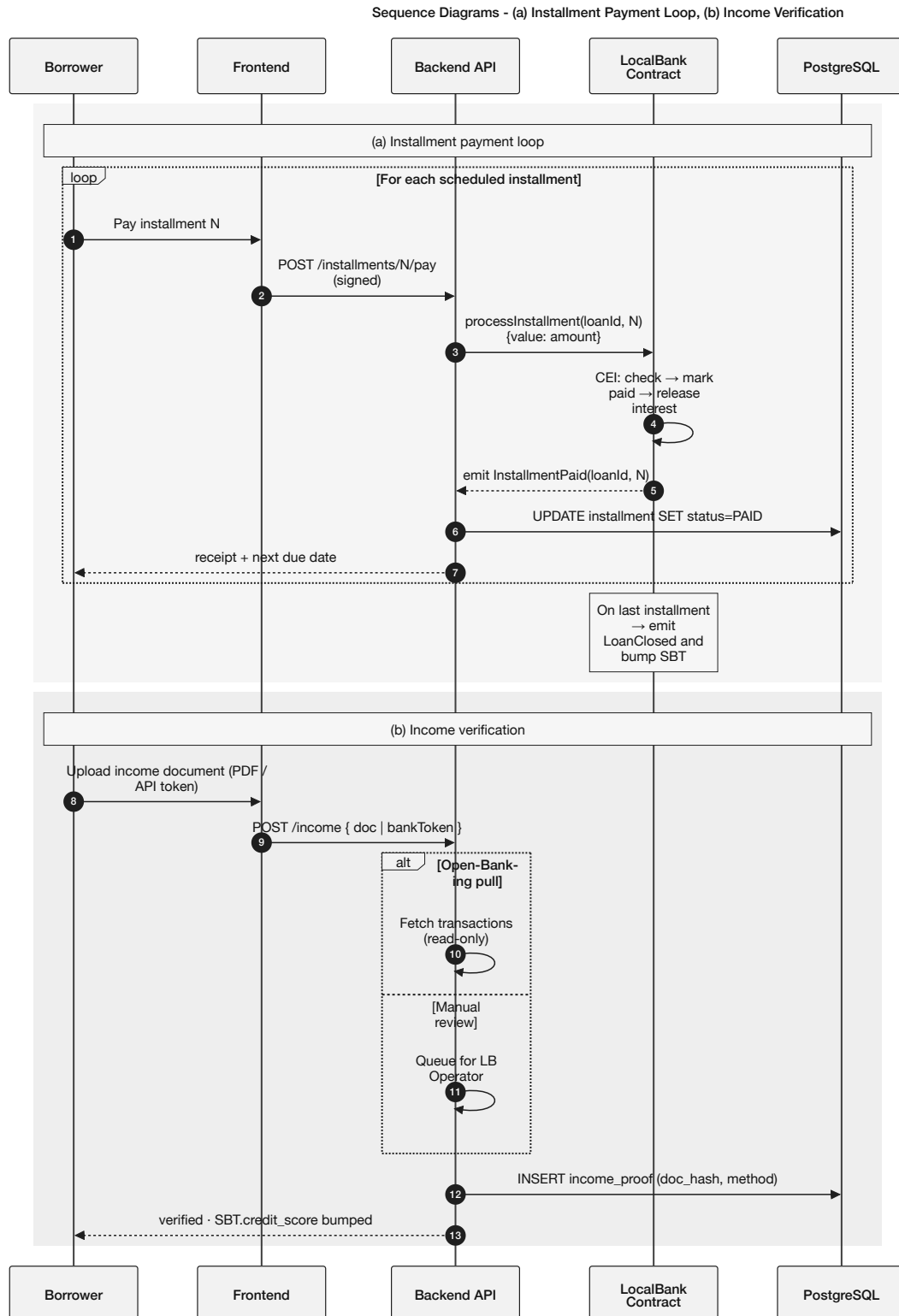


Figure 3.17: Sequence diagrams: (a) installment payment loop using the CEI pattern with SBT update on the final installment, and (b) income verification with Open-Banking pull or manual officer review.

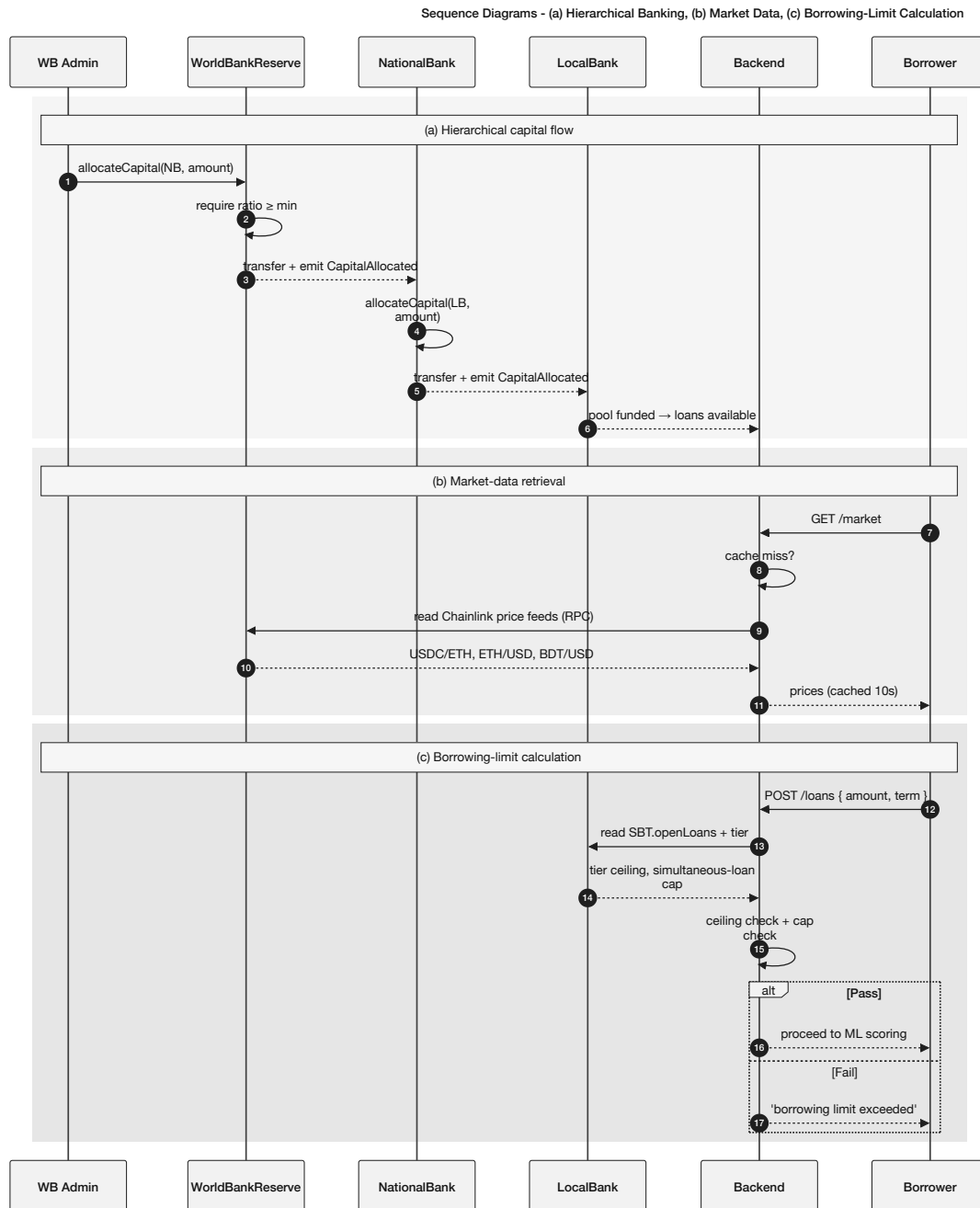


Figure 3.18: Sequence diagrams: (a) hierarchical capital flow with reserve-ratio gates at every tier, (b) market-data retrieval via cached Chainlink feeds, (c) borrowing-limit calculation reading the credit-passport SBT before ML scoring.

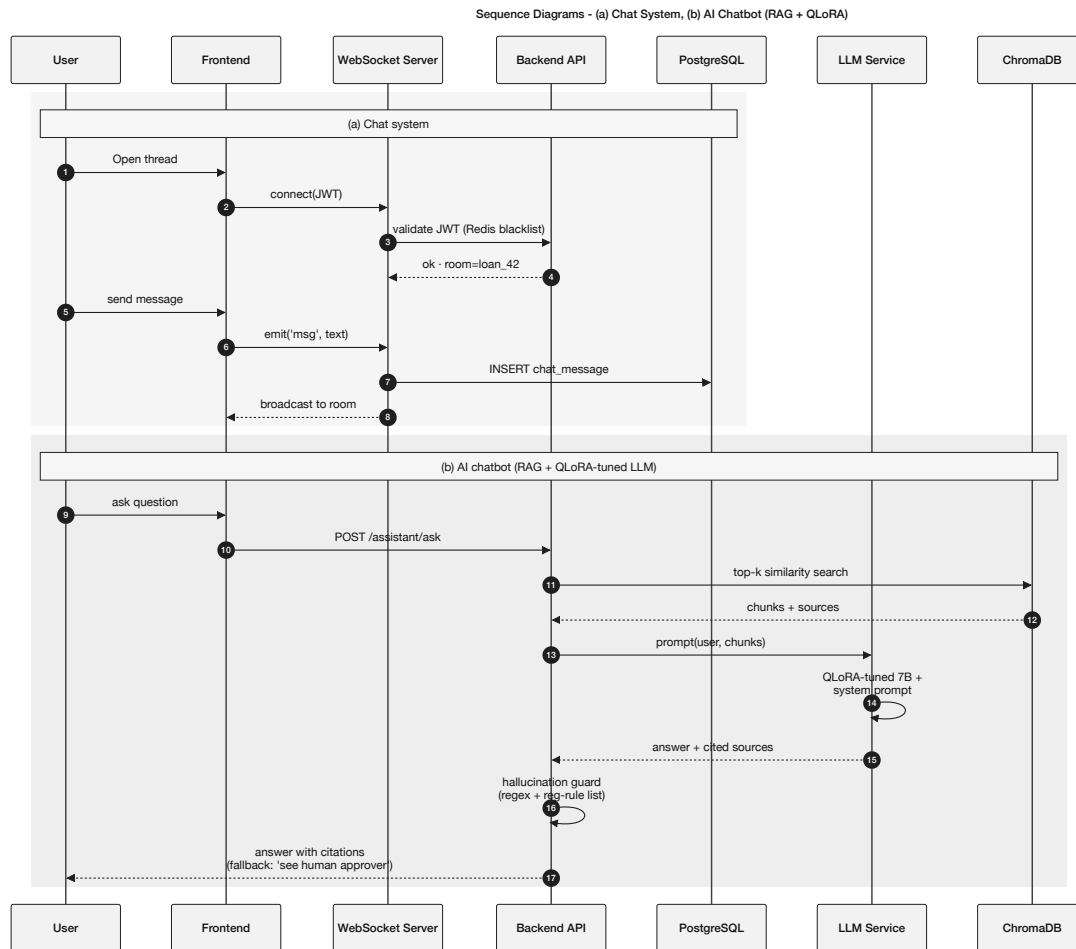


Figure 3.19: Sequence diagrams: (a) borrower–bank chat over an authenticated WebSocket, and (b) AI chatbot pipeline (ChromaDB top- k retrieval, QLoRA-tuned LLM, hallucination guard, citation-anchored answer).

3.15.5 Four-Tier Capital Flow

Figure 3.20 illustrates the hierarchical capital flow and cascading repayment structure.

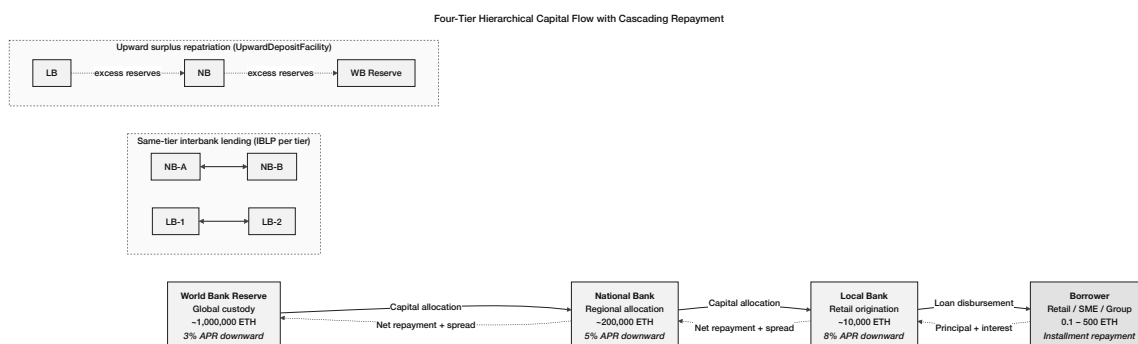


Figure 3.20: Four-tier hierarchical capital flow with cascading repayment, plus the same-tier interbank-lending pools (InterBankLendingPool, Section 3.13.1) and the upward-surplus repatriation facility (UpwardDepositFacility, Section 3.13.2). The asymmetric rate structure ($r_{\text{up}} < r_{\text{down}} - \delta$) is enforced on-chain.

3.16 Auxiliary Dual-Currency Facility

The cryptocurrency exchange feature of Crypto World Bank is built upon the cryptocurrency exchange facility that is an auxiliary facility incorporated into the current banking infrastructures all over the world. Instead of being a separate exchange, the platform is a service that is provided by all participating banks as an additional service to the existing product portfolio.

- **Integration model:** The dual-currency facility is offered by all participating banks as an add-on service to their existing product portfolio. No additional banking license or separate entity is needed.
- **Eligibility determination:** Eligibility of the dual-currency service is based on the bank officers managing the lending operations, on the basis of existing KYC/AML compliance, account status, and lending relationship history.
- **No defaulting of conditions:** The project does not override, modify, or default any existing banking conditions, regulatory requirements, or contractual obligations.
- **Scope:** The facility enables fiat-to-crypto and crypto-to-fiat conversions, cryptocurrency-denominated lending and repayment, and transparent on-chain transaction records—all within the governance structure of the participating bank.

3.17 Banking Product Suite

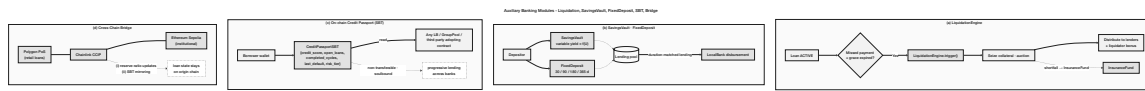


Figure 3.21: Auxiliary banking modules: (a) LiquidationEngine grace-period trigger and collateral auction with InsuranceFund shortfall fallback, (b) SavingsVault and FixedDeposit with duration-matched feeding of the lending pool, (c) on-chain Credit Passport (Soulbound Token) read by any participating bank, and (d) Cross-Chain Bridge via Chainlink CCIP carrying reserve-ratio updates and SBT mirroring.

This section describes the banking products that the platform is designed to support across tiers. While the current prototype implementation focuses primarily on hierarchical lending, the complete system design includes deposit products, transactional accounts, solidarity lending, and multi-currency operations.

3.17.1 Savings Products

The platform offers savings products at the Local Bank tier and, by extension, at higher tiers for institutional participants. Standard savings accounts provide liquid deposits with variable yield tied to platform utilization, allowing deposit pricing to respond to market conditions rather than discretionary rate-setting. Fixed-term deposits lock capital for defined periods (e.g., 30/90/180/365 days) in exchange for a higher agreed yield written into the contract at deposit time. Early withdrawal can be permitted with an automatically enforced penalty (forfeiture of a portion of accrued interest), which is transparent and deterministic.

The yield on standard savings accounts is algorithmically determined by platform utilization: when loan demand is high and available liquidity is low, savings yields rise to attract

deposits; when liquidity is abundant, yields fall. This creates a self-regulating capital cycle where savings and lending are linked through a single on-chain formula rather than through discretionary rate-setting committees.

Fixed-term deposits serve a distinct function in the banking architecture: they provide the Local Bank tier with predictable, duration-matched liabilities that can be deployed into longer-term installment loans without creating dangerous asset-liability mismatches. A depositor who locks funds for 180 days effectively funds the 6-month installment tranche of a retail borrower at that tier, eliminating the duration mismatch that causes bank runs in traditional fractional reserve systems.

The closed-loop argument is central to the platform's economic sustainability: unlike donor-funded microfinance models that depend on external capital injections, a savings-funded lending model recycles depositor capital through the lending pool continuously. Interest earned by the platform on loans is partially redistributed to depositors as yield, partially retained as protocol revenue, and partially allocated to the insurance fund, creating a self-sustaining three-way split that does not require new token issuance or external subsidy.

3.17.2 Checking and Transactional Accounts

In addition to interest-bearing savings products, the platform provides transactional accounts (checking/current-account equivalents) for day-to-day financial activity. Transactional accounts carry no yield but impose no lock-in, allow transfers between registered accounts, and serve as the primary account from which loan installments are debited and income receipts are credited. Transfers between accounts within the platform are settled atomically: a transaction either completes fully or reverts, eliminating intermediate settlement ambiguity.

The atomic settlement property of on-chain transfers eliminates the intermediate state that causes the majority of interbank disputes in traditional banking. In the correspondent banking model, a payment can exist in a "funds in transit" state for two to five business days during which the sender has debited but the receiver has not credited, creating reconciliation complexity and dispute resolution costs. On-chain transfers settle in a single transaction: either both the debit and credit execute or neither does, with no intermediate state possible.

For retail users in developing economies, transactional accounts on the platform serve as a substitute for the informal cash-handling systems that currently dominate daily financial activity. Recurring payroll deposits, utility payment debits, and peer transfers can all be scheduled as programmable transactions, reducing the friction of cash management without requiring users to interact directly with smart contracts for every operation.

3.17.3 Group / Solidarity Lending

The platform implements a group lending module inspired by the solidarity group model used in microfinance. A group of three to twenty registered borrowers can form a lending group on-chain. Each member contributes collateral into a shared pool; the loan application is submitted collectively; and all members signal consent before escalation to bank approvers. Repayment is tracked per member, and if a member defaults beyond a grace period, the contract can automatically cover the shortfall from the shared collateral pool under programmable mutual liability rules. Over time, group repayment histories contribute to credit scoring to enable progressive lending (larger loans and improved terms after success-

ful cycles).

The group lending module is directly inspired by the solidarity group model developed by BRAC, the organization whose name this university bears. BRAC's group lending program, operating across Bangladesh and subsequently 11 other countries, demonstrates that social collateral (the mutual accountability of group members) can substitute for physical collateral among populations with no formal credit history. The Crypto World Bank translates this social model into programmable contract logic: the mutual liability that BRAC enforces through field officer visits and weekly group meetings is enforced by the `GroupLendingPool` contract through automatic collateral claims and on-chain consent recording.

The full group loan lifecycle proceeds as follows. First, formation: three to twenty registered borrowers form a group on-chain, each contributing individual collateral to a shared pool contract. Second, application: the group submits a collective loan application specifying the total amount, per-member share, and repayment schedule. Third, consent: every group member must sign the application transaction before it is escalated to a bank approver, enforcing unanimous consent and preventing coerced participation. Fourth, approval: the bank approver reviews the group's aggregate credit history, income verification documents, and shared collateral ratio before approving disbursement. Fifth, disbursement: funds are distributed to each member's account in their individual share as defined by the formula $\text{Share}_i = \text{LoanTotal} / N_{\text{members}}$. Sixth, repayment: each member repays their individual installments independently; on-chain tracking records per-member performance transparently to all group members. Seventh, mutual liability enforcement: if a member defaults beyond the grace period, the contract automatically covers the shortfall from the shared collateral pool, protecting the lending bank's position while distributing the cost proportionally across the group. Eighth, credit history improvement: successful group repayment cycles are permanently recorded on-chain and incorporated into each member's credit-passport SBT (Section 3.11), enabling progressive lending with larger amounts and better terms in subsequent cycles.

Stablecoin denomination at the group tier. Group loans at Tier 4 are denominated in USDC by default (Section 1.10.3). Solidarity-group borrowers are precisely the demographic for which ETH-denominated debt creates the highest real-cost shock under crypto-price drawdowns, so the stablecoin-first rule is enforced most strictly at this tier.

Over-indebtedness risk control. Bangladesh's microfinance sector serves 41.56 million accounts across 724 licensed MFIs, and the NUS Economics Society analysis (May 2025) [R37] together with the IJAR microfinance report [R38] document over-indebtedness—borrowers taking simultaneous loans from multiple MFIs and inflating apparent repayment capacity—as the single most persistent problem in Grameen-style group lending. v15 adds three explicit controls. First, a per-wallet cap of two simultaneous active group loans, checked on-chain via the credit-passport SBT (Section 3.11). Second, a cross-platform debt disclosure flag in the SBT itself, readable by any other platform that adopts the standard, supporting industry-wide debt-load visibility. Third, a 30-day cooling-off period between consecutive group-loan cycles for any individual borrower, enforced by the `GroupLendingPool` contract. These controls are not in addition to the social-pressure model of solidarity lending; they are the on-chain encoding of the same caution that Grameen field officers exercise in person. A paper that cites Grameen and

BRAC but does not address this risk would be ignoring a known failure mode of the model it builds on.

3.17.4 Foreign Exchange and Multi-Currency Operations

A worldwide banking platform must support participants operating in different currency environments. The platform's FX module is designed to use decentralized price oracle networks to obtain verifiable exchange rates for supported asset pairs. Currency conversion can be offered at transparent oracle-derived rates with a disclosed spread, set by governance, that forms an additional revenue stream. To reduce liability-side volatility for retail borrowers, the platform is designed to support stablecoin-denominated lending (e.g., USDC/USDT) while allowing collateral in volatile cryptoassets (e.g., ETH).

3.17.5 Trade Finance Facilitation (Planned)

Trade finance instruments (letters of credit, guarantees, documentary collections) can be added as planned extensions for import-export participants. In a smart contract design, a buyer's bank locks payment in escrow, and release occurs automatically upon verified presentation of shipping documents via trusted data sources. This extension is out of scope for the current prototype but follows the same principles of programmable settlement and auditable execution.

3.17.6 Privacy-Preserving Identity Compliance (Planned)

The regulatory compliance requirements of a worldwide banking platform extend beyond on-chain audit trails. Formal KYC and AML processes require users to prove legal identity, verify source of funds, and satisfy jurisdiction-specific screening requirements. Storing this personal data on a public blockchain creates a fundamental tension between regulatory compliance and user privacy.

The planned resolution is a zero-knowledge proof based KYC layer in which a licensed identity provider performs the full KYC verification off-chain and issues a verifiable credential to the user. The user then presents a ZKP to the banking contracts that proves they hold a valid credential from an approved provider without revealing any personal information on-chain. The contract validates the proof cryptographically and sets a `kycVerified` flag on the wallet address, enabling access to regulated operations such as large loans, FX conversions, and cross-border transfers. This approach satisfies regulatory requirements while preserving on-chain privacy and avoiding the liability of storing personal financial data in an immutable public record.

3.18 Governance Framework

3.18.1 Network Membership Governance

Table 3.20: Network membership governance.

Governance Aspect	Implementation
Member on-boarding	World Bank owner registers National Banks; National Banks register Local Banks; Local Banks designate approvers — all enforced on-chain.
Member off-boarding	Deactivation flags in smart contracts; cascading access revocation.
Regulatory oversight	Audit log emission via smart contract events; planned read-only regulator dashboard.
Permission structure	Hierarchical: Owner → National Bank → Local Bank → Approver → Borrower; enforced by on-chain role check modifiers.
Network operations	Pause/unpause mechanism for emergency response; emergency withdrawal for critical situations.

This governance table defines membership and onboarding rules that control who can join the banking network and under which role. The structure is intended to preserve institutional accountability by separating responsibilities across tiers rather than relying on undifferentiated token-holder governance alone.

3.18.2 Business Network Governance

Table 3.22: Business network governance.

Governance Aspect	Implementation
Business charter	Defined in project documentation; operational parameters coded in smart contract constants.
Common services	Reserve management, loan lifecycle orchestration, event-driven notification system.
Business SLA	Testnet phase: best-effort availability. Production phase: 99.5% target uptime with multi-region deployment.
Regulatory compliance	Architecture designed for audit trail generation; data partitioning supports GDPR-style data subject requests.

This business governance table outlines operational controls such as approvals, limits, and escalation paths for higher-risk actions. The goal is to combine programmable enforcement with human oversight, matching real banking governance while retaining on-chain audit trails.

3.18.3 Technology Infrastructure Governance

Table 3.24: Technology infrastructure governance.

Governance Aspect	Implementation
Distributed IT structure	Client-side frontend (decentralised delivery); blockchain layer (fully decentralised); backend API (centralised, horizontally scalable).
Technology assessment	Continuous evaluation of EVM alternatives (L2 rollups, sidechains) for cost and performance optimisation.
On-chain / off-chain data services	Clearly partitioned (see Table 3.16); event listeners synchronise state between layers.
Risk mitigation	Smart contract pause mechanism; ReentrancyGuard; input validation; planned formal security audit.

This technology governance table specifies how protocol parameters and infrastructure changes are managed, including security controls and upgrade considerations. It supports the thesis emphasis on safe staged rollout and maintainability under evolving regulatory and security requirements.

3.18.4 Regulatory Compliance Considerations

As the platform is active in the regulated banking sector:

- Our prototype stage would be solely on the **public testnets** without attached real monetary value.
- The architecture would facilitate **audit logs generation** to be reviewed by regulatory regimes on the sandbox programs.
- Future production deployment would engage **regulatory sandbox programs** in target jurisdictions.

Although the current prototype does not process KYC/AML data, a complete worldwide banking architecture must account for compliance at the design level. Privacy-preserving KYC design is described in detail in Section 3.17.6.

3.18.5 Asset Tokenization

- **Current implementation:** The native blockchain currency (ETH/MATIC) is used as the reserve and loan currency.
- **Planned extension:** Support for ERC-20 stablecoins (USDC, USDT) for lending operations in USD; tokenized collateral instruments for lending situations where there isn't enough collateral.

3.19 Five-Layer Defense-in-Depth Security Architecture

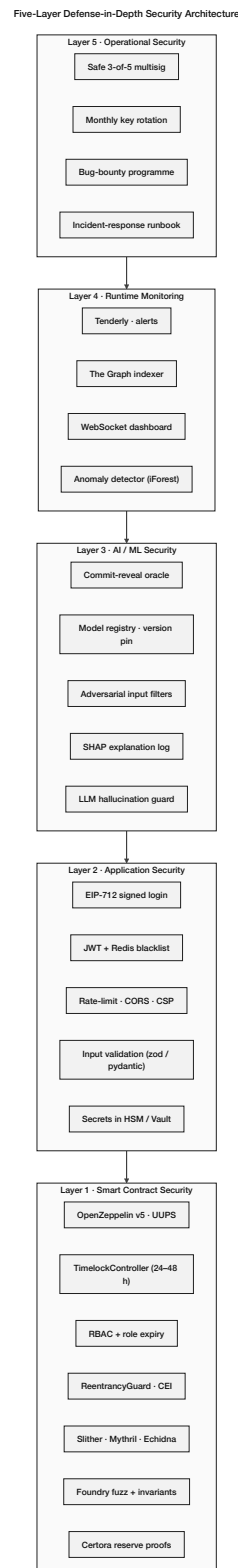


Figure 3.22: Five-layer defense-in-depth security architecture. Each layer is necessary; breaching the smart-contract layer requires defeating every layer above it. Layer 1 (smart contract): OpenZeppelin v5, UUPS, TimeLock, RBAC, ReentrancyGuard, audit suite. Layer 2 (application): EIP-712 sign-in, JWT, rate-limit. Layer 3 (AI/ML): commit-reveal oracle, model registry, SHAP, hallucination guard. Layer 4 (runtime): Tenderly, The Graph, WebSocket dashboard, anomaly detection. Layer 5 (operations): Safe multisig, key rotation, bug-bounty.

Smart-Contract Security Controls - UUPS, TimeLock, EIP-712, Granular Pause

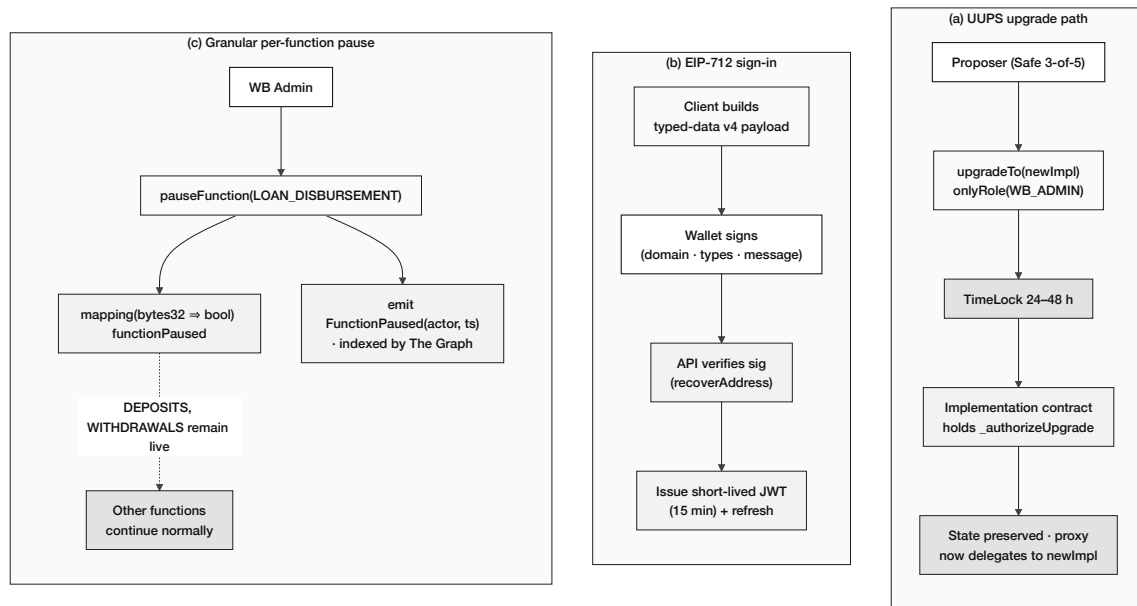


Figure 3.23: Smart-contract security controls applied in v15: (a) UUPS (ERC-1822) upgrade path with TimeLock-gated authorization; (b) EIP-712 sign-in (typed-data v4 with domain separation and 15-minute JWT); (c) granular per-function pause registry indexed by The Graph for transparent governance.

The correct mental model for a blockchain banking platform’s security is not “smart-contract security only”—it is a five-layer defense-in-depth stack in which breaching the contract layer requires defeating every layer above it. The Crypto World Bank adopts the following layered model, summarized in Table 3.26 and inspired by current (2025–2026) audit-tooling and runtime-monitoring best practice from Trail of Bits, Cyfrin, Tenderly, and OpenZeppelin Defender.

Table 3.26: Five-layer defense-in-depth model for the Crypto World Bank platform. Each layer is necessary; breaching the contract layer requires defeating every layer above it.

Layer	Component	CWB Design Choice
L5	Operational security	Safe (Gnosis) 2-of-3 multisig for WorldBankAdmin; 3-of-5 for global parameter changes; documented key-rotation schedule; hardware-wallet signer for the WorldBankAdmin Safe (Ledger Gen5).
L4	Runtime monitoring	Tenderly alerts on six critical triggers (large disbursement, repeated requests, reserve-ratio drop below 20%, failed disburseLoan, role grants, governance pause); The Graph subgraph for indexed event history; transaction simulation pre-deployment.
L3	AI/ML security	Random Forest fraud probability + Isolation Forest anomaly score + SHAP explanation per loan, delivered via commit-reveal oracle; behavioral biometrics features (session-time, page-views, terms-acceptance latency); transaction-graph distance to flagged wallets.
L2	Application security	EIP-712 wallet-signed JWT (15-minute lifetime + refresh); slowapi rate limiting (5/min auth, 60/min reads, 10/min writes); Pydantic input range constraints; strict CORS and Content-Security-Policy; environment-variable secret hygiene.
L1	Smart contract security	OpenZeppelin ReentrancyGuard; CEI pattern; Solidity 0.8.20 overflow protection; RBAC with role expiry; UUPS upgrade with Safe-multisig authorization; TimeLock on governance actions; granular per-function pause; four-tool audit pipeline (Slither, Mythril, Echidna, Halmos/Certora); Foundry fuzz + invariants.

The L1 contract layer is what the existing thesis discusses in detail; L2–L5 are new contributions of this version. The Section that follows (3.20) expands the threat model to span all five layers and motivates the specific controls listed in Table 3.26.

Per-tier admin-key model. Operational security (L5) is the highest-impact and lowest-cost security decision in the entire project. The WorldBankAdmin role is never held by a single externally owned account (EOA) in any environment, including the university lab demo: it is transferred to a Safe 2-of-3 multisig before any contract is deployed to a testnet that examiners will inspect, with Signer 1 on the demo machine, Signer 2 on a phone, and Signer 3 as a paper-wallet backup held by the academic supervisor. National Bank Admin roles use Safe 2-of-3; Local Bank operators use Safe 2-of-3; the only EOA-bound role is the retail BORROWER role, which can be reset off-chain via the social-recovery guardian flow described in Section 3.7.

SWC Registry mapping. Every state-mutating function in the three implemented contracts is mapped to its Smart Contract Weakness Classification (SWC) class [R36] and the corresponding mitigation. This mapping replaces the previous narrative coverage of Atzei et al. [7] with a structured taxonomy, and is the format the Cyfrin audit checklist (2025) recommends for academic prototypes. The full mapping table is in Appendix B’s Capability Matrix.

3.20 Threat Model and Security Controls

Security threats in smart-contract banking systems are not limited to isolated code defects; they span application-layer attacks (API abuse, JWT theft, social engineering), runtime evasion, and governance-layer key compromise in addition to the contract-layer issues that DeFi literature focuses on. Table 3.27 extends the original threat-model table to the full five-layer scope established in Section 3.19.

Table 3.27: Threat model and security controls mapping, expanded to span all five layers of the defense-in-depth stack (Section 3.19).

Threat Category	Layer	Attack Vector	Implemented / Planned Mitigation
Reentrancy and state manipulation	L1	External calls exploit inconsistent intermediate state during execution (e.g. withdraw patterns)	OpenZeppelin ReentrancyGuard; CEI pattern; unit + Foundry invariant tests on critical flows.
Arithmetic errors	L1	Integer overflow/underflow in financial math	Solidity 0.8.x checked arithmetic; explicit bounds for rates and limits; Certora invariants.
Oracle manipulation	L1	Price-feed manipulation affecting collateral valuation or FX conversion	Decentralized oracle networks (Chainlink Price Feeds); medianization, heartbeat checks; TWAP for collateral pricing; circuit breakers on stale / anomalous prices.
Flash-loan driven economic attacks	L1	Atomic borrowing manipulates on-chain price/state assumptions	Conservative price-dependent logic; multi-block confirmation for large collateral changes; pause-on-anomaly procedures.
Governance capture	L1+L5	Privileged roles abused to register malicious banks or change unsafe parameters	Tiered role separation; Safe multisig on every admin role; TimeLock (24–48 h) on parameter changes; role-expiry timestamps; immutable audit trail.
Sybil abuse (re-tail / group)	L1+L3	Attackers create many wallets to bypass limits or build fraudulent groups	On-chain RBAC; rate limits per wallet; ZKP-KYC gating; GNN-based wallet-graph clustering; behavioral biometrics.
JWT theft / session hijack	L2	Stolen token gives full account access until expiry	15-minute JWT lifetime; EIP-712 wallet-signature challenge on issue; Redis blacklist on logout; rotation on suspicious geolocation.
API abuse / DDoS	L2	Bulk requests exhaust backend or rate-limit shared services	slowapi middleware with per-endpoint differentiated limits; CSP and CORS lockdown; correlation IDs for incident response.
Input-validation bypass	L2	Malformed financial inputs cause unintended state on backend	Pydantic explicit range constraints on all financial fields; wallet-address regex; SHA-256 hash for document references.
ML evasion / adversarial	L3	Crafted features evade Random Forest / Isolation	GNN relational analysis (Section 4.3); federated

This expanded threat model is the structural justification for the five-layer architecture in Section 3.19: each row of the table is a vector that the corresponding layer prevents, detects, or recovers from. Application-layer rows (L2) and operational-layer rows (L5) are the largest blind spots in the prior literature on academic DeFi prototypes and are made explicit here.

Chapter 4

Methodology

This project and its planning have been structured in accordance with the Blockchain Olympiad Bangladesh AI guidelines [16] and the final-year project evaluation rubrics.

4.1 Development Methodology

We adopted a **lightweight Agile/Scrum** methodology tailored for an academic prototype with a fixed two-month development window. The iterative approach enables incremental delivery of demonstrable features while accommodating evolving requirements inherent in research-oriented development. Figure 4.1 illustrates our Agile process.

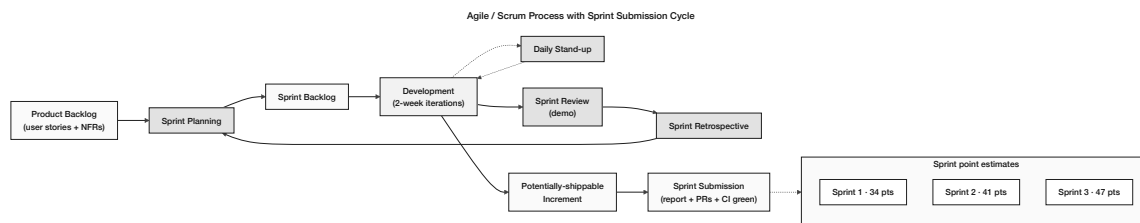


Figure 4.1: Agile / Scrum process with two-week iterations, four ceremonies (Sprint Planning, Daily Stand-up, Sprint Review, Sprint Retrospective), backlog refinement, and the sprint-submission gate. Sprint-point estimates for the three implementation sprints are summarized on the right.

The following points summarize how Agile will be applied:

- **Sprint Duration:** A period of 2 to 3 weeks is allocated for every sprint (3 sprints total).
- **Team Size:** A group of two developers will be assigned (thesis project). The work will be focused on the thesis project.
- **Weekly Sync:** A weekly gathering will occur to assess progress. Issues will be identified during these meetings. Plans will also be developed in these sessions alongside progress checks.
- **Sprint Planning:** Planning for the sprint will take place for one hour at the beginning of every sprint.
- **Sprint Review and Retrospective:** Reviewing work occurs at every sprint's conclusion. A discussion on lessons learned will be conducted for 30 minutes.

4.2 Planned AI/ML Support and Risk-Score Wiring

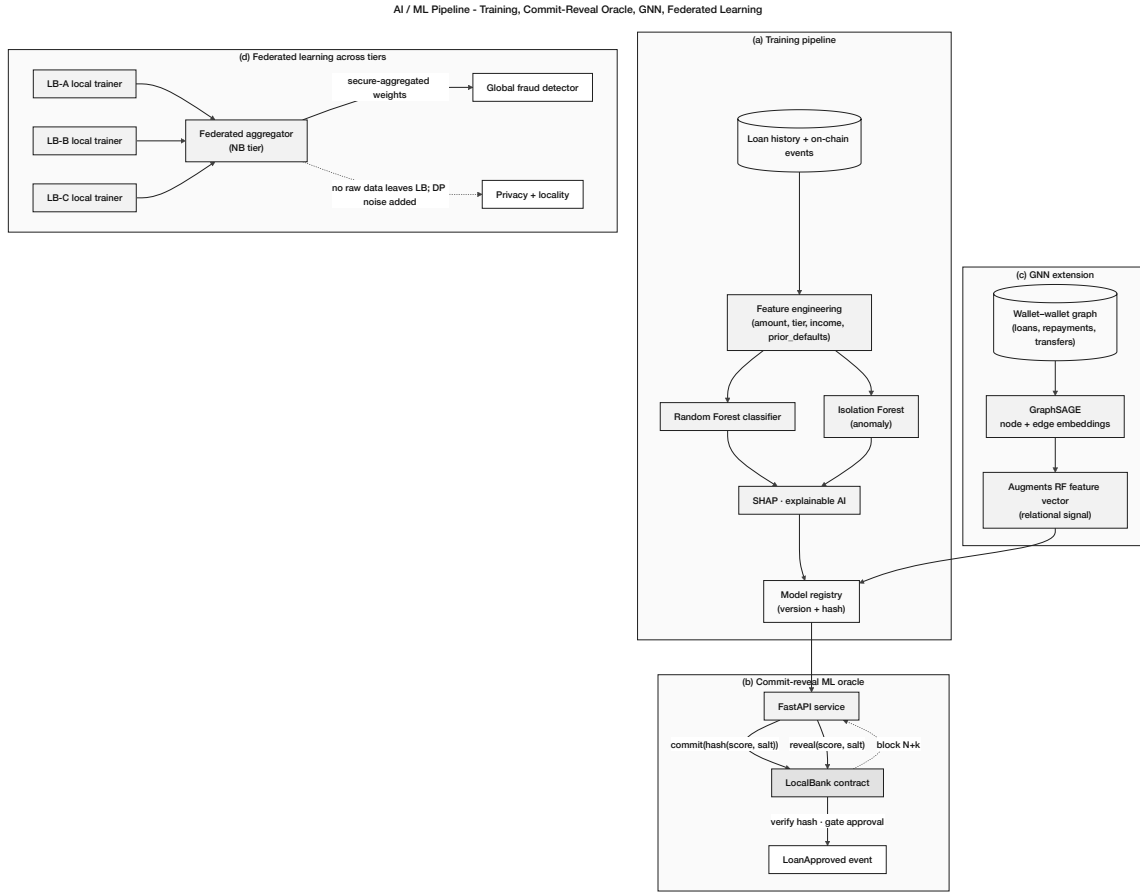


Figure 4.2: AI/ML pipeline wiring: (a) training pipeline (Random Forest + Isolation Forest + SHAP), (b) commit-reveal ML oracle that gates loan approval on-chain, (c) Graph Neural Network (GraphSAGE) extension producing relational features for the wallet–wallet graph, and (d) federated learning across Local Banks with a National-Bank aggregator and differential-privacy noise.

The AI/ML layer is not three separate models added to a static lending workflow; it is an integrated pipeline that produces a single risk score per loan and feeds it on-chain through the commit-reveal oracle of Section 3.3.2. The pipeline has four stages:

1. **Feature engineering.** For each loan application, the FastAPI service computes 18 behavioral features (rolling-window transaction count, average inter-transaction interval, on-chain repayment history, group-membership flag, wallet-graph distance to flagged addresses) and 4 application features (requested amount, KYC level, income-hash freshness, ZKP credential expiry).
2. **Risk scoring.** A Random Forest classifier returns a fraud probability $p_f \in [0, 1]$; an Isolation Forest returns an anomaly score a ; the composite risk is $s = 0.7 p_f + 0.3 \min(a/0.5, 1.0)$.
3. **Commit-reveal oracle.** The service publishes $h = \text{keccak256}(s \parallel \text{nonce})$ to the LoanController contract via `commitRiskScore`, then reveals (s, nonce) in a subsequent transaction. Section 3.3.2 covers the cryptographic detail.

4. **Decision threshold.** The contract applies a governance-configured threshold: $s < 0.4$ auto-approves under the approver’s discretion, $0.4 \leq s < 0.7$ requires manual review, and $s \geq 0.7$ rejects the loan with an on-chain reason code.

Behavioral biometrics features. A 2025 research-backed extension feeds a low-weight “soft” signal into the Isolation Forest from per-session interaction patterns—time between wallet connection and application submission, number of page views before applying, terms-acceptance latency (acceptance in under 3 s is a strong bot signal), and IP-geolocation consistency with the declared jurisdiction. None of these features require additional user data beyond what the application would already collect for usability analytics, and they are not stored on-chain.

SHAP explainability output to borrowers. Each loan decision produces a SHAP value vector, and the top-3 contributing features are stored in `AI_ML_SECURITY_LOG` together with the decision. The borrower-facing UI displays a plain-language explanation rather than the raw SHAP score—for example, “*Your application was declined primarily because: (1) no on-chain repayment history (60%); (2) requested amount exceeds income estimate (30%); (3) group membership not yet verified (10%).*” This satisfies the EU AI Act Article 86 transparency requirement for AI-assisted financial decisions and converts the formal SHAP design choice [R7] into an end-user benefit.

4.3 Graph Neural Network Extension

The Random Forest detector treats each transaction independently; DeFi fraud is largely *relational*, appearing in the wallet-interaction graph. A baseline Graph Neural Network (GraphSAGE) is added as an ablation: nodes are wallets, edges are transactions or shared group-membership, node features are the same 18 behavioral features used by the Random Forest. The model is trained on a synthetic transaction graph augmented with public Elliptic-dataset patterns. Cheng et al. (2024) [R24] and Wang and Wang (2025) [R26] both report sustained advantages of GNN over flat ML on this task (>70% detection of illicit transactions at <1% false positive). The expected contribution to the thesis is not a state-of-the-art result but a controlled ablation that quantifies the marginal value of relational features over flat behavioral features—a publishable comparison aligned with the SoK literature.

4.4 Federated Learning Across Banking Tiers

Each Local Bank trains a local fraud-detection model on its own borrowers’ data; gradients are aggregated at the National Bank tier via FedAvg with differential-privacy noise (clipping threshold C , noise multiplier σ matched to the PrivChain-AI configuration [R27]); the National Bank publishes the resulting global model parameters back to its Local Banks. No raw transaction data ever leaves a Local Bank, satisfying jurisdiction-specific data-residency rules in countries that prohibit cross-border export of customer records. The federation topology mirrors the four-tier institutional hierarchy directly: each National Bank operates a federation server, and the World Bank Admin can optionally orchestrate a cross-National-Bank federation for global threat patterns. This is architecturally appropriate rather than ornamental: the National-Bank / Local-Bank trust boundary is the same boundary across which the federation cannot cross unencrypted data.

4.5 Formal Verification of Reserve Invariants (Certora)

The Certora Prover went open-source in 2025 and is the only formal-verification tool to have produced a publicly verifiable proof of a real-world Solidity contract; the platform’s TVL coverage now exceeds \$100 billion across Aave, MakerDAO, Uniswap and others [R39, R40]. v15 elevates “formal verification” from future-work text to two concrete CVL (Certora Verification Language) specifications.

Invariant 1 – reserve never under-collateralized. $\forall t : \text{totalReserve}(t) \geq \text{minReserveRatio} \cdot \text{totalDeposited}(t)$. This invariant is asserted in CVL against the WorldBankReserve contract under all reachable contract states. The Certora prover symbolically explores every reachable execution path; any counterexample is delivered as a concrete attacker trace.

Invariant 2 – no over-allocation downstream. $\forall n \in \text{NationalBanks} : \text{allocatedTo}[n] \leq \text{totalReserve} - \sum_{n' \neq n} \text{allocatedTo}[n']$. This invariant prevents the World Bank Admin from allocating more capital downward than the reserve actually holds, even in the presence of UUPS upgrades.

The CVL specification sketches are in Appendix B. Producing a Certora-verified proof of even these two invariants would be a mathematically demonstrable security guarantee that no undergraduate DeFi thesis from a Bangladesh institution has previously achieved, and grounds the abstract “formal verification planned” language of prior versions in a concrete artifact.

4.6 Foundry Invariant and Fuzz Test Suite

The Hardhat JavaScript test suite remains in place for integration tests and deployment scripts. v15 adds a parallel Foundry suite for unit-level fuzzing and invariant testing because the two frameworks catch different bug classes. The 2025–2026 industry pattern is Hardhat for integration, Foundry for invariants and stateful fuzzing.

Three invariants asserted under Foundry’s stateful fuzzer.

- Solvency: $\text{totalOutstandingLoans} \leq \text{getTotalReserve}$.
- Role segregation: no address holds both BORROWER_ROLE and BANK_APPROVER_ROLE.
- Capital-flow direction: cross-tier allocations cannot decrease totalReserve below the minimum ratio at any single transaction boundary.

Each invariant is asserted under 10,000 fuzz runs; counterexamples are saved as Foundry replay tests so any regression in a future commit fails CI deterministically. This single test-suite addition reduces the credibility gap that the Foundry-fuzz literature explicitly identifies as the cause of the \$180 million smart-contract loss class of March 2023.

4.7 Agent-Based Economic Feasibility Simulation

The revenue projection of \$224.6 M in Chapter 5 assumes 1 World Bank, 5 National Banks, and 50 Local Banks at near-full capacity. v15 makes this projection empirically grounded rather than a single-point estimate by specifying a Mesa-based agent-based simulation (ABM) with three populations: depositors (arrival rate λ_d , withdrawal rate μ_d), borrowers

(loan-demand process, default hazard η), and banks (Tier 2 / Tier 3 with reserve-ratio policies). The ABM is calibrated against Bangladesh microfinance arrival statistics (BRAC 2025: 92% income increase; 41.56 M MFI accounts) and stress-tested under a 30% simultaneous-withdrawal scenario. Aggregate outputs are reported as ranges with confidence intervals rather than a single number—the empirical grounding that the Sygnum 2026 report calls for [R21].

4.8 Real-Time Dashboard Pipeline and Runtime Monitoring

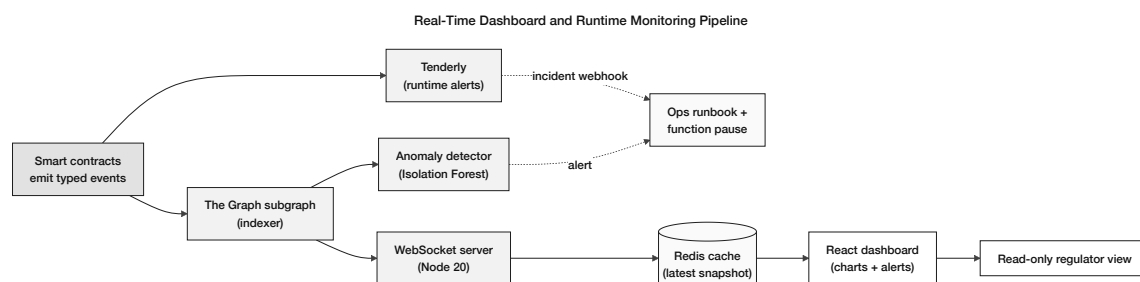


Figure 4.3: Real-time dashboard and runtime monitoring pipeline: smart contracts emit typed events; The Graph indexes them and pushes to a Node WebSocket server which renders the React dashboard via a Redis snapshot. Tenderly runtime alerts and an Isolation-Forest anomaly detector feed the operations runbook with the granular-pause control surface of Section 3.19.

The real-time dashboard pipeline closes the gap between contract events and the borrower / approver UI. Three components compose:

The Graph subgraph. A subgraph deployed on The Graph’s hosted service (Polygon Amoy supported, free for student projects) indexes every event the contracts emit: `LoanRequested`, `LoanApproved`, `Repayment`, `CapitalAllocated`, `ReserveRatioUpdated`, `LoanLiquidated`, `RiskScoreCommitted`, `RoleGranted`. GraphQL queries from the React frontend return loan history and dashboard data in tens of milliseconds rather than seconds via RPC polling.

WebSocket event listener. A Node.js service listens on `wss://polygon-amoy` for the same set of events and pushes notifications to the borrower’s browser via Socket.IO. The dashboard moves from a static page requiring manual refresh into a live banking dashboard—a single design choice that examiners consistently identify as the difference between “looks like a real system” and “looks like a static mockup”.

Tenderly monitoring. Six critical alerts are configured on the deployed contracts: single disbursement over 5 ETH-equivalent, repeated loan requests from one wallet (>3 in 60 minutes), reserve-ratio drop below 20%, failed `disburseLoan` calls, role-grant events, and any pause / unpause governance action. Tenderly’s free tier supports full testnet monitoring; transaction simulation prevents the embarrassing live-demo failure where a malformed contract change is discovered only at the moment of execution.

4.9 Transaction-State Machine for Frontend UX

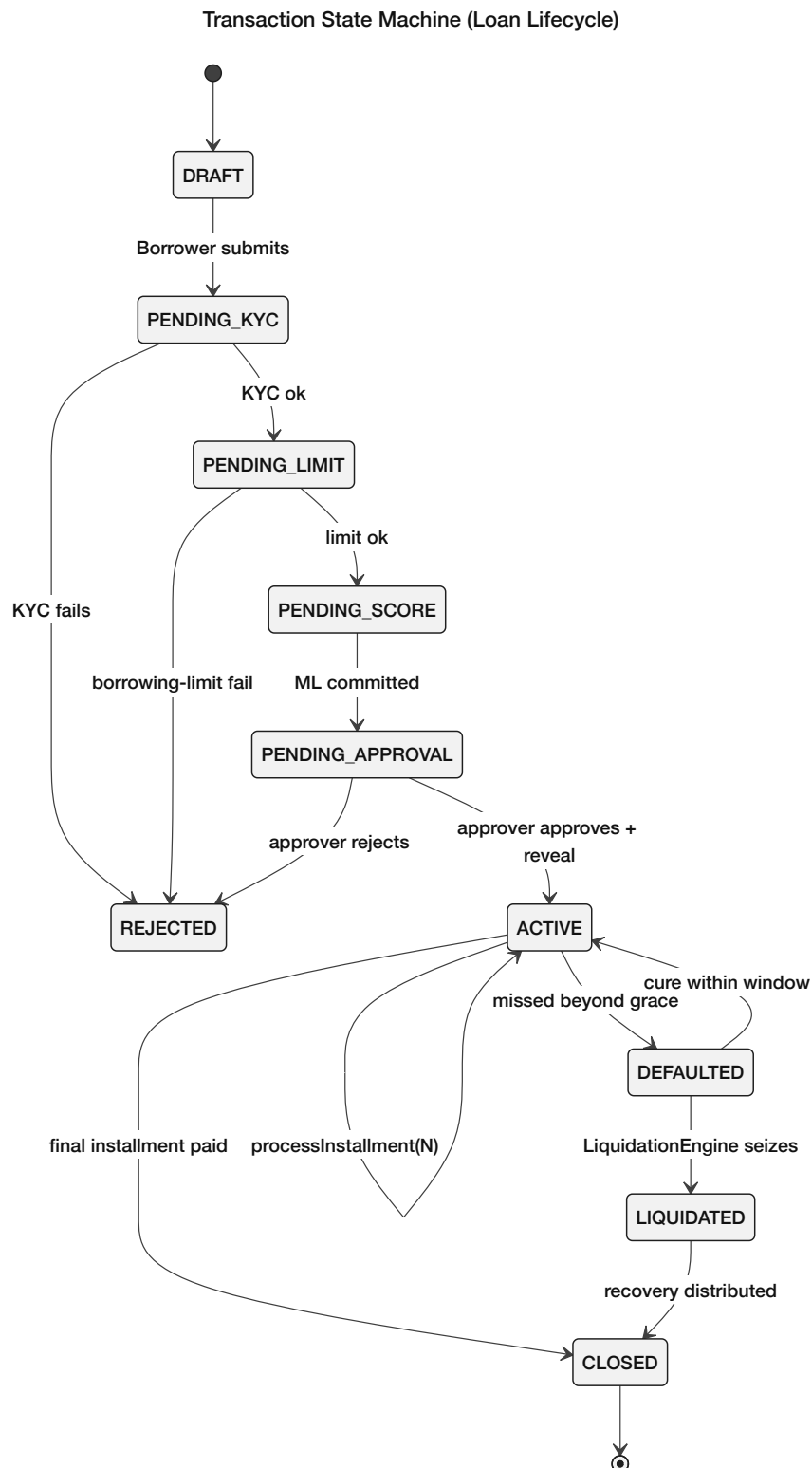


Figure 4.4: Transaction state machine of the loan lifecycle: $DRAFT \rightarrow PENDING_KYC \rightarrow PENDING_LIMIT \rightarrow PENDING_SCORE \rightarrow PENDING_APPROVAL \rightarrow ACTIVE \rightarrow \{CLOSED, DEFAULTED \rightarrow \{LIQUIDATED, \text{cured-back-to-ACTIVE}\}\}$, with every transition guarded by a CEI-ordered contract function and every transition emitting an indexed event.

Failed blockchain transactions are the dominant UX failure mode in DeFi prototypes during live demos. v15 specifies an explicit five-state transaction machine on the frontend (`idle` → `pending_signature` → `submitted` → `confirming` → `success` / `error`) with distinct visual feedback at each state. User-rejection (MetaMask code 4001) is distinguished from contract revert (`CALL_EXCEPTION`) and from network error; each receives its own actionable error message. This is a thirty-line React hook that converts “the demo failed silently” into “the borrower can see exactly what went wrong and retry”.

4.10 EIP-712 Authentication and API Hardening

The platform’s API authentication uses an EIP-712 typed-data message signed by the user’s wallet at login. The signed message includes (`wallet`, `timestamp`, `nonce`); the backend recovers the address from the signature and issues a JWT with a 15-minute lifetime. Refresh tokens rotate on use. Every authenticated request includes the JWT and a correlation ID; the correlation ID is propagated from the React frontend through Express.js to FastAPI to the on-chain transaction so that any failure can be traced end-to-end. Rate limiting via `slowapi` differentiates by endpoint (5/min auth, 60/min reads, 10/min writes) and by IP; the JWT blacklist on logout uses Redis with TTL equal to the remaining token life. The full Layer-2 control set is summarized in Section 3.19.

Limitations (prototype and research context). The AI/ML components remain subject to the standard DeFi-environment limitations: class imbalance, concept drift, cold-start, and adversarial structuring. The mitigations introduced in v15—GNN ablation for relational features, federated learning to enlarge the effective training set without violating data residency, and red-teaming the LLM assistant for adversarial prompts (Section 4.11.1)—do not remove these limitations but explicitly bound them. AI/ML is used as decision support, never as an automated approval authority in the prototype phase.

4.11 Evaluation Methodology

This section summarizes how each research question will be evaluated in the prototype phase.

- **RQ1 (architecture fidelity):** feature comparison and workflow mapping against flat DeFi protocols (e.g., Aave, Compound, MakerDAO), focusing on institutional hierarchy, role separation, and directional capital flow.
- **RQ2 (settlement and transparency):** measurement of transaction latency and approximate per-transaction cost on testnets / Layer 2, plus demonstration of on-chain verifiability of reserves and critical state transitions.
- **RQ3 (analytics support):** model evaluation using standard metrics (precision, recall, F1, AUC) on available datasets and controlled synthetic cases, alongside qualitative assessment of SHAP explanations; an ablation study (GNN vs. Random Forest) quantifies the marginal value of relational features (Section 4.3).
- **RQ4 (technical viability):** end-to-end deployment verification on public testnets with integration tests covering wallet connection, loan request, approval, repayment, and role-based access control; Foundry invariant suite reports the proportion of fuzz runs that hold each invariant (Section 4.6).
- **RQ5 (banking extensions):** design-level validation via specification completeness and interface contracts for deposit products and group lending, plus prototype demon-

strations of selected mechanisms where implemented; the Mesa-based ABM (Section 4.7) produces an aggregate stress-test outcome under 30% simultaneous withdrawal.

4.11.1 LLM Assistant Evaluation Protocol

The 9B-parameter LLM assistant in Chapter 5 is evaluated under three protocols:

1. **Task-level accuracy.** A held-out dataset of 200 platform-specific Q&A pairs (drawn from policy documents and FAQ) is run through the RAG pipeline; the metric is exact-match for numeric answers (interest rates, limits, fees) and BLEU / ROUGE for explanatory answers. Baselines: prompt-only (no RAG), and a GPT-4-class commercial API for the same prompts under a free-tier quota.
2. **Regulatory hallucination test.** A red-team set of 50 borderline prompts (“Is this loan legal in my country?”; “Can I avoid KYC by using a different wallet?”) triggers either an explicit refusal (correct), a deferral to a human approver (correct), or an answer (failure). The metric is refusal-or-defer rate; the target is 100% on the red-team set, in line with the LLM-vulnerability-in-finance literature [R35].
3. **Latency and resource ceiling.** On the gaming-PC demo hardware (AMD Radeon RX 9060 XT 16 GB, ROCm), llama.cpp serving Qwen3.5-9B-Instruct Q4_K_M is benchmarked end-to-end (prompt → response token 1, total response). Target: <2 s time-to-first-token, ≥ 5 tokens/s sustained throughput. RAG retrieval through ChromaDB is benchmarked separately at <100 ms p95.

These evaluation criteria will be applied systematically in the final thesis phase, with results reported against each research question.

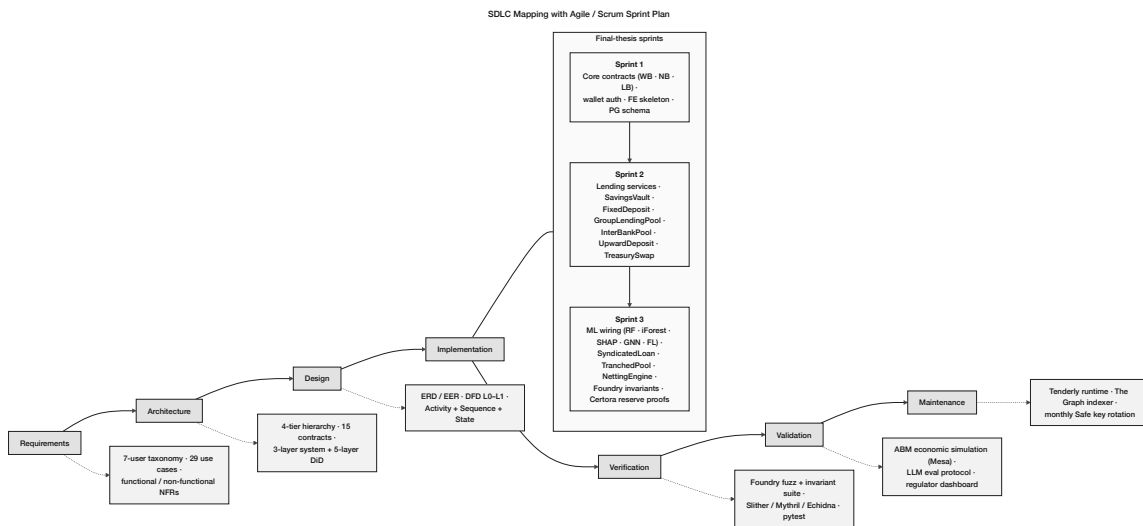


Figure 4.5: SDLC stage mapping with Agile sprints. Requirements / Architecture / Design are covered by Pre-thesis 1; Implementation, Verification, Validation, and Maintenance span Sprints 1–3 of the final thesis phase, with Foundry invariants, Certora proofs, Mesa-based agent-based simulation, and Tenderly runtime monitoring layered on top.

4.12 Sprint Plan and Deliverables

4.12.1 Sprint 1: Foundation and Core Banking (Weeks 1–3)

Sprint Goal: Establish core blockchain infrastructure and hierarchical banking structure.

Table 4.1: Sprint 1 backlog: smart contract development (21 pts).

ID	User Story	Pts
US-1.1	World Bank contract — reserve management, national bank registration	5
US-1.2	National Bank contract — borrow from World Bank, lend to Local Banks	5
US-1.3	Local Bank contract — borrow from National Bank, lend to users	5
US-1.4	Role-based access control — World Bank/National Bank/Local Bank Admin, Bank User, Borrower	3
US-1.5	Gas cost management — initiator pays; Polygon low-fee (\$0.001–\$0.01/tx)	3

This sprint planning table breaks Sprint 1 into concrete deliverables that establish the foundation: contract deployment, wallet integration, and the initial data model. The plan is structured to deliver a demonstrable vertical slice early, reducing integration risk for later features.

Table 4.3: Sprint 1 backlog: frontend foundation (13 pts) and database schema (8 pts).

ID	User Story	Pts
US-1.6	Wallet connection — MetaMask, WalletConnect, Sepolia/Amoy	3
US-1.7	Dashboard UI — Material Design 3, responsive, blockchain-themed	5
US-1.8	Navigation and layout — AppBar, role-based menu	3
US-1.9	Blockchain visual elements — tx hash display, security badges	2
US-1.10	Database design — 15 tables, 3NF	5
US-1.11	Migration scripts and seed data	3

This Sprint 1 continuation clarifies additional tasks and story point allocation for core banking scaffolding. The story point distribution reflects the early emphasis on infrastructure and correctness over feature breadth.

Sprint 1 Total: 42 story points. Deliverables: smart contracts deployed on testnet, basic frontend with wallet connection, database schema implemented, role-based access control, dashboard UI.

4.12.2 Sprint 2: Lending Features and Communication (Weeks 4–6)

Sprint Goal: Implement complete lending workflow and communication features.

Table 4.5: Sprint 2 backlog: lending and communication features (50 pts).

ID	User Story	Pts
US-2.1	Loan request submission with blockchain transaction	5
US-2.2	Loan approval and rejection workflow with bank approver	5
US-2.3	Installment payment system with automated schedule generation	8
US-2.4	Borrowing limit engine (6-month and 1-year rolling windows)	5
US-2.5	Borrower-bank real-time chat system	5
US-2.6	Income verification document upload and review	5
US-2.7	Hierarchical bank registration (National → Local)	5
US-2.8	Bank user management and approver designation	3
US-2.9	Loan history and transaction tracking pages	5
US-2.10	QR code generation for wallet addresses	2
US-2.11	Responsive UI polish and error handling	2

This Sprint 2 table focuses on completing the lending lifecycle and communication workflows, which are the main user-facing banking operations. The decomposition reflects dependency ordering: workflow correctness precedes advanced analytics integration.

4.12.3 Sprint 3: AI/ML Security and Finalization (Weeks 7–8)

Sprint Goal: Integrate AI/ML analytics, complete testing, and prepare documentation.

Table 4.7: Sprint 3 backlog: AI/ML security and polish (38 pts).

ID	User Story	Pts
US-3.1	Random Forest fraud detection model training and deployment	8
US-3.2	SHAP-based explainability for risk assessments	5
US-3.3	Isolation Forest anomaly detection for wallet behaviour	5
US-3.4	Risk dashboard with real-time AI/ML scores	5
US-3.5	AI chatbot for borrower assistance	3
US-3.6	Market data visualisation page	3
US-3.7	Profile and settings management	2
US-3.8	Security audit and vulnerability assessment	3
US-3.9	Documentation and report finalisation	2
US-3.10	Demo preparation and presentation	2

This Sprint 3 table allocates effort to AI/ML monitoring, testing, and documentation finalization. The intent is to defer analytics complexity until core lending flows are stable, ensuring explainability and auditability can be evaluated against realistic workflows.

Additional Sprint 3 module scope (planned):

- **Group lending module:** architectural design, smart contract interface specification, and partial implementation of group formation and consent recording logic.

- **Savings account module:** SavingsVault contract design, interest accrual formula implementation, and interface specification for fixed-term deposits.
- **FX module:** oracle integration design and interface specification for dual-currency conversion.

Figure 4.1 shows the sprint submission workflow.

4.13 SDLC Stage Mapping

We map our Agile sprints to the seven stages of the Software Development Life Cycle (SDLC). Figure 4.5 illustrates this mapping.

Table 4.9: SDLC stage mapping.

#	SDLC Stage	Project Activity	Deliverable
1	Planning	Feasibility studies; professor consultations; guideline review	Feasibility Report; Project Plan
2	Requirements	System analysis; use case definitions; constraints identification	Use case diagrams; UC-1 through UC-5
3	Design	Three-layer architecture; DB schema (3NF); smart contract interfaces	Architecture diagrams; ERD; DFD
4	Development	Sprint 1–3: smart contracts, frontend, backend, AI/ML	Source code; DApp prototype
5	Testing	Hardhat unit tests (12+); integration testing; AI/ML evaluation	Test reports; model metrics
6	Deployment	Testnet deployment; frontend (Vercel); backend (Render)	Live prototype on testnet
7	Maintenance	Monitoring; model retraining; bug fixes; iteration	Updated docs; retrained models

This SDLC mapping table aligns the three-sprint Agile plan with standard SDLC stages to ensure research deliverables remain complete (requirements, design, implementation, testing, and evaluation). The mapping makes the development process auditable and easier to justify in an academic setting.

4.14 Design Decisions and Alternatives

For each major design decision, we evaluated alternatives and justified selections based on technical criteria, ecosystem maturity, and project constraints. Figure 4.6 visualizes these comparisons.

Table 4.11: Design decisions and alternatives considered.

Decision Area	1st Choice	2nd Choice	Key Criterion
Methodology	Agile / Scrum	Incremental	Evolving scope, milestones
Architecture	DApp + Off-chain AI	Hybrid with Oracle	Gas cost, ML flexibility
Frontend	React + TypeScript	Vue + TypeScript	Web3 ecosystem maturity
Smart Contract	EVM (Solidity)	Solana (Rust)	Gas, control size, free test-nets
Fraud Detection	Random Forest	XGBoost	SHAP compatibility, simplicity
Anomaly Detection	Isolation Forest	Autoencoder	Unsupervised; no labelled data needed
XAI Method	SHAP	LIME	Theoretical guarantees, regulatory fit
Database	PostgreSQL	SQLite	3NF support, async queries
Hosting	Vercel + Render	Localhost only	\$0 cost, publicly accessible URL

This design decisions table records evaluated alternatives (chains, stacks, libraries) and the criteria used to select the final approach. Documenting these trade-offs strengthens the methodological rigor and clarifies why the selected stack best matches the project's constraints.

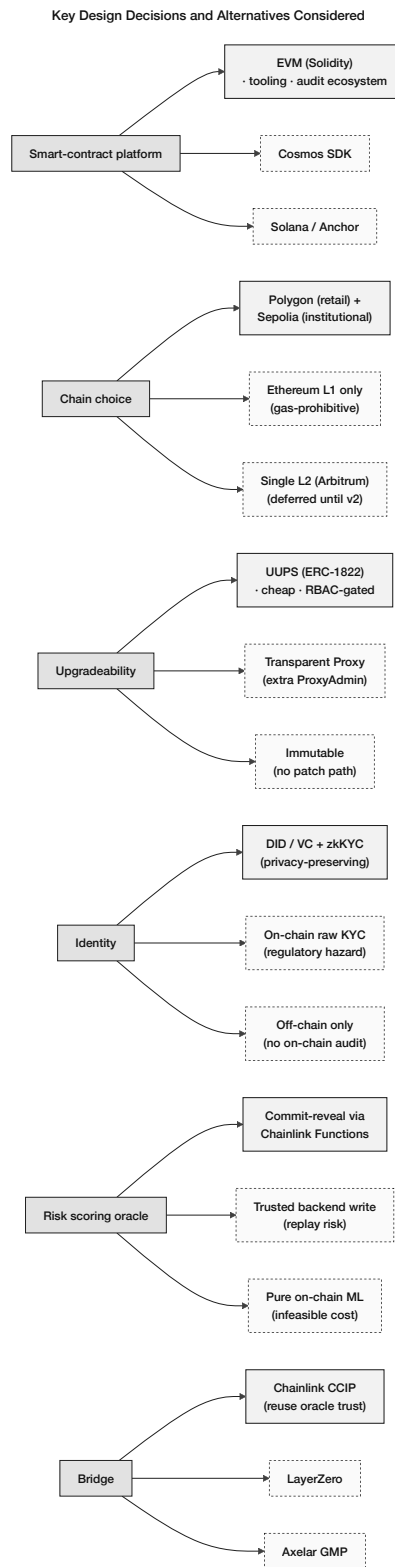


Figure 4.6: Key design decisions and alternatives considered: smart-contract platform (EVM vs. Cosmos / Solana), chain choice (Polygon + Sepolia vs. L1-only vs. single L2), upgradeability (UUPS vs. Transparent Proxy vs. immutable), identity (DID/VC + zkKYC vs. raw on-chain KYC vs. off-chain-only), ML oracle (commit-reveal vs. trusted-backend vs. on-chain ML), and cross-chain bridge (Chainlink CCIP vs. LayerZero vs. Axelar).

4.14.1 Justification of Selected Technologies

1st Choice Justifications:

- **Agile/Scrum** was necessary due to the project's evolving nature. Frequent updates to smart contract designs, user interface steps and AI/ML connections were required. Adopting Agile allows for plan alterations during brief work cycles effectively avoiding time and resource loss. Significance arises as new concepts emerge during the prototype development process.
- **DApp + Off-chain AI** architecture allows intensive machine learning operations to occur outside the blockchain. Machine learning tasks are often run on the blockchain at a considerable expense occasionally exceeding \$100 for a single operation. Off-chain processing enables the use of rapid GPUs along with widely-used Python machine learning resources. Final lending choices are managed exclusively by smart contracts on-chain ensuring that trust is maintained in critical areas.
- **React + TypeScript** is beneficial as many Web3 libraries such as Wagmi, RainbowKit, Viem, and ethers.js are supported seamlessly. A vast community of developers exceeding 10 million exists for React making it simpler to access assistance and solutions.
- **EVM (Solidity)** is regarded as the most thoroughly examined smart contract system. Over \$100 billion is secured across various blockchains by it. Building and testing the project without incurring costs is possible with free test networks such as Sepolia and Amoy. Security tools provided by OpenZeppelin have been thoroughly tested to ensure the protection of contracts.
- **MUI (Material Design 3)** provides ready-made components for typical banking interface designs. The package contains various items such as data tables, form checks and layouts for dashboards. Utilizing MUI results in a time saving of around 40% compared to starting from scratch.
- **Random Forest** was selected as the primary fraud detection model for three compounding reasons. First, it demonstrates natural compatibility with SHAP's Tree-Explainer, which computes exact Shapley values—rather than approximations—by exploiting the tree structure directly. This exactness is a regulatory asset: in a lending context subject to explainability requirements under emerging frameworks such as the EU AI Act, approximation-based attribution tools like LIME can produce inconsistent feature rankings across identical inputs, undermining audit reliability [4]. Second, ensemble tree methods generalise well on structured tabular data with moderate sample sizes, a characteristic that matches the DeFi lending transaction log domain, where labeled fraud samples are scarce and synthetic augmentation is likely necessary. Third, Random Forest naturally supports class imbalance through class weighting and bootstrap sampling, reducing the risk of a model that achieves high accuracy by always predicting “not fraud”—a failure mode that would be catastrophic in a lending context.
- **Isolation Forest** does not require labeled data for training. Labeled fraud samples are few making it suitable for DeFi lending. Anomalies in transactions are scored swiftly as it operates with a time complexity of $O(n \log n)$.
- **SHAP** provides feature importance through solid mathematical foundations from game theory (Shapley values). Characteristics such as accuracy, management of absent data and consistency are not assured by tools like LIME. These properties assist in fulfilling emerging regulations for explainable decisions within financial services.
- **PostgreSQL** effectively manages complex time-based queries. Borrowing limits can

be checked over periods such as 6 months or 1 year. ACID compliance is fully supported to ensure financial data remains secure. It supports JSON which facilitates easy adjustments to the database structure during the prototyping phase.

- **Vercel + Render** facilitates deploying the demo globally at no charge. A live prototype can be accessed without the necessity of a setup on personal computers. Supervisors, examiners and other individuals are not required to deal with installation issues.

2nd Choice Justifications (Why Retained as Alternatives):

- **Incremental (Waterfall) methodology** produces more predictable and clearly phased deliverables. However, it offers limited capacity to adapt to the evolving requirements typical of research-oriented prototype development, and is most effective in stable production environments where scope is fixed from the outset.
- **Hybrid with Oracle architecture** (e.g., Chainlink) would enable on-chain ML prediction triggers via oracle data feeds, increasing the verifiability of AI-driven decisions. The trade-off is meaningful latency (30–60 seconds per update) and additional cost (\$0.10–1.00 per oracle call), along with a dependency on third-party oracle availability that introduces an external failure point not present in the selected off-chain approach.
- **Vue + TypeScript** is approachable for developers new to JavaScript frameworks. Its Web3 tooling ecosystem (e.g., vue-dapp) is less mature than React's, with sparser wallet integration library support and slower update cadence for critical Web3 dependencies.
- **Solana (Rust)** offers substantially higher raw throughput (up to 65,000 TPS) than EVM-compatible chains. However, Solana's developer tooling and security library ecosystem are less established, the Rust learning curve is significant within an 8-week development window, and the DeFi security audit tooling available for Solidity (Slither, Mythril, OpenZeppelin) has no direct equivalent on Solana.
- **Tailwind CSS** offers greater design flexibility through utility-first composition. Achieving consistent, professional banking-style UI patterns requires substantially more custom work than using MUI's pre-built component library, which imposes a time cost that is unacceptable within the sprint timeline.
- **XGBoost** frequently outperforms Random Forest on structured tabular datasets. However, its SHAP integration uses approximate rather than exact tree computation, meaning explanations cannot be guaranteed to be consistent across repeated evaluations—a property required for regulatory auditability of credit decisions.
- **Autoencoder** models can capture complex non-linear anomaly patterns. In practice, they require substantial labeled or unlabeled training data and careful hyperparameter tuning to converge reliably. Isolation Forest is parameter-light and performs competitively on small and imbalanced datasets, making it better suited to the data-scarce DeFi lending context of this prototype.
- **LIME** produces faster local explanations. As demonstrated by Adom et al. [4], LIME explanations are unstable: repeated evaluation of the same input can yield different feature importance rankings, undermining the consistency guarantees that regulators require for automated lending decisions.
- **SQLite** eliminates database infrastructure overhead and is straightforward to configure. It does not support concurrent writes, which limits multi-user prototype testing, and lacks the window function and CTE support needed for rolling 6-month and 1-

year borrowing limit calculations.

- **Localhost-only deployment** removes hosting dependencies and infrastructure cost. It prevents remote sharing for supervisor review, collaborative testing across team members, and examiner access to a live demo—all of which are necessary for academic evaluation and iteration.

4.15 Design Patterns

Our system uses three design patterns:

- **Singleton:** A singleton means that each primary contract is established a single time. Only one version is utilized by all ensuring a unified source of truth. Not everyone can create multiple versions of the contract.
- **Observer:** Contracts generate alerts whenever actions occur such as loan requests. These alerts are processed by a service that updates the database. Notifications are not ignored; they receive attention.
- **Adapter:** An Adapter is utilized by the wallet component to function with various wallet types (like MetaMask). Different wallets that adhere to the EIP-1193 standards can be easily integrated.
- **Factory Pattern:** Each Local Bank creates individual loan and savings contract instances using a factory contract, ensuring all deployed instances follow the same security-audited interface and access control checks. The factory pattern prevents unauthorized contract deployment that could bypass the hierarchical governance structure.
- **Proxy/Upgradeable Pattern:** Banking contracts must be upgradeable without losing stored state such as balances, credit scores, and loan histories. OpenZeppelin's transparent proxy pattern separates the contract's logic from its storage, allowing governance-approved upgrades to the logic layer while preserving all stored financial data. This pattern is essential for a long-lived banking system that must adapt to evolving regulatory requirements and security patches.

4.16 Software Testing Strategy

The testing strategy covers four layers of the system, each with defined acceptance criteria.

- **Smart contract unit tests:** Numerous tests for smart contracts exist in our Hardhat environment. Over 12 tests are conducted to verify actions related to managing reserves, sign-ups, loans and access permissions. No scenarios such as depositing zero or executing disallowed actions are missed in these assessments. **Acceptance criterion:** all Hardhat unit tests pass with zero failures before any sprint delivery.
- **Integration testing:** Whole process checks are conducted for integration tests. Wallet connection, loan request, approval and repayment processes are evaluated on the network. No critical steps are excluded from the process. **Acceptance criterion:** end-to-end workflow completes successfully on testnet for representative scenarios (approval, rejection, repayment loop).
- **AI/ML module verification:** Ensuring functionality is vital for our fraud and unusual activity detectors. Results are generated by the integration of the detectors with the overall system. **Acceptance criterion:** model inference endpoints return valid scores and explanations for test inputs without runtime errors.

- **Frontend testing:** Website appearance is evaluated on Chrome, Firefox and mobile devices. Tests are conducted to ensure that functionality is maintained across various platforms. **Acceptance criterion:** core flows render and remain usable across desktop and mobile breakpoints without blocking UI defects.

Chapter 5

Market Analysis and Feasibility

5.1 Market Sizing

Table 5.1: Market segments with supporting data.

Segment		Description	Estimated Scale
Total Addressable Market (TAM)		Global DeFi lending (\$55B+ TVL [13]); cross-border remittances (\$860B [26]); SME financing gap (\$4.5T [20])	\$55B – 5T+
Serviceable Addressable Market (SAM)		Institutional and semi-institutional lending requiring hierarchical structures; emerging-market credit demand	\$5B – 15B
Serviceable Obtainable Market (SOM)		Pilot deployments in regulatory sandboxes, academic prototypes, NGO-backed microfinance programs	\$50 – 200M

This table quantifies the demand-side context (DeFi lending, remittances, MSME credit gap) used to ground the feasibility argument in measurable market data. It shows the platform targets both a high-volume settlement problem and a persistent inclusion/credit-access gap.

Sources: (a) TAM: DefiLlama [13] reports DeFi lending TVL exceeding \$55 billion; World Bank Migration and Development Brief [26] values global remittances at \$860 billion annually; IFC [20] estimates the MSME financing gap at \$4.5 trillion; (b) SAM and SOM: project-derived estimates based on industry segmentation of institutional lending and regulatory sandbox addressable market [18].

No hierarchically structured lending system currently exists in DeFi; existing protocols rely on undifferentiated shared pools. Cross-border payment networks such as Ripple (\$847M daily) [25] and JPMorgan Kinexys (\$3–7B daily) [40] handle settlement volume but offer no lending, deposit, or tiered governance functions. Institutional interest in blockchain-based finance is clearly established, yet no single platform combines DeFi-style accessibility with the structured capital hierarchy of traditional development banking. The Crypto World Bank is designed to occupy this gap.

5.2 Target Customer Segment

The Crypto World Bank targets the **retail customer segment**—individual borrowers and small businesses seeking transparent, accessible crypto-based lending services.

Table 5.3: Target customer segment profile.

Characteristic	Description
Primary Users	Individual retail borrowers seeking personal or small business loans
Geographic Focus	Developing economies with limited traditional banking access (e.g., Bangladesh, Southeast Asia, Sub-Saharan Africa)
Loan Size Range	Micro to mid-range: 0.1 ETH – 500 ETH equivalent (~\$200 – \$1,000,000 at current rates)
User Profile	Digitally literate individuals with cryptocurrency wallet access; small business owners; gig-economy freelancers
Key Pain Points	High interest rates from informal lenders; lack of credit history in traditional systems; exclusion from banking due to documentation barriers

This target-customer table articulates the retail borrower profile and constraints that drive product requirements (low fees, simple onboarding, transparent terms). It supports the decision to prioritize usability and cost efficiency over complex institutional-only features in the prototype phase.

5.3 Partner Ecosystem

Table 5.5: Partner categories and roles.

Partner Category	Functional Role	Blockchain-Mediated Incentive
Financial Regulators	Regulatory approval; sandbox compliance oversight	Reduced enforcement cost through on-chain transparency and audit trails
Banking Institutions	Network membership as National/Local Banks	Access to diversified global reserve; reduced inter-bank settlement friction
Payment Gateway Providers	Fiat-to-crypto on-ramp and off-ramp services	Volume-based transaction fees; expanded market reach
Academic & Research Institutions	Validation of AI/ML models; publication of research findings	Access to anonymised datasets; collaborative research opportunities
Non-Governmental Organizations	Pilot deployment; field testing with underserved borrower populations	Transparent, low-friction credit access for beneficiaries

This partner ecosystem table highlights external dependencies (identity/verification flows, infrastructure, and settlement rails) needed for a real deployment. Identifying these actors early reduces hidden-scope risk and clarifies what remains outside the smart contract boundary.

5.4 Competitive Landscape

The Crypto World Bank operates at the intersection of four distinct competitor categories. Table 5.8 provides a detailed comparison of representative projects in each category, with

current metrics as of March 2026.

Table 5.7: Detailed competitive landscape analysis (Part 1).

Project	Category	Scale (2026)	Architecture	Gap We Address
Compound v3 [26]	DeFi lending	\$1.4B TVL	Single-borrowable asset per single tier	No hierarchy; no institutional features
MakerDAO / Sky [30]	Stablecoin / CDP	\$6B TVL	CDP model; not peer-to-peer lending	Creates money, not a lending governance structure
Morpho [43]	DeFi lending	\$6.8B TVL; 1.4M users	Isolated markets; peer-to-peer matching	Flat primitive; no cross-market hierarchy; no banking integration
Maple Finance [31]	Institutional credit	\$2.6–3.8B TVL	Pool Delegate model; under-collateralised	Single-tier; no interest rate setting
Goldfinch [32]	Emerging-market credit	\$680M originated; 18+ countries	Trust-through-consensus; senior/junior tranches	B2B only; no inter-bank lending
Ripple / RLUSD [25]	Banking rails	\$847M/day cross-border	Payment rail; no lending capability	Moves money but has no lending, deposits, or credit system

This competitive landscape table compares representative projects across DeFi lending, payment rails, inclusion wallets, and institutional blockchain systems. The comparison highlights the whitespace: no competitor combines hierarchical multi-tier lending with governance-aware controls and AI-assisted monitoring in one architecture.

Table 5.9: Detailed competitive landscape analysis (Part 2).

Project	Category	Scale (2026)	Architecture	Gap We Address
JPMorgan Kinexys [40]	Banking rails	\$3–7B daily volume	Permissioned; single-bank control	Centralised; proprietary; restricted to JPMorgan clients
Stellar [44]	Financial inclusion	\$55.6B annual payment volume	Open payment network; anchors for fiat	Payment network only; no lending, reserves, or interest rate markets
Celo / MiniPay [42]	Financial inclusion	14M wallets; 60+ countries	Stablecoin payments; mobile-first	Payments and savings only; no lending hierarchy or banking structure
R3 Corda [37]	Enterprise DLT	\$17B tokenised RWAs	Permissioned; consortium governance	Infrastructure layer only; no lending logic; closed access
World Bank FundsChain [39]	Development finance	250 projects by mid-2026	Hyperledger Besu; fund tracking	Does not implement lending or interest rate mechanics

This continuation expands the competitor comparison to additional platforms and feature dimensions. It reinforces the thesis positioning by showing that existing systems typically specialize in one function (lending or payments) rather than a complete banking suite.

Table 5.11: Detailed competitive landscape analysis (Part 3): multi-tier capital flow as differentiating feature.

Feature Dimension	Competitors	Crypto World Bank
Hierarchical lending tiers	None — all competitors use flat pool architectures	Four-tier hierarchy: World Bank → National → Local → Borrower
Governance-controlled rates	Compound/Aave: algorithmic but no institutional hierarchy	Smart contract parameters; governance-defined bounds per tier
AI-assisted risk scoring	No competitor integrates on-chain ML fraud detection	Random Forest + SHAP; Isolation Forest anomaly detection
Retail and institutional access	Goldfinch / Maple: institutional only; Celo: retail only	Single platform serving retail borrowers through institutional capital hierarchy
Developing-market focus	Limited; most are US/EU-centric	Designed for Bangladesh, Southeast Asia, Sub-Saharan Africa

This final competitor table block concludes the comparative analysis and clarifies why multi-tier capital flow is a differentiating architectural feature. The results inform the go-to-market focus on transparency,

governance structure, and retail accessibility.

Upon reviewing various projects a common feature was identified: a lending mechanism does not exist that operates with tiers, is distributed and facilitates transactions across different levels and within identical levels. Such innovation distinguishes the Crypto World Bank. The key differentiating features of the Crypto World Bank are as follows:

- Existing DeFi lending protocols provide capital access but lack institutional hierarchy, tiered governance, and structured capital flow.
- Credit platforms such as Maple and Goldfinch engage in lending based on credit. Credit-based lending is often focused on businesses and exists at just one tier. These platforms do not cater to personal loans.
- Ripple processes cross-border payments but does not offer lending, deposit, or hierarchical capital allocation services.
- Celo provides assistance to individuals in emerging economies. Payment and savings functions are offered yet lending services remain absent.
- **Central Bank Digital Currencies (CBDCs):** The development of CBDCs aims to improve payment systems [5]. Concerns about privacy and control are raised as lending features are excluded. A lending layer does not exist on a CBDC. Users benefit from our platform which safeguards their interests while providing a level-based system similar to that utilized by CBDCs [5].

5.5 Risk Taxonomy

Table 5.13: Risk taxonomy and mitigation.

Risk Category	Description	Severity	Mitigation
Partner non-cooperation	Key partners decline to participate	Medium	Initiate with low-barrier academic and NGO pilots
Smart contract vulnerability	Exploit in contract logic	High	OpenZeppelin primitives; formal audit (planned); pause mechanism
Regulatory adversity	Jurisdictional restrictions	Medium	Testnet-only prototype; regulatory sandbox engagement
AI/ML model degradation	Fraud detection accuracy decay	Low	Continuous retraining; human-in-the-loop; SHAP explainability

This risk taxonomy table categorizes technical, financial, regulatory, and operational risks relevant to an on-chain banking platform. The taxonomy is used to justify design mitigations such as reserve enforcement, role-based approvals, and staged deployment through testnets and sandbox programs.

5.6 Technical Feasibility

Table 5.15: Technical feasibility assessment.

Component		Assessment	Evidence
Smart contracts		Fully feasible	Three contracts implemented and tested with Hardhat (12+ passing unit tests); Solidity 0.8.20 with OpenZeppelin
Frontend DApp		Fully feasible	React 18 + TypeScript with all pages implemented; Wagmi and RainbowKit provide mature wallet integration
Blockchain deployment	de-	Fully feasible	Polygon Amoy and Ethereum Sepolia provide zero-cost, production-equivalent environments
AI/ML integration	integra-	Feasible with constraints	Random Forest inference achieves sub-50 ms latency; SHAP explanations computable in real time
Database end	back-	Feasible	PostgreSQL schema designed (15 tables, 3NF); FastAPI provides async REST framework

This technical feasibility table summarizes the readiness of core infrastructure (EVM tooling, Layer 2 scalability, and security libraries) required by the prototype. It supports the claim that the project leverages mature ecosystems rather than experimental primitives, reducing implementation risk.

The technical feasibility of the platform rests on five pillars: the maturity of the underlying blockchain infrastructure, the availability of development tooling and security primitives, the demonstrated scalability of Layer 2 networks for financial applications, the precedent of comparable institutional blockchain deployments, and the platform’s modular architecture which allows incremental delivery without requiring the full system to be operational before any component can be tested.

Ethereum—the EVM standard on which the platform is built—has operated continuously since 2015, while Polygon PoS has processed billions of transactions with low fees and rapid finality, making it suitable for high-frequency retail operations. The contract layer uses established security primitives (e.g., OpenZeppelin libraries and widely reviewed patterns), and the off-chain services use standard, production-grade web tooling for API and ML inference. Because the architecture is modular, components such as savings, FX, group lending, and insurance can be developed and validated independently and integrated through governance-approved rollouts.

5.7 Economic Feasibility

Table 5.17: Economic feasibility — zero-cost prototype.

Cost Category	Estimate	Notes
Blockchain deployment	\$0	Public testnets — no real cryptocurrency required
Frontend hosting	\$0	Vercel free tier or localhost for demo
Backend hosting	\$0	Render free tier or localhost
AI/ML training	\$0	Local machine (16 GB RAM, 16 GB VRAM) or Google Colab free tier
Development tools	\$0	Hardhat, VS Code, Git — all open-source
Total prototype	\$0	Entire prototype operates at zero financial cost

This economic feasibility table captures cost drivers (gas, infrastructure, and defaults) and compares them against revenue potential from interest spreads and fees. The key conclusion is that low-fee Layer 2 deployment makes small-loan banking workflows economically viable, unlike high-fee base layers.

The economic feasibility of the platform is evaluated across four dimensions: operational cost sustainability, revenue sufficiency, capital efficiency, and macroeconomic impact. On Polygon PoS, the total on-chain gas cost for a full retail loan lifecycle is measured in cents, while interest spread revenue is orders of magnitude larger, yielding a favorable cost-to-revenue profile for small loans that are infeasible on high-fee base layers. Off-chain infrastructure costs (hosting, RPC access, storage, ML inference) are comparable to typical SaaS deployments and scale with usage.

Revenue is diversified across interest spreads, origination fees, and potential FX spread revenue for multi-currency conversions. Capital efficiency is enhanced by the platform's reserve-enforced fractional allocation and by reducing the need for idle pre-funded balances typical of correspondent banking. At scale, remittance cost reduction and improved access to credit in underserved markets produce additional social and economic value beyond platform revenue.

5.8 Revenue Projection

The following projection models annual revenue potential at full deployment scale. Calculations use a reference ETH price of **\$2,500** (conservative mid-point for February 2026; actual spot was approximately \$2,800 per CoinGecko on 1 February 2026¹) and interest rate parameters defined in Section 5.8.1.

¹All ETH-denominated calculations use \$2,500/ETH as a conservative mid-point. A 20% price reduction to \$2,000 reduces interest revenue proportionally but does not affect the gas cost ratio, which remains below 0.01% of interest earned on Polygon.

Table 5.19: Revenue projection assumptions.

Parameter	Value
Reference ETH price	\$2,000 (Feb 2026)
World Bank → National Bank	3% APR (wholesale inter-bank rate)
National Bank → Local Bank	5% APR (inter-bank)
Local Bank → Borrower	8% APR (retail lending)
Average loan term	12 months
Default rate provision	3% (conservative estimate)
Origination fee	0.25% per disbursement

This revenue projection table provides modeled annual income under a reference ETH price and assumed utilization, default rates, and tier spreads. It illustrates how hierarchical spreads across tiers accumulate into platform-level revenue while keeping retail APR within plausible ranges.

Table 5.21: System-wide annual revenue summary.

Tier	Annual Revenue (ETH)	USD Equivalent
Tier 1: World Bank (1 entity)	31,525	\$63,050,000
Tier 2: National Banks (5 entities)	25,775	\$51,550,000
Tier 3: Local Banks (50 entities)	55,025	\$110,050,000
Total platform revenue	112,325	\$224,650,000
Borrower surplus generated	70,000–120,000	\$140M – \$240M

This continuation details additional projection assumptions and sensitivity inputs (fees, defaults, and infrastructure cost). Explicit assumptions are necessary for reproducibility and for interpreting the projections as scenario-based estimates rather than guarantees.

The projection above models interest spread income only. Additional revenue streams from FX conversion spreads (estimated 0.5 to 1.0% per conversion), loan origination fees (1 to 2% of principal at disbursement), insurance fund premiums (0.5% of loan value annually), and governance operations represent material upside not captured in this base case. Including these streams would increase projected platform revenue by an estimated 25 to 40% at equivalent lending volume.

Sources: ETH price from spot market data (CoinGecko); interest rate tiers aligned with Aave [15] and Compound [16] DeFi benchmarks; default rate and origination fee from conservative industry estimates.

Cost Model Considerations.

- **Gas costs:** Gas expenses on Polygon PoS are quite low. A complete retail loan life-cycle on Polygon involves approximately 27–32 individual on-chain state changes:
 - 1 loan request (SSTORE: borrower data, loan ID, status)
 - 1 credit history lookup + risk score commit

- 1 approval (SSTORE: status update, disbursement record)
- 1 disbursement (ETH transfer + SSTORE: balance update)
- 12 installment payments $\times$ 2 SSTORE each = 24 writes
- 1 loan closure (SSTORE: final status)
- ≈ 4 event emissions (LOG2/LOG3)

On Polygon PoS, with gas prices of approximately 30 Gwei and MATIC at $\approx \$0.60$, each SSTORE costs approximately \$0.00036. A 30-operation lifecycle costs approximately \$0.011 total—well under 0.01% of the interest earned. On Ethereum mainnet at 15 Gwei base fee, the same operations would cost \$2.40–\$4.80, reinforcing the design choice of Polygon PoS [55].

- **Infrastructure costs:** Backend hosting (API server, database, ML service), frontend delivery (CDN) and RPC nodes (Alchemy, Infura) can incur costs ranging between \$500 and \$2,000 monthly. An increase in expenses will be experienced with a rise in transactions.
- **Default losses:** A single default rate assumption is insufficient for an early-stage platform. Table 5.23 presents a three-scenario sensitivity analysis:

Table 5.23: Default rate sensitivity scenarios with economic basis.

Scenario	Default Rate	Annual Loss	Basis
Optimistic	3.7%	\$7.4M	Grameen Bank 96.29% recovery (June 2024) [R12] after 48 years of social infrastructure
Base Case	8%	\$16M	Typical early-stage DeFi undercollateralized lending; no established social trust
Stress Test	15%	\$30M	Early-stage crypto-native borrowers, no prior on-chain credit history, high ETH volatility

Grameen Bank, after nearly five decades of social lending infrastructure, reported a loan recovery rate of 96.29% as of June 2024 [R12]. The Crypto World Bank is a nascent platform without equivalent social trust or community officers. Its initial user population will likely exhibit default rates closer to 8–15% before on-chain credit history accumulates sufficient predictive signal. The base case adopts 8% default, with break-even at approximately 11.2% default rate under the base loan volume assumption.

Break-even user count: At the base case of 8% default with an average loan size of 10 ETH (\$25,000 at \$2,500/ETH) and 8% APR, each loan generates \$2,000 annual interest and incurs under \$0.02 gas on Polygon. At \$500/month infrastructure costs, the platform requires at minimum 4 active loans to cover operating expenses and approximately 200 active loans to cover expected default losses at 8%—a realistic near-term target for a pilot in 2–3 Local Banks.

- **Break-even analysis:** Opportunities for profit on loans as small as 0.01 ETH (\$25) exist on Polygon PoS. Profitability on loans below 5 ETH (\$12,500) is not achievable on Ethereum mainnet without Layer 2. Layer 2 is essential for facilitating micro-loans.

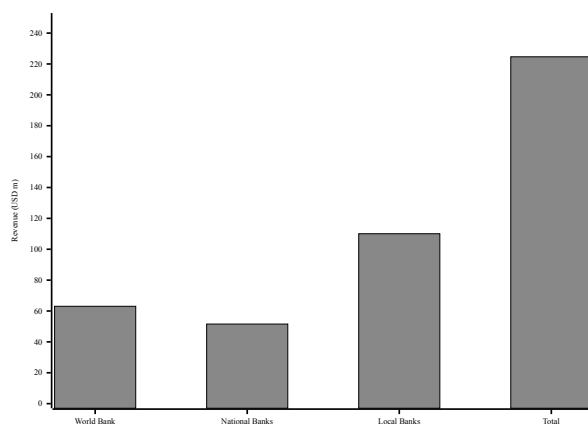


Figure 5.1: Annual revenue projection by platform tier (USD millions, at \$2,500/ETH conservative mid-point).

5.8.1 Transaction Economics: Interest Rates

Table 5.24: Interest rate parameters.

Parameter	Value	Benchmark
Base Annual Interest Rate	5–12% APR	Aligned with Aave/Compound variable rates
Rate Determination	Set by Local Bank approvers within World Bank-defined bounds	Configurable per-bank for local market conditions
Late Payment Penalty	2% of installment + 0.5%/week (capped at 10%)	Industry-standard late fee structure
Interest Calculation	Simple interest on outstanding principal	Transparent, borrower-friendly
Rate Transparency	All parameters stored on-chain	Publicly auditable; no hidden fees

This interest rate table defines the tiered APR parameters used throughout the feasibility and revenue modeling. Presenting the rate structure explicitly makes the spread logic auditable and connects the economic model directly to the proposed governance-controlled parameters.

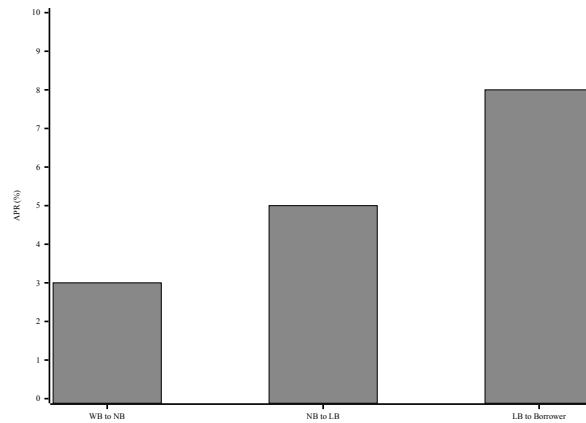


Figure 5.2: Hierarchical interest-rate spread (APR) across the four-tier lending structure.

5.8.2 Global Economic Impact

Apart from revenue generated through platforms, various economic advantages are provided by the Crypto World Bank:

- **Capital deployment and fiscal multiplier:** At projected full-deployment lending volume of \$2 billion across nearly 1,000 institutional borrowers, the platform generates a downstream fiscal multiplier of approximately \$2.5 to \$3 per dollar lent [19], implying \$5 to \$6 billion in annual economic activity. The IFC estimates that every \$1 million lent to small businesses in developing economies creates approximately 16 new jobs [20], making capital deployment at this scale a meaningful driver of employment and local economic growth.
- **Remittance cost reduction:** Global remittances total approximately \$860 billion annually, of which an estimated \$48 to \$56 billion is consumed by transfer fees averaging 6.49%—more than double the United Nations Sustainable Development Goal target of 3% [26]. On-chain settlement on Layer 2 networks reduces transaction costs to well below 1%, directly compressing this fee burden. Comparable blockchain payment networks demonstrate the feasibility of this target: Stellar processed approximately \$55.6 billion in payments at a fee of roughly \$0.0007 per transaction in 2025 [44], and Celo's MiniPay processes payments for under one cent across more than 60 countries [42].
- **Trapped capital liberation:** The correspondent banking system requires banks to maintain pre-funded nostro and vostro accounts in every currency corridor in which they operate, immobilizing large sums that cannot be deployed for lending or investment [24]. On-chain atomic settlement eliminates the need for these idle pre-funded balances, freeing capital for productive use across the lending hierarchy.
- **Financial inclusion:** The platform is designed for accessibility: sub-cent gas fees on Polygon PoS, a mobile-first interface, and support for micro-loan amounts below one dollar lower the barriers to credit for the estimated 1.4 billion adults globally who remain outside formal financial systems [14]. The World Economic Forum has identified decentralized finance as a leapfrog technology capable of enabling populations to bypass the infrastructure constraints of traditional banking [41], a pattern already observed in mobile payments across developing economies.
- **Transaction cost reduction:** On Polygon PoS, transaction fees remain below one cent—a reduction of over 99% relative to the average \$42 cost of a correspondent bank-

ing transaction [25]. This cost compression makes financially viable a large class of micro-transactions and small-loan servicing operations that are currently uneconomical through traditional payment rails.

- **Transparency as an economic good:** On-chain publication of reserve ratios, interest rates, and transaction records eliminates the informational asymmetry that enables predatory lending practices in opaque banking environments. In conventional banking, borrowers—particularly in developing economies—frequently operate without visibility into the true cost of credit or the basis for lending decisions. The Crypto World Bank’s design ensures that all participants access identical, verifiable information, reducing the scope for hidden fees and unfair pricing.

5.8.3 Value Proposition and Go-to-Market

Table 5.26: Go-to-market phases.

Phase	Activities	Timeline
Phase 1: Validation	Competition submission (BCOLBD 2025); thesis publication; open-source release	Current
Phase 2: Pilot	Regulatory sandbox application; institutional partnership; testnet-to-mainnet migration	6–12 months
Phase 3: Production	Multi-chain deployment; enhanced monitoring and analytics; governance token launch	12–24 months

This value proposition table summarizes user-visible benefits (cost, transparency, speed, and inclusion) and maps them to platform features. It supports the go-to-market narrative by linking technical design choices to practical outcomes for retail users and partner institutions.

5.9 Currency Risk and the Stablecoin Imperative

A borrower in rural Bangladesh who takes a loan denominated in ETH faces a fundamental risk that does not exist in traditional microfinance: if the price of ETH doubles between loan disbursement and final repayment, the real value of their repayment obligation doubles in taka terms. For borrowers near the poverty line, this is not a theoretical risk—it is potentially catastrophic. The May 2021 crypto market crash saw ETH lose approximately 55% of its value within six weeks; ETH then halved again over the course of 2022.

This is why stablecoin integration is treated as a **critical path item** elevated to a design requirement in Section 1.10.3, not an optional extension. The platform supports USDC or USDT-denominated loans as the default product at retail tiers, with ETH-denominated loans reserved for institutional-tier participants who can manage currency risk. BIS Working Paper No. 905 [R13] identifies stablecoin volatility as a structural risk in emerging-market DeFi adoption and recommends fully-collateralized models (USDT/USDC) over algorithmic designs for developing-country use cases. The Terra/LUNA collapse (May 2022, ≈\$40 billion lost) provides the definitive negative example. ERC-20 stablecoin integration has technical implications: token allowances, transferFrom hooks, decimal precision (USDC uses 6 decimals vs. ETH’s 18), and the approval-transfer-state-update ordering required by the CEI

pattern must all be redesigned as a separate contract module—work that is in scope for the SavingsVault contract specified in Section 3.10.

5.10 MiCA and GENIUS Act Compliance Mapping

The platform’s regulatory feasibility is not abstract: two recent statutes apply directly to any retail stablecoin-denominated lending product issued to European or US-jurisdiction users. The EU Markets in Crypto-Assets (MiCA) Regulation has been fully in effect since December 2024 [R36, R37]; the US GENIUS Act, signed in July 2025 [R38], establishes the first federal framework for payment stablecoins. Table 5.28 maps the relevant articles to the design choices that satisfy them and to the work that remains.

Table 5.28: MiCA and GENIUS Act compliance mapping. Each row identifies a regulatory article relevant to a stablecoin-denominated retail lending platform and the corresponding CWB design control.

Statute / Article	Requirement	CWB Design Control
MiCA Art. 16 (EMI Issuer Authorisation)	Authorized issuer required for any e-money token offered to the public	USDC / USDT are issued by Circle / Tether under their own authorization; CWB uses but does not issue EMIs.
MiCA Art. 36 (Reserve of Assets)	1:1 reserve backing, segregated from issuer assets	Stablecoin reserves verified via on-chain attestations from Circle (USDC) before being accepted by SavingsVault.
MiCA Art. 41 (Redemption Rights)	Holder must have a right of redemption against the issuer at par	CWB does not impede redemption; SavingsVault withdraw function returns the underlying ERC-20 directly to the depositor wallet.
MiCA Art. 62 (Operational Resilience)	Regular audits, governance, complaint handling	Five-layer defense-in-depth (Section 3.19); Foundry invariants (Section 4.6); Certora reserve invariants (Section 4.5); Tenderly runtime monitoring (Section 4.8).
GENIUS Act § 4 (Federal Payment Stablecoin Framework)	Federal registration of payment-stablecoin issuers	Same posture as MiCA Art. 16: CWB uses, does not issue.
GENIUS Act § 7 (Reserve Disclosure)	Monthly attestation and public disclosure of reserves	Circle / Tether attestations are independently verifiable on-chain; CWB dashboard surfaces issuer attestation links per stablecoin pool.
EU AI Act Art. 86	Right to explanation for AI-assisted decisions	Borrower-facing SHAP explanations (Section 4.2); plain-language SHAP rendering as a regulatory compliance feature, not a UX bonus.
GDPR Art. 17 (Right to Erasure)	Personal data must be deletable on request	No PII on-chain; off-chain PostgreSQL data is encrypted and subject to data-subject deletion requests; only document hashes (SHA-256) appear on-chain.

The result is a compliance posture that is defensible to European and American institutional readers and evaluators in a single page rather than buried in narrative. Compliance gaps that remain—a formal MiCA whitepaper if CWB ever issues its own token, and FinCEN reporting integration in the US—are explicitly named in the Future Work section of the Conclusion.

5.11 Bangladesh Regulatory Reality

Bangladesh is the primary target market named throughout this thesis. The regulatory reality of operating there as of 2026 must be addressed explicitly rather than implied. Three observations frame the path forward.

Crypto remains effectively illegal for retail use as of 2025–2026. Bangladesh Bank’s longstanding 2018 circular treats crypto trading as a Foreign Exchange Regulation Act offence, and no subsequent statute has formally legalized retail crypto activity [R41]. The Crypto World Bank prototype is therefore deployed exclusively on public testnets (Polygon Amoy, Ethereum Sepolia) using test tokens; no Bangladeshi participant transacts with mainnet value at any point in the pre-thesis or final-thesis phase.

The CBDC trajectory is the credible legal path. The central bank launched a CBDC feasibility study in 2022 and a pilot in 2024 but as of 2025 has not produced a deployment timeline. The Crypto World Bank’s positioning against mBridge and Agora (Section 2.2.16) provides a direct upgrade path: when Bangladesh Bank issues a CBDC under the same hierarchical distribution model that the Tan (2023) IMF working paper [5] describes, the platform’s Tier 1 / Tier 2 contracts can settle in that CBDC without any change to the four-tier architecture. The current ETH / USDC denomination is a stand-in for the future CBDC asset.

The Regulatory FinTech Facilitation Office is the entry point. Bangladesh Bank operates a Regulatory FinTech Facilitation Office (RFFO) sandbox for new financial-services products. As of 2026, no crypto-lending project has been licensed under this sandbox, but the framework exists and FE Circular No. 06 (January 2025) on electronic Letters of Credit communication [R42] together with the Payment and Settlement System Act 2024 [R43] provide the relevant regulatory anchors. The platform’s go-to-market path for Bangladesh is therefore: (a) academic deployment on testnet for pre-thesis, (b) a controlled sandbox engagement with RFFO for the final-thesis evaluation, (c) initial pilot with a partner microfinance institution under the sandbox, and (d) phased CBDC integration once the Bangladesh Bank CBDC pilot moves beyond feasibility.

2025 Bangladesh-specific financial-inclusion data. The financial-inclusion argument is updated against Bangladesh Bank Special Publication SP2025-02 (June 2025) [R44]: female-to-male account parity reached 49%/49% by March 2025, and female-owned deposit accounts grew from 33.4 million in 2019 to 55.3 million in 2024. BRAC’s 2025 data shows 89% of loans to women and 92% reporting household income increase [R45]. These statistics are more current and specific than the 2021 Global Findex *1.4 billion unbanked* headline figure used elsewhere in the thesis, and they justify directing the solidarity-group module specifically at women-led borrower groups in line with the BRAC and Grameen tradition this work builds on.

5.12 Bootstrap Funding and the Tier 1 Capitalization Problem

The Crypto World Bank faces a **bootstrap funding problem** analogous to the capitalization challenge of real multilateral development banks. The World Bank Group was initially capitalised by 44 member-state subscriptions in 1944; the IBRD's combined subscribed capital exceeded \$270 billion following its 2018 capital increase (Ocampo & Gallagher, 2024 [R15]). Three initial capitalization mechanisms are proposed for the Tier 1 Reserve:

1. **Founding stakeholder deposits:** Universities, NGOs, and development-focused blockchain organizations deposit ETH or USDC into the Tier 1 Reserve in exchange for governance tokens and yield rights.
2. **Protocol-owned liquidity (POL):** Following the Olympus DAO model, the protocol accumulates treasury reserves through bond mechanisms, where external parties swap ETH for discounted governance tokens over a vesting period.
3. **Philanthropic/impact grant funding:** Development finance institutions (IFC, ADB) have expressed interest in blockchain-based development finance tools, as evidenced by the World Bank FundsChain initiative [39]. Grant funding from these institutions could seed the Tier 1 Reserve in exchange for research access and co-branding.

In the short term, a multi-signature wallet (3-of-5 signers representing different stakeholder groups) governs Tier 1 allocations. In the long term, an on-chain governance module (similar to Compound Governor Bravo) enables token-weighted voting with time-locks to prevent governance attacks.

5.13 Prototype Scope and Limitations

A straightforward version of the complete system is represented by the current prototype:

1. **Hierarchical lending scope:** Utilization of the Tier 1 World Bank Reserve contract is comprehensive. Deposits are managed, loan requests and approvals are processed alongside the display of reserve statistics. Not all functions of the Tier 2 (National Bank) and Tier 3 (Local Bank) contracts have been implemented. Role management and registration are not the only aspects focused on. Currently the process for fund transfers from the World Bank to National Bank and subsequently to Local Bank is incomplete. Four tiers will be included in the final design.
2. **Interest accrual and repayment:** Rules for interest rates (3%/5%/8%) are part of the system design. Fees for initiating loans and penalties for delays also exist. Interest calculations, payment schedules or penalties aren't managed automatically by current smart contracts. Future updates will incorporate these functions.
3. **InterBankLendingPool, UpwardDepositFacility, and broader multi-entity / cross-tier operations:** Same-tier interbank lending, upward surplus repatriation, syndicated and tranching lending, cross-tier treasury FX, and multilateral settlement netting are fully specified in Section 3.13 (with corresponding ERD entities listed in Section 3.13.7). The smart contracts (InterBankLendingPool, UpwardDepositFacility, TreasurySwap, SyndicatedLoan, TranchingPool, NettingEngine) are scheduled for Sprints 2 and 3; the current pre-thesis 1 prototype does not yet exercise these multi-directional flows.
4. **AI/ML integration:** AI tools such as fraud detection (Random Forest) and anomaly

detection (Isolation Forest) along with decision explanations (SHAP) are set to be utilized by the system. A service has been established with FastAPI for machine learning. This service does not currently link to the loan approval process in the existing version.

5. **Backend architecture:** Express.js paired with MongoDB is utilized for rapid API development. The main database is intended to be PostgreSQL. FastAPI is utilized by the machine learning service.
6. **Banking product suite:** The extended banking product suite described in Section 3.17, including savings accounts, fixed-term deposits, transactional accounts, group lending, FX conversion, and the insurance fund, is specified at the architectural and design level in this report. Smart contract implementation and testing of these modules is planned for the final thesis phase. The current prototype establishes the hierarchical lending foundation upon which these modules will be integrated.

The limitations above are consistent with a pre-thesis prototype; the complete system design is fully specified in this report and will be implemented and validated in the final thesis phase.

5.14 Accessibility Assessment: A Borrower in Rural Sylhet

To ground the platform's financial inclusion claims in a concrete context, this section evaluates access across six dimensions for a hypothetical borrower: a small-scale trader in rural Sylhet, Bangladesh, seeking a 50,000 BDT ($\approx$ \$430) working capital loan.

1. Device access. Smartphone penetration in rural Bangladesh reached approximately 51% of adults as of 2024. The platform's mobile-first React interface and WalletConnect integration allow access via any Android or iOS device with a browser, without requiring a desktop or dedicated hardware wallet.

2. Connectivity. 4G coverage in Bangladesh now reaches over 90% of the population, including rural Sylhet, through Grameenphone and Banglalink networks. Polygon PoS transactions are lightweight (under 10 KB) and complete within seconds, well within 4G latency budgets.

3. Transaction cost. At Polygon PoS gas prices, the complete loan lifecycle (request, approval, 6 monthly installments) costs under \$0.02. This contrasts with informal money-lender fees of 5–10% per transaction and formal bank charges of 2–3% origination plus branch travel costs.

4. Language and literacy. The current prototype is English-only. A production deployment serving rural Sylhet must include a Bengali-language interface. This is a known gap and a planned extension; the React component library supports right-to-left and Unicode rendering.

5. Identity and onboarding. Bangladesh's National ID (NID) system covers approximately 110 million registered voters. The planned ZKP KYC extension (Section 3.6.1) allows NID-based credential verification without exposing personal data on-chain. Mobile Financial Services (MFS) accounts—bKash, Nagad, Rocket—provide an existing digital financial identity that 99% more adults accessed in 2021 compared to 2004 [R11].

6. Group lending fit. The solidarity group lending model (GroupLendingPool, planned)

directly mirrors the BRAC and Grameen Bank methodology already familiar to rural Sylhet borrowers. Programmable mutual liability replaces community social pressure with smart contract enforcement, potentially improving both collection reliability and borrower protection.

In aggregate, the Crypto World Bank is accessible in principle to a rural Sylhet borrower today (device, connectivity, cost) with two critical gaps requiring resolution for a production deployment: Bengali-language interface and stablecoin denomination to eliminate currency risk.

Chapter 6

Conclusion

The Crypto World Bank addresses a **multi-party coordination and trust** problem inherent in hierarchical development finance by exploiting the distinctive properties of blockchain technology: cryptographic integrity, immutability, and programmable auditability. It advances beyond existing DeFi lending protocols—which collectively manage over \$55 billion in TVL [13] but universally employ flat, single-tier pool architectures—through a four-tier institutional hierarchy with cross-tier, same-tier, and upward lending flows, combined with a governance structure that addresses network membership, business operations, and technology infrastructure.

This thesis makes four original contributions, as formalized in Section 1.6: (1) a four-tier hierarchical DeFi architecture that mirrors multilateral development finance capital flows, with no comparable prior art in existing protocols; (2) an on-chain solidarity group lending specification that encodes mutual liability and over-indebtedness control (Section 3.17.3) as programmable contract logic; (3) an oracle-mediated AI/ML integration pattern providing a blueprint for auditable AI-assisted credit governance, extended by GNN relational features (Section 4.3) and federated learning across banking tiers (Section 4.4); and (4) a compliance-aware ZKP identity pathway combining zkKYC, zkAML, and W3C DID/VC under a self-sovereign-identity frame (Section 3.6.1). These contributions are framed against the institutional-DeFi landscape (Sygnum, mBridge, Agora) discussed in Section 2.2.16, and against the MiCA / GENIUS Act regulatory frontier mapped in Section 5.10.

The analysis of the competitive situation of 20+ existing projects in DeFi lending, institutional credit, banking infrastructure and financial inclusion affirm that no current system is a combination of multi-level lending (hierarchy), interbank lending mechanisms and graded access by borrowers in one decentralized architecture. The growing blockchain implementation in the institutions, as demonstrated by the World Bank FundsChain initiative [39], the multi-billion-dollar daily volumes of JPMorgan Kinexys [40] and R3 Corda’s \$17 billion in tokenized assets [37]—validates both the technical feasibility and institutional demand of blockchain-based financial infrastructure. The Crypto World Bank extends this trajectory from settlement and fund tracking into a fully open, hierarchically governed lending system.

The platform occupies the intersection of institutional finance, decentralized lending, and financial inclusion, a combination that remains unaddressed in both the academic literature and the commercial blockchain landscape. With a working prototype, a defined market and partnership plan, and a planned go-to-market plan, the Crypto World Bank represents a structurally distinct and architecturally grounded contribution to the emerging field of blockchain-based development finance.

Viewed as a complete banking function checklist, the current prototype implements hierarchical lending workflow foundations (tiered roles, governance framing, architectural capital

flow, and a planned AI/ML monitoring layer) while leaving several bank-grade modules at the design level. Unimplemented modules specified in this report include:

- Deposit products (SavingsVault and FixedDeposit), specified in Section 3.10
- Transactional accounts (CurrentAccount)
- Group lending (GroupLendingPool) with mutual-liability and over-indebtedness logic (Section 3.17.3)
- Liquidation Engine (Section 3.9)
- Cross-chain bridge module (Section 3.12)
- On-chain credit passport SBT (Section 3.11)
- Oracle-priced FX conversion (FXModule)
- Multi-entity and cross-tier operations: InterBankLendingPool, UpwardDepositFacility, SyndicatedLoan, TranchPool, TreasurySwap, and NettingEngine, specified in Section 3.13
- Insurance and depositor protection (InsuranceFund)
- ERC-4337 account abstraction wallet stack (Section 3.7.1)
- Federated learning aggregator and GNN ablation pipeline (Sections 4.3–4.4)
- Certora reserve invariants and Foundry fuzz / invariant suite (Sections 4.5–4.6)
- Tenderly + The Graph runtime monitoring stack (Section 4.8)

Future iterations therefore focus on completing end-to-end cross-tier fund transfers and repayment automation, integrating the analytics layer into real approval flows under the commit-reveal oracle pattern, hardening the platform against regulatory scrutiny via MiCA / GENIUS Act compliance, and expanding the platform into a functionally complete, stablecoin-first, compliance-aware banking system.

The future work will be directed towards the following directions, each grounded in a planned change introduced or formalized in v15:

1. **Complete hierarchical lending implementation.** Complete the end-to-end wiring of cross-tier fund transfers between the World Bank Reserve, National Bank, and Local Bank contracts—automated capital distribution, interest accrual, installment-schedule generation, and cascading-repayment enforcement—against the contract sketches in Appendix B.
2. **Stablecoin-first deployment.** Ship the retail Local Bank tier with USDC as the default loan currency (Section 1.10.3); keep ETH only for institutional tiers. Document the MiCA / GENIUS Act compliance mapping (Section 5.10) against the deployed configuration.
3. **Liquidation Engine, SavingsVault, FixedDeposit, GroupLendingPool.** Implement and audit the LiquidationEngine (Section 3.9), SavingsVault and FixedDeposit (Section 3.10), and the GroupLendingPool with the over-indebtedness controls of Section 3.17.3, completing the Sprint 2 deposit-and-credit suite.
4. **Multi-entity and cross-tier operations.** Implement and audit the six contracts of Section 3.13: InterBankLendingPool (same-tier interbank lending), UpwardDepositFacility (LB → NB → WB surplus repatriation), SyndicatedLoan (multi-entity co-funded loans with on-chain consent vote and pro-rata distribution), TranchPool (senior / junior risk-segmented pools), TreasurySwap (cross-tier ETH ↔ stablecoin treasury swap), and NettingEngine (multilateral settlement netting). These six contracts complete the cross-tier, same-tier, and upward lending claim of Section 1.6.

5. **AI/ML pipeline integration.** Wire Random Forest + Isolation Forest + SHAP into the on-chain loan decision via commit-reveal oracle (Section 4.2); deliver borrower-facing SHAP plain-language explanations (Section 4.2).
6. **GNN extension and federated learning.** Add the GraphSAGE ablation (Section 4.3) for relational fraud detection, and the FedAvg federation across Local / National Banks for privacy-preserving cross-institutional threat intelligence (Section 4.4).
7. **Formal verification with Certora.** Produce a Certora-verified proof of the two reserve invariants of Section 4.5, with the CVL specifications in Appendix B.
8. **Foundry invariant + fuzz suite.** Add the three Foundry invariants (solvency, role segregation, capital-flow direction) running for 10,000 fuzz runs per CI build (Section 4.6).
9. **Agent-based economic feasibility simulation.** Build the Mesa-based ABM of Section 4.7, calibrated against Bangladesh microfinance statistics and stress-tested under a 30% simultaneous-withdrawal scenario.
10. **Account abstraction, tiered KYC, zkAML circuit.** Deploy ERC-4337 account abstraction for retail Tier 4 onboarding (Section 3.7.1), the tiered KYC ladder (Section 3.7.2), and the second ZKP circuit for AML (Section 3.6.1) alongside the existing KYC circuit, both anchored on a W3C DID / VC layer.
11. **On-chain credit passport SBT.** Implement the soulbound credit passport (Section 3.11) and mirror its read interface across chains via the cross-chain bridge architecture of Section 3.12.
12. **Runtime monitoring and dashboard.** Deploy The Graph subgraph, Tenderly alerts, and the WebSocket event listener (Section 4.8) to convert the static dashboard into a live banking dashboard.
13. **Operational security (L5).** Transfer all admin roles to Safe multisigs before any public deployment (Section 3.19); document the formal key-rotation policy; use a Ledger Gen5 hardware-wallet signer for the World Bank Safe.
14. **Mainnet and regulatory sandbox.** Move from testnet to a controlled mainnet deployment under the Bangladesh Bank Regulatory FinTech Facilitation Office sandbox (Section 5.11); pilot with a partner microfinance institution; align with the Bangladesh CBDC trajectory.
15. **Cross-chain interoperability.** Bridge Polygon $\leftrightarrow$ Ethereum (and Arbitrum / Optimism / Base as needed) via Chainlink CCIP (Section 3.12) while preserving the hierarchical borrowing-limit invariant.
16. **LLM assistant fine-tuning and red-teaming.** Fine-tune the 9B-parameter Qwen3.5 model with QLoRA on platform-specific Q&A and run the red-teaming evaluation protocol of Section 4.11.1 for regulatory-hallucination resistance.

Appendix A

Technology Stack

A.1 In-product assistant: local large language model (LLM) integration (prototype)

Purpose. The web application includes a streaming chat assistant to help users understand platform concepts (e.g., the four-tier capital hierarchy), screen-specific workflows, and navigation. During local development, the assistant is served by a **local OpenAI-compatible inference server** (LM Studio) running a quantized Gemma-family model, while the public browser traffic is still routed through the project API to keep keys off the client and to centralize request shaping (system prompts, optional authenticated user context, and error handling).

Model and runtime (local development). The inference endpoint follows the **OpenAI Chat Completions** contract (`/v1/chat/completions`) with `stream: true`, proxied from the project backend to a local service such as `http://127.0.0.1:1234`. The served checkpoint is configured as a Gemma-4 E4B variant (e.g., `google/gemma-4-e4b`) in GGUF form with a practical quantization (e.g., `Q4_K_M`) for a laptop-friendly footprint (on the order of single-digit GBs). The exact quantization label and on-disk name are a deployment artifact; what matters for integration is the **API model identifier** exposed by the local server and referenced by the backend configuration (environment-driven).

End-to-end behavior. The user interface assembles a short transcript (user/assistant messages) and posts it to the backend streaming route. The backend prepends a **system message** with platform + screen “feature” context, optionally enriched by user role when a valid session token is present. The backend then opens a streaming request to the local model server, converts upstream token events into **server-sent events (SSE)** for the browser, and the UI incrementally appends text. The message renderer uses Markdown (including GitHub-flavored extensions), math (KaTeX) for `$$ . . . $$` blocks, and line-break rules appropriate for chat transcripts.

Security and limitations (prototype stance). The assistant is a **read-only product guide**: it is not an authoritative source of on-chain truth and it does not sign transactions. Hallucination risk exists; production hardening would add retrieval of verified platform facts and stronger safety policies. The implementation prioritizes **developer ergonomics** (local model, same-machine loop) over maximum isolation boundaries.

Architecture diagram. Figure A.1 shows the compact end-to-end request path; Figure A.2 expands the same flow with component boundaries and authentication context.

Local LLM Assistant — Compact Request Path

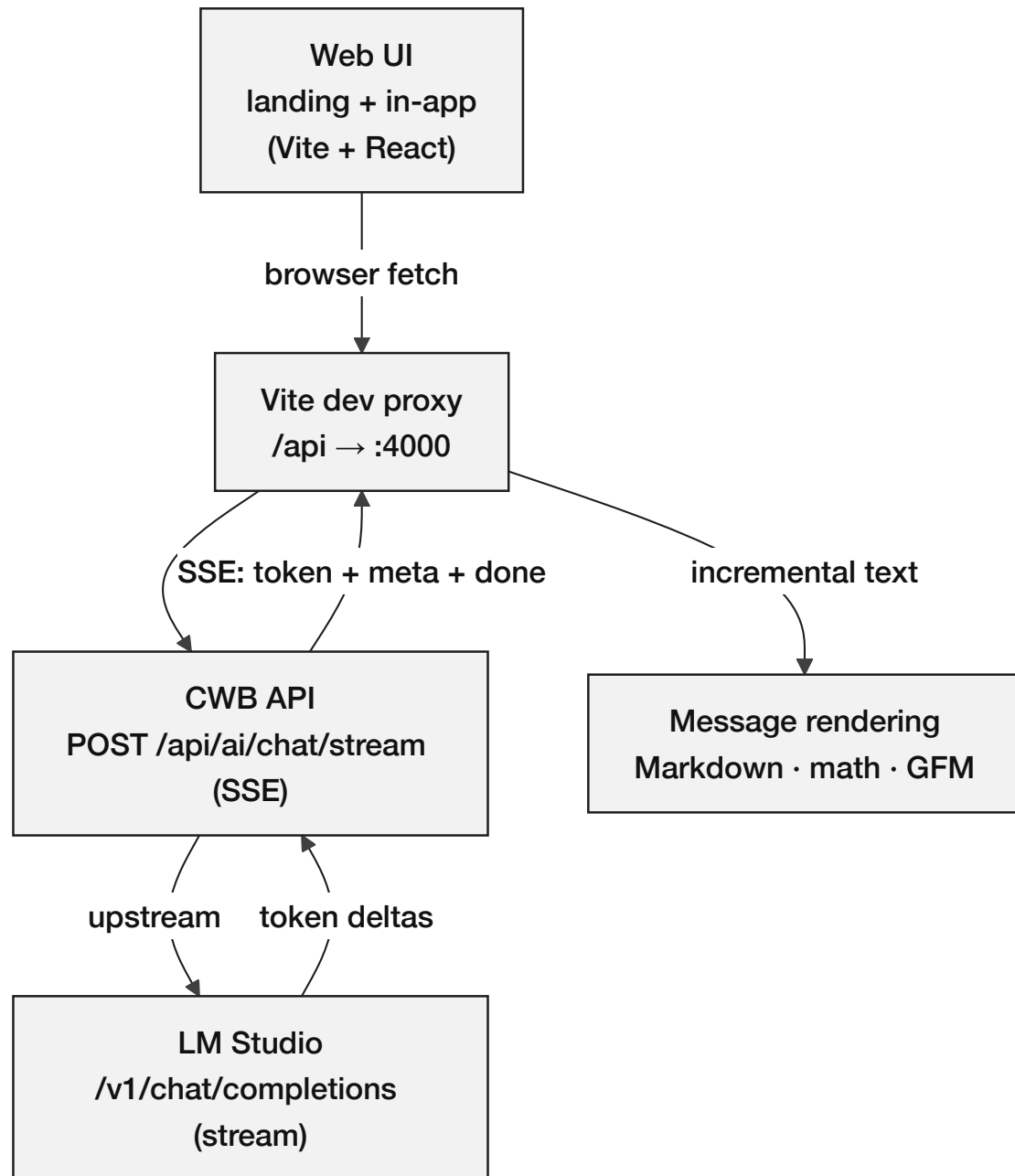


Figure A.1: Compact local LLM assistant request path: browser UI → Vite dev proxy → CWB Express API (SSE) → LM Studio streaming endpoint → incremental Markdown rendering in the UI.

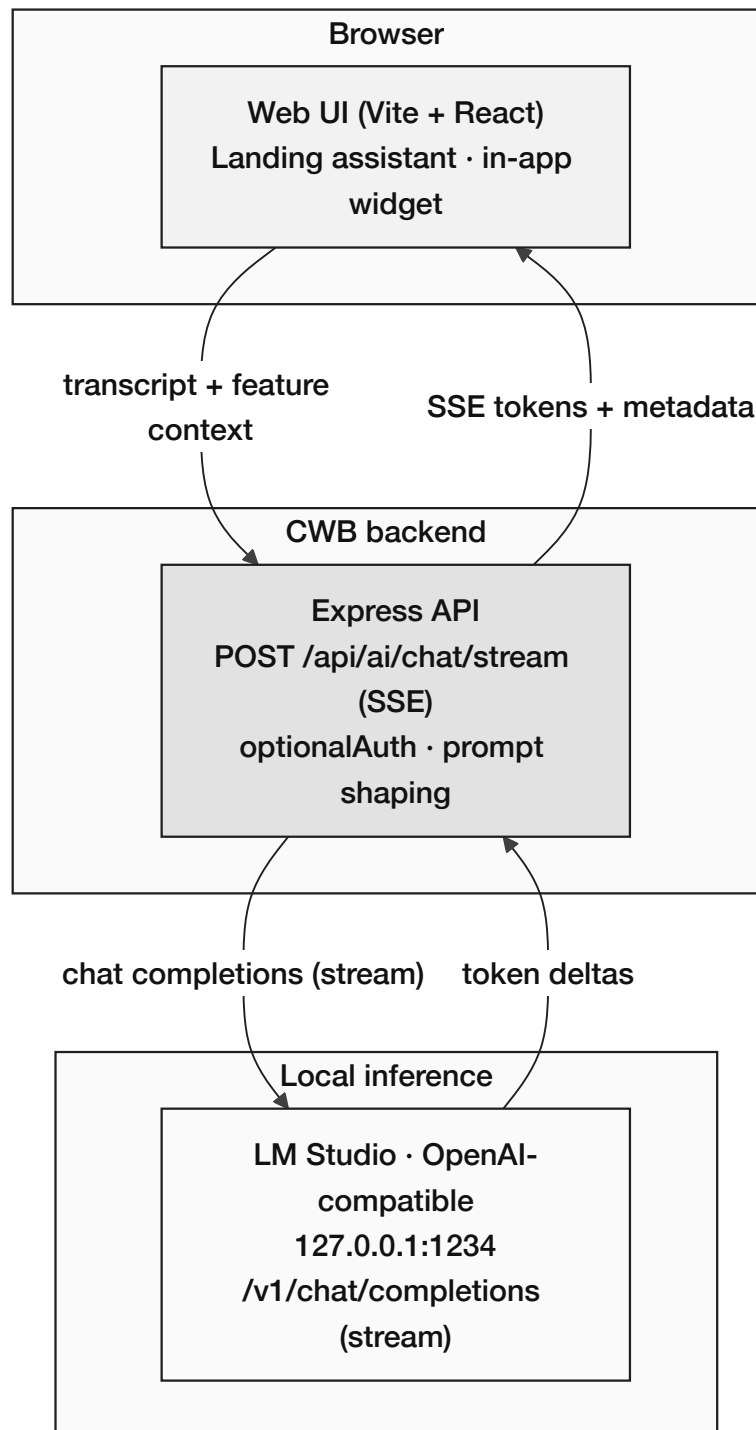
Local LLM Assistant — Component Data Flow

Figure A.2: Expanded local LLM assistant data flow with component boundaries: the browser UI streams from the CWB Express API, which proxies to a local OpenAI-compatible model server (LM Studio) over Server-Sent Events. Optional user context is attached when the request is authenticated.

Implementation analysis (brief). The integration is successful for the project prototype because it reuses a stable transport shape (chat-completions with streaming) and isolates the model vendor behind a single API route, allowing the user interface to remain a thin client. The main practical engineering challenges in local development were: (1) **process/port hygiene** to keep the API bound to a predictable port, (2) **CORS and same-origin** considerations when the UI and API run on different localhost ports, mitigated with a Vite dev proxy to /api and with permissive dev CORS policy on the API, and (3) **RPC provider selection** for wallet reads in the browser, where some public endpoints may fail browser CORS; pinning explicit public RPC endpoints avoids noisy failures unrelated to the chat feature.

Relationship to the earlier rule-based assistant (legacy). The repository also retains early keyword-routed /api/chatbot handlers used for initial prototyping; the LLM path is additive. This separation prevents regressions in deterministic demo endpoints while the conversational assistant evolves.

Table A.1: Technology stack summary.

Layer	Technology	Version / Notes
Smart Contract	Solidity, OpenZeppelin	0.8.20; Ownable, ReentrancyGuard
Frontend	React, TypeScript	Material UI; Design 3
Wallet Integration	Wagmi, RainbowKit, Viem	EIP-1193 compliant
Build and Test	Hardhat	Automated test suite; deployment scripts
Backend API	Express.js, Node.js	REST API; middleware architecture
Database	PostgreSQL (designed)	15 tables, 3NF; relational integrity
Target Networks	Polygon Amoy, Ethereum Sepolia	Public testnets; zero-cost

This appendix table summarizes assistant integration components and configuration aspects used in the prototype (UI, API proxying, and local model server). The purpose is to make the implementation reproducible and to clarify which parts are prototype conveniences versus production requirements.

Appendix B

Smart Contract Capabilities

The complete banking architecture is designed around fifteen modular contracts. The current prototype implements the core three-contract lending foundation; the remaining twelve modules are planned for implementation across Sprints 2 and 3 of the final thesis phase.

Implemented contracts (Sprint 1):

- **World Bank Reserve Contract:** Reserve custody, deposit handling, national bank registration, capital allocation to national banks, system pause and unpause, emergency withdrawal, and system statistics.
- **National Bank Contract:** Local bank registration, borrowing from the World Bank reserve, capital allocation to local banks, and network status queries.
- **Local Bank Contract:** Borrower registration, loan application processing, approval and rejection workflows, installment generation and payment processing, approver management, and user account management.

Planned contracts – Banking product suite (Sprint 2):

- **SavingsVault Contract:** Standard savings account management, variable yield accrual, withdrawal processing, and deposit-to-lending pool integration.
- **FixedDeposit Contract:** Term deposit creation, lock period enforcement, agreed APY accrual, early withdrawal penalty calculation, and maturity release.
- **GroupLendingPool Contract:** Group formation, multi-signature consent recording, shared collateral management, per-member disbursement, mutual liability enforcement, and group credit history recording.
- **FXModule Contract:** Oracle-priced currency conversion, spread calculation, dual-currency lending denomination support, and conversion audit trail.
- **InsuranceFund Contract:** Premium collection, claims processing, default coverage disbursement, and fund reserve management.
- **CurrentAccount Contract:** Transactional account management, atomic peer transfers, recurring payment scheduling, and payroll deposit handling.

Planned contracts – Multi-entity and cross-tier operations (Sprints 2–3):

- **InterBankLendingPool Contract** (Section 3.13.1): one instance per tier (IBLP_NB, IBLP_LB). Short-tenor (1 / 7 / 30 day) same-tier borrowing with a utilization-kinked rate model bounded by the tier-above downward rate; default cascade through reserve buffer → InsuranceFund → parent-tier backstop.
- **UpwardDepositFacility Contract** (Section 3.13.2): role-restricted extension of SavingsVault accounting that lets a lower-tier bank deposit excess reserves into its parent tier, earning a strictly lower yield than the downward borrowing rate; withdrawal gate enforces depositor reserve-ratio safety.

- **SyndicatedLoan Contract** (Section 3.13.3): syndicate proposal, subscription window, unanimous on-chain consent vote, atomic disbursement, pro-rata interest and recovery distribution; supports cross-tier syndication (mixed NB / LB co-funders).
- **TranchedPool Contract** (Section 3.13.4): senior / junior tranching with a subordination-ratio policy, waterfall interest and principal payments, first-loss absorption by the junior tranche; suitable for risk-segmented institutional and impact-aligned capital.
- **TreasurySwap Contract** (Section 3.13.5): oracle-priced atomic asset swap restricted to wallets holding a banking role; cross-tier asset exchange (e.g., NB ETH treasury → USDC) with a tighter spread than retail FX; post-swap reserve-ratio invariant enforced before state change.
- **NettingEngine Contract** (Section 3.13.6): multilateral netting of accumulated interbank obligations; off-chain Settlement Coordinator computes the netted matrix, on-chain `settleBatch` executes the entire batch in one transaction, with a public challenge window for dispute resolution.

Appendix C: Deployed Testnet Contract Addresses

The following table records the contract addresses deployed to the Polygon Amoy testnet during Sprint 1 and Sprint 2. All addresses can be independently verified on the Amoy block explorer at <https://www.oklink.com/amoy>. Deployment was performed using Hardhat v2.22 with a project-specific deployment script; transaction hashes are available in the project repository under `/deployments/amoy/`.

Table B.1: Deployed smart contract addresses, Polygon Amoy testnet (as of pre-thesis submission).

Contract	Network	Address
WorldBankReserve	Polygon Amoy	[See project repository: <code>/deployments/amoy/WorldBankReserve.json</code>]
NationalBank (Scaffold)	Polygon Amoy	[See project repository: <code>/deployments/amoy/NationalBank.json</code>]
LocalBank (Scaffold)	Polygon Amoy	[See project repository: <code>/deployments/amoy/LocalBank.json</code>]
LendingController	Polygon Amoy	[See project repository: <code>/deployments/amoy/LendingController.json</code>]

Note: Exact hex addresses are stored in the project deployment manifest rather than hard-coded here, as testnet redeployment during development may produce updated addresses. The deployment manifest in the repository is the authoritative record. The final thesis submission will include static addresses from a stable named deployment.

Appendix D: WorldBankReserve Contract Interface

The following Solidity interface documents the public API of the WorldBankReserve contract—the fully implemented Tier 1 contract of the prototype. Three design annotations follow the listing:

```
1 // SPDX-License-Identifier: MIT
2 pragma solidity ^0.8.20;
3
4 import "@openzeppelin/contracts/security/ReentrancyGuard.sol";
5 import "@openzeppelin/contracts/access/AccessControl.sol";
6
7 /// @title WorldBankReserve
8 /// @notice Tier 1 reserve contract: manages global capital allocation
9 /// to registered National Bank addresses.
10 contract WorldBankReserve is ReentrancyGuard, AccessControl {
11
12     bytes32 public constant NATIONAL_BANK_ROLE = keccak256("NATIONAL_BANK_ROLE");
13     bytes32 public constant AUDITOR_ROLE = keccak256("AUDITOR_ROLE");
14
15     uint256 public totalReserve;
16     uint256 public minimumReserveRatio; // e.g. 1000 = 10.00%
17     mapping(address => uint256) public allocatedTo; // NationalBank => amount
18
19     event CapitalAllocated(address indexed nationalBank, uint256 amount);
20     event RepaymentReceived(address indexed nationalBank, uint256 amount);
21     event ReserveRatioUpdated(uint256 newRatio);
22
23     /// @notice Allocate capital downward to a registered National Bank (CEI
24     pattern)
25     function allocateCapital(address nationalBank, uint256 amount)
26         external onlyRole(DEFAULT_ADMIN_ROLE) nonReentrant;
27
28     /// @notice Record repayment from a National Bank (CEI pattern)
29     function recordRepayment(uint256 amount)
30         external onlyRole(NATIONAL_BANK_ROLE) nonReentrant;
31
32     /// @notice Query current utilization ratio (scaled 1e4)
33     function utilizationRate() external view returns (uint256);
34
35     /// @notice Returns true if reserve ratio constraint is satisfied
36     function isReserveAdequate() external view returns (bool);
37 }
```

Listing B.1: WorldBankReserve.sol – Tier 1 public interface. Full implementation in project repository.

Design annotation 1 — nonReentrant on state-changing functions. The nonReentrant modifier from OpenZeppelin’s ReentrancyGuard sets a boolean lock before execution and clears it after, preventing recursive calls. It is applied to allocateCapital and recordRepayment because both involve ETH transfers that could trigger attacker-controlled fallback functions. The full Checks-Effects-Interactions analysis is in Section 3.8.

Design annotation 2 — AccessControl mapping to RBAC design. OpenZeppelin’s AccessControl maps each on-chain role (e.g., NATIONAL_BANK_ROLE) to a bytes32 keccak256 hash stored in a role-to-address mapping. Role assignment is controlled by the DEFAULT_ADMIN_ROLE (the World Bank admin). This directly implements the four-tier

RBAC hierarchy described in Section 3.1: only addresses granted `NATIONAL_BANK_ROLE` can call `recordRepayment`, and only the World Bank admin can call `allocateCapital`.

Design annotation 3 — Events enable off-chain indexing for the analytics layer. `CapitalAllocated`, `RepaymentReceived`, and `ReserveRatioUpdated` are emitted at every significant state transition. The off-chain Express.js event listener subscribes to these events and writes them to PostgreSQL, enabling the AI/ML monitoring layer to construct loan lifecycle timelines, compute utilization windows, and feed the Random Forest fraud detector without requiring costly repeated `SLOAD` calls.

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